Data Analytics

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Experience	30+ years in IT as Head-Technology, Project Manager, Team Leader, Developer Worked with Atos-Syntel, American Express (UK), Deutsche Bank (Germany), LTI-Mindtree, i-flex Solutions/Oracle
Expertise	AI & Gen AI, Machine Learning, Deep Learning, Data Analytics Java, C/C++, Python, Databases, Cloud & DevOps Cryptography & Information Security Parallel Programming (CUDA), Hadoop, Apache Spark, MongoDB, Kafka, IBM Mainframe
Teaching	5 years at C-DAC 22 years at Symbiosis
Teaching This Batch	HPCSA (Computer Architecture, Cloud Computing) • DAC (Operating Systems, DevOps) AI (Data Analytics) • DBDA (Linux & Cloud Computing, Data Analytics) DITISS (Public Key Infrastructure & ITIM-DevOps)
Author	80 books – 25 on IT (English) and 65 Non-fiction (Marathi)

Data Analytics

Definition	Process of collecting, cleaning, transforming, and analyzing raw data to uncover useful insights, patterns, and trends.
Purpose	Supports data-driven decision-making for future strategies and actions.
Inputs	Raw data from logs, databases, sensors, files, surveys, APIs, etc.
Outputs	Insights, reports, dashboards, predictions, recommendations.
Techniques Used	Descriptive analytics, Diagnostic analytics, Predictive analytics, Prescriptive analytics.
Common Tools	Excel, SQL, Python, R, Power BI, Tableau, Spark, Hadoop.
Who Uses It	Analysts, data scientists, business leaders, product teams, operations teams.

Key Steps in Data Analytics



Data Analytics Project Lifecycle

Phase	Description	Key Activities
1. Problem Definition	Clearly define the business problem or objective.	Identify goals, KPIs, and success criteria; stakeholder discussions.
2. Data Acquisition	Gather data from relevant sources.	Databases, APIs, IoT devices, web scraping, surveys, logs.
3. Data Preparation / Cleaning	Prepare raw data for analysis.	Handle missing values, remove duplicates, normalize/standardize data, feature engineering.
4. Exploratory Data Analysis (EDA)	Understand the dataset and discover patterns.	Summary statistics, correlation analysis, visualizations, detecting outliers.
5. Modeling / Analysis	Apply analytical or machine learning techniques to extract insights.	Statistical modeling, regression, classification, clustering, forecasting.
6. Evaluation and Validation	Assess model performance and analysis accuracy.	Cross-validation, metrics evaluation (accuracy, RMSE, F1-score), sensitivity analysis.
7. Visualization and Reporting	Present findings in an understandable form.	Dashboards, charts, graphs, executive summaries, interactive visualizations.
8. Deployment / Action	Implement insights in business decisions or automated systems.	Integrate models into apps, business process automation, decision support systems.
9. Monitoring and Maintenance	Track performance and update models as needed.	Monitor outcomes, retrain models, update data pipelines, feedback loops.

Types of Data Analytics

Type	Purpose / Question Answered	Examples
Descriptive Analytics	Answers “What happened?” — summarizes past data to identify patterns or trends.	Monthly sales reports, website traffic summaries, customer feedback trends.
Diagnostic Analytics	Answers “Why did it happen?” — finds causes of past outcomes.	Identifying why sales dropped in a particular region, analyzing customer churn reasons.
Predictive Analytics	Answers “What is likely to happen?” — uses historical data to forecast future outcomes.	Predicting product demand, forecasting stock prices, credit risk modeling.
Prescriptive Analytics	Answers “What should we do?” — recommends actions for desired outcomes.	Suggesting optimal pricing strategies, supply chain optimization, personalized marketing offers.

Python Libraries for Data Analytics

Library	Purpose / Use	Typical Applications
NumPy	Numerical computing	Scientific computing, data manipulation, numerical simulations
Pandas	Data manipulation and analysis	Data cleaning, preprocessing, EDA, time series analysis
Matplotlib	Data visualization	Visualizing trends, reporting, dashboards
Seaborn	Statistical data visualization	Visualizing distributions, correlations, statistical patterns
scikit-learn	Machine learning and modeling	Predictive modeling, classification, regression, clustering, feature selection

NumPy

Feature	Description	Example / Notes
Library Purpose	Used for creating and working with N-dimensional arrays	Core library for numerical computing in Python
Efficiency	More efficient than lists due to homogeneous data and contiguous memory storage	Faster computations and lower memory usage
Homogeneity	All elements in a NumPy array are of the same data type	Unlike Python lists which allow mixed types
Vectorization	Perform operations on entire arrays without loops	result = arr + 5
Broadcasting	Perform operations on arrays of different shapes	result = arr_2d + arr_1d
Linear Algebra	Supports matrix operations, eigenvalues, eigenvectors, etc.	np.dot(), np.linalg.eig()
Statistical Functions	Provides functions like mean, median, standard deviation, covariance	np.mean(), np.std()
Random Number Generation	Generates random numbers from various distributions	np.random.normal(), np.random.uniform()
Missing Data Handling	Can handle missing data (using np.nan)	Useful in data preprocessing
Compatibility	Works well with Pandas, Matplotlib, SciPy, TensorFlow, etc.	Backbone of Python data ecosystem

Various Ways to Create a NumPy Array

Method	Code	Output
1. From a Python list	np.array([1, 2, 3])	[1 2 3]
2. From a nested list	np.array([[1, 2, 3], [4, 5, 6]])	[[1 2 3] [4 5 6]]
3. Using np.arange()	np.arange(0, 10, 2)	[0 2 4 6 8]
4. Using np.linspace()	np.linspace(0, 1, 5)	[0. 0.25 0.5 0.75 1.]
5. Using np.zeros()	np.zeros((2, 3))	[[0. 0. 0.] [0. 0. 0.]]
6. Using np.ones()	np.ones((2, 3))	[[1. 1. 1.] [1. 1. 1.]]
7. Using np.eye()	np.eye(3)	[[1. 0. 0.] [0. 1. 0.] [0. 0. 1.]]
8. Using np.full()	np.full((2, 2), 7)	[[7 7] [7 7]]
9. Using np.random.rand()	np.random.rand(2, 2)	[[0.19 0.86] [0.43 0.72]] (random)

Python List vs NumPy Array

Python List

```
a = [1, 2, 3]
```

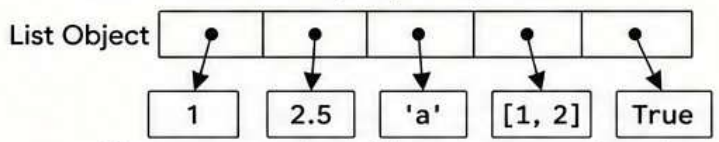
Flexible
Easily grow, shrink, insert, delete

Basic Python
Built-in Python data type

Heterogeneous
Allows mixed data types

Memory & Storage

✗ **Not Stored Sequentially**
Each element is a PyObject* pointer



Elements stored at different memory locations

Operations

```
[1, 2, 3] * 2
```

↓

```
[1, 2, 3, 1, 2, 3]
```

- Repeats the list
- Not element-wise multiplication

⚠ **Not Element-wise**

NumPy Array

```
a = np.array([1, 2, 3])
```

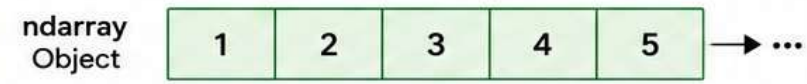
Fast
Optimized for performance

Numerical Computing
Designed for mathematical operations

Homogeneous
All elements must be same data type

Memory & Storage

✓ **Stored Sequentially**
Contiguous memory block



Elements stored next to each other in memory

Operations

```
np.array([1, 2, 3]) * 2
```

↓

```
[2, 4, 6]
```

- Element-wise multiplication
- Vectorized computation

✓ **Element-wise (Vectorized)**

Slower	Performance	Faster (Vectorized)
More Memory	Memory Usage	Less Memory
General Purpose	Use Case	Scientific & Numerical Computing
Built-in (No extra library)	Dependency	Requires NumPy

NumPy Arrays: np.arange() vs np.linspace()

np.arange(start, stop, step)

arr = np.arange(0, 10, 1)
→ [0 1 2 3 4 5 6 7 8 9]

- Starts at start; stops before stop
- Third parameter = step

arr[4] = 88
→ [0 1 2 3 88 5 6 7 8 9]

- Use for integer sequences with a step

np.linspace(start, stop, num)

arr = np.linspace(0, 10, 5)
→ [0. 2.5 5. 7.5 10.]

- Includes both start AND end value
- Third parameter = number of values (not step)
- Returns float values

arr[1] = 99
→ [0. 99. 5. 7.5 10.]

- Use when you need a fixed number of evenly spaced values (including the end) – graphs, ML, data sampling

Array Filtering in NumPy

Step	Code	Result	Explanation
1. Create array	arr = np.arange(55, 65)	[55 56 57 58 59 60 61 62 63 64]	Creates values from 55 to 64
2. Apply condition	arr > 58	[F F F F T T T T T T]	Checks which values are greater than 58
3. Boolean mask	boolean_arr = arr > 58	[False False False False True True True True True True]	Stores True/False result
4. Filter values	arr[boolean_arr]	[59 60 61 62 63 64]	Selects only True values

In one line: arr[arr > 58] → [59 60 61 62 63 64]

Shallow Versus Deep Copy

Feature	Shallow Copy (b = a[0:5])	Deep Copy (b = a.copy())
Memory	Shares memory with a	Separate memory
Relationship	b is a **view** of a	b is **independent** of a
Change in b	Reflects in a	Does NOT affect a
Change in a	Reflects in b	Does NOT affect b
Use Case	When we need a **view**	When we need a **full copy**

Random Numbers in NumPy

Function	Description	Example	Sample Output
randint	Random integers in a range	np.random.randint(1, 10, 5)	3 7 2 9 5
rand	Uniform random numbers (0 to 1)	np.random.rand(5)	0.23 0.89 0.45 0.12 0.67
uniform	Random floats in given range	np.random.uniform(10, 50, 5)	12.5 33.2 41.8 25.6 18.9
normal	Normal distribution (mean, std)	np.random.normal(50, 5, 5)	48.2 52.1 55.3 46.7 49.8
seed	Fix random output (reproducible)	np.random.seed(42)	Same output every run

Pandas

Feature	Description	Example / Notes
Purpose	Library for data analysis	Built on top of NumPy
Core Structure	DataFrame (table-like structure)	Rows and columns with labels
Performance	Computationally efficient	Optimized for large datasets
Read/Write Data	Supports multiple formats	CSV, Excel, JSON, SQL
Data Operations	Apply logic and subset data	Filtering, conditions
Missing Data	Handles missing values easily	NaN, fillna(), dropna()
Data Restructuring	Modify and reshape data	groupby(), pivot()

Series

What is it? A data structure that holds an array of information along with a labelled (named) index.

NumPy Array – has a numeric index

Index	Data
0	67
1	36
2	81

- We can still use the numeric index, if we want

Pandas Series – has a labelled index

Labelled Index	Data
Alice	67
Bob	36
Charlie	81

- Finding data using this index is very easy

In short: Series = Data + Meaningful (labelled) Index

NumPy Array versus Pandas Series

Step	NumPy Array	Pandas Series
Create Data	import numpy as np arr = np.array([100, 200, 300])	import pandas as pd s = pd.Series([100, 200, 300], index=['A','B','C'])
Output	[100 200 300]	A 100 B 200 C 300
Index	0 1 2	A B C
Access	arr[1] → 200	s['B'] → 200
Add 50	[150 250 350]	A 150 B 250 C 350
Filter (>150)	[200 300]	B 200 C 300
Mean	200.0	200.0

Pandas Series Features

Feature	Explanation	Example
1. 1D labeled array	Holds one-dimensional data with **labels (index)**	s = pd.Series([10, 20, 30], index=['a','b','c'])
2. Heterogeneous data	Can store different data types (int, float, string, objects)	pd.Series([10, "hello", 3.14])
3. Indexing	Access values via index labels or positions	s['a'], s.iloc[0]
4. Vectorized ops	Supports element-wise operations like NumPy arrays	s + 10, s * 2
5. Alignment	Auto-aligns by index when operating on multiple Series	s1 + s2 (aligns on index labels)
6. Missing data support	Uses NaN for missing values	pd.Series([1, None, 3])
7. Name attribute	Both Series and its index can have names	s.name = "marks"; s.index.name = "students"
8. Conversion	Can be converted to/from lists, dicts, NumPy arrays	s.to_numpy(), pd.Series(dict_obj)
9. Functions	Rich set of functions: sum(), mean(), value_counts(), etc.	s.mean(), s.value_counts()
10. Slicing	Works with both label-based (loc) and position-based (iloc) indexing … Note: Label-based indexing is inclusive at both ends (a, b, c) but position-based is exclusive at the end (0, 1 here)	s.loc['a':'c'], s.iloc[0:2]

DataFrame

A **DataFrame** is a table (2D data structure) with rows and columns.
It is made up of multiple **Series** that share the same index.

1 Series

Marks

Index	Marks
Alice	67
Bob	36
Charles	81

2 Multiple Series (same index)

Marks

Index	Marks
Alice	67
Bob	36
Charles	81

Course

Index	Course
Alice	DBDA
Bob	DAC
Charles	DBDA

Rank

Index	Rank
Alice	2
Bob	3
Charles	1

3 DataFrame

Index	Marks	Course	Rank
Alice	67	DBDA	2
Bob	36	DAC	3
Charles	81	DBDA	1

DataFrame = Several Series that share the same index (like a spreadsheet)

Why use DataFrame?

- Easy to filter and select data
- Easy to clean and transform data
- Easy to analyze and visualize data
- Works like a spreadsheet

Pandas DataFrame: Basic Operations

Operation	Code Example	Notes
Create a DataFrame	df = pd.DataFrame({'Sales':[100,200,300], 'Profit':[20,50,70]}, index=['TCS','Infosys','Wipro'])	Creates DataFrame with company names as row labels
Select one column	df['Sales']	Returns a Series
Select multiple columns	df[['Sales','Profit']]	Returns a DataFrame
Select a row by label	df.loc['Infosys']	Uses row label (company name)
Select a row by position	df.iloc[1]	Uses row number
Select one value	df.loc['Infosys','Profit']	Gets a single cell value
Add a new column	df['Margin'] = df['Profit']/df['Sales']*100	Adds new column
Add a new row	df.loc['HCL'] = [150,30,20]	Adds new company row
Update a value	df.loc['TCS','Sales'] = 120	Modifies existing value
Delete a column	df = df.drop('Margin', axis=1)	Removes a column

Series vs DataFrame

Series – 1-D labeled data

Index	Sales	Index	Profit
TCS	100	TCS	20
Infosys	200	Infosys	50
Wipro	300	Wipro	70

- One-dimensional data (single column)
- Each Series has its own index
- Need MULTIPLE objects to represent multiple columns

  sales_s['Infosys'] -> 200; profit_s['Infosys'] -> 50
- Good for: single column data, time series, simple lookups

DataFrame – 2-D labeled tabular data

Company	Sales	Profit
TCS	100	20
Infosys	200	50
Wipro	300	70

- Two-dimensional data (rows & columns)
- All data in a SINGLE structured object
- Easy to add, remove or modify columns
- Built-in alignment by index

  df.loc['Infosys'] -> Sales: 200, Profit: 50
- Good for: tabular data, analysis, ML features, reporting

Bottom line: Use Series for single-column data. Use DataFrame for structured, multi-column data.

loc Versus iloc

Player	Match 1	Match 2	Match 3
Virat	45	67	82
Rohit	55	72	60
MS Dhoni	30	40	50

Operation	.loc[] (Label-based)	Output	.iloc[] (Position-based)	Output
First row	df.loc['Virat']	45 67 82	df.iloc[0]	45 67 82
Second row	df.loc['Rohit']	55 72 60	df.iloc[1]	55 72 60
Specific cell	df.loc['Virat','Match 2']	67	df.iloc[0,1]	67
Multiple rows	df.loc[['Virat','Rohit']]	2 rows	df.iloc[0:2]	First 2 rows
Subset	df.loc[['Virat','MS Dhoni'], ['Match 1','Match 3']]	45 82 / 30 50	df.iloc[[0,2],[0,2]]	Same values
Slicing	df.loc['Virat':'Rohit']	Includes Rohit	df.iloc[0:2]	Excludes index 2

Pandas DataFrame: Advanced Operations

Operation	Code	Use
Indexing	df.set_index('A')	Set column as index
Filtering	df[df['B'] > 3]	Filter rows
Check missing	df.isnull()	Find missing values
Fill missing	df.fillna(0)	Replace NaN
Drop missing	df.dropna()	Remove missing rows
Grouping	df.groupby('A')['B'].mean()	Aggregate data
Merge (join)	pd.merge(df1, df2, on='key')	Add new columns
Concat	pd.concat([df1, df2])	Add new rows

CONCATENATE

Stack DataFrames vertically (row-wise)

df1

Employee	Role
John	Manager
Alice	Analyst
Bob	Developer

+

df2

Employee	Role
Kiran	Designer

→

Combined DataFrame

Employee	Role
John	Manager
Alice	Analyst
Bob	Developer
Kiran	Designer

```
pd.concat([df1, df2], ignore_index=True)
```

Key Point: Appends rows. Columns must match.

MERGE

Join DataFrames horizontally (column-wise) on a key

df1

Employee	Role	Department
John	Manager	Sales
Alice	Analyst	Finance
Bob	Developer	IT

+

df2

Employee	Salary
John	70000
Alice	65000
Kiran	60000

→

Merged DataFrame

Employee	Role	Department	Salary
John	Manager	Sales	70000
Alice	Analyst	Finance	65000
Bob	Developer	IT	NaN
Kiran	Designer	NaN	60000

```
pd.merge(df1,df2, on='Employee', how='outer')
```

Key Insights

- **CONCATENATE** = Stack rows (same columns)
- **MERGE** = Join columns (using a key)

Notes

- Use `ignore_index=True` in concat to reindex rows.
- Use `how='inner' | 'left' | 'right' | 'outer'` in merge.

Pandas Searching/Filtering

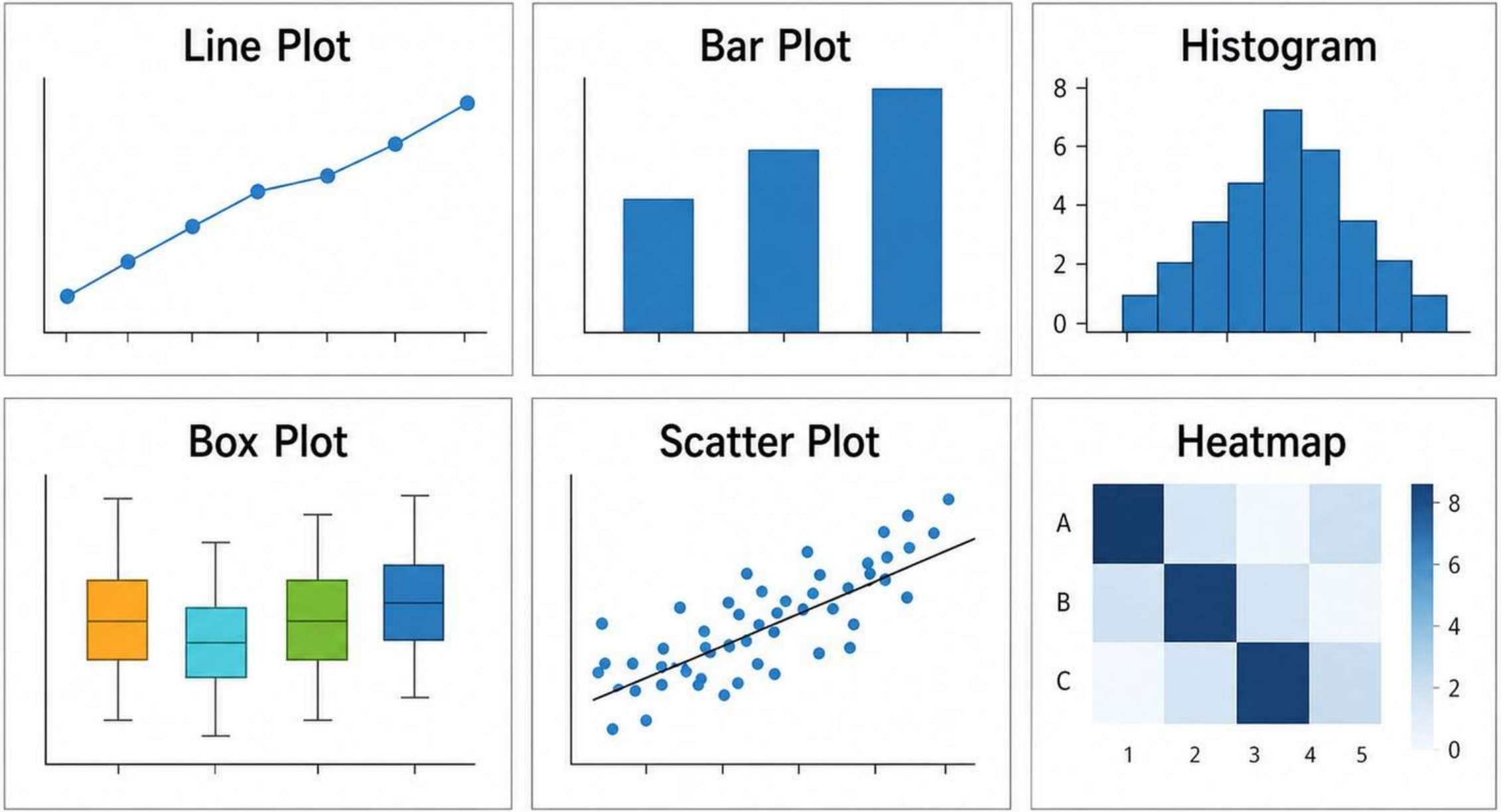
Method	Example	Use
Boolean	df[df['A'] > 150]	Filter rows
Multiple conditions	df[(df['A'] > 100) & (df['B'] < 30)]	Combine conditions
.loc[]	df.loc[df['B'] == 30, ['A','B']]	Filter + select columns
query()	df.query("A > 150 and B < 30")	Clean readable filtering
isin()	df[df['A'].isin([100,300])]	Match values in list
between()	df[df['A'].between(150,300)]	Range filtering
String filter	df[df['name'].str.contains("Ali")]	Text search
Top values	df.nlargest(2,'A')	Top N rows

Python Pandas Crosstab

- Pandas crosstab() method computes the cross tabulation of multiple column variables

Age	Female	Male
20	0	2
40	1	0
50	0	1

Matplotlib and Seaborn


Line Plot
Bar Plot
Histogram
8
6
4
2
0
Box Plot
Scatter Plot
Heatmap
A
B
C
1
2
3
4
5
8
6
4
2
0

Important Plots & Their Purpose

Plot	Purpose	Example from Tips Dataset
Line Plot	Show trend over order or time	Average tip across row index / customer order
Bar Plot	Compare values across categories	Average total_bill for each day
Histogram	Show distribution of one numeric column	Distribution of total_bill values
Box Plot	Show spread, median, outliers	Spread of tip for each day
Scatter Plot	Show relationship between two numeric columns	total_bill vs tip
Heatmap	Show matrix values / correlation	Correlation between total_bill, tip, and size

Matplotlib

Why Matplotlib?	Turns data into clear visuals to find trends, patterns, peaks, and anomalies.
Input Data	Works with lists, NumPy arrays, Pandas Series, and DataFrames.
Programming Styles	Functional: plt.plot() OOP: fig, ax = plt.subplots() OOP is best for professional, multi-plot dashboards.
Real-Life Examples	Sales trend • Marks distribution • Salary comparison • ML accuracy vs epochs

Matplotlib Subplots – 2x2 Grid

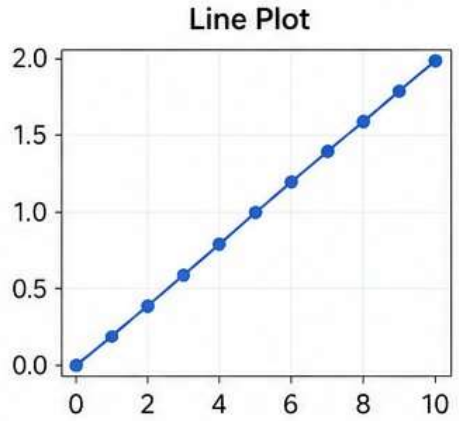
Figure size: (2, 2) → 2 rows, 2 columns

Line Plot

```
F: plt.subplot(2, 2, 1)
OO: axes[0, 0]
```

What it means

- Row: 0 (first row)
- Column: 0 (first column)

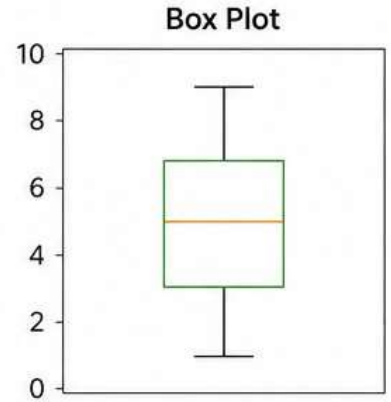


Box Plot

```
F: plt.subplot(2, 2, 2)
OO: axes[0, 1]
```

What it means

- Row: 0 (first row)
- Column: 1 (second column)

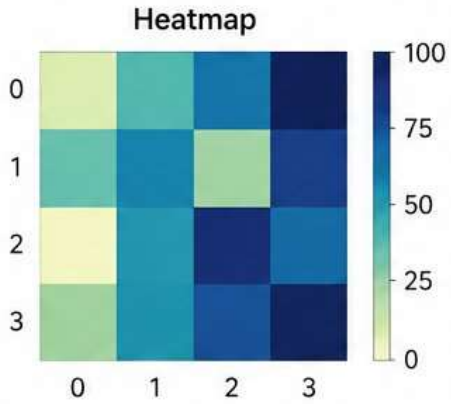


Heatmap

```
F: plt.subplot(2, 2, 3)
OO: axes[1, 0]
```

What it means

- Row: 1 (second row)
- Column: 0 (first column)

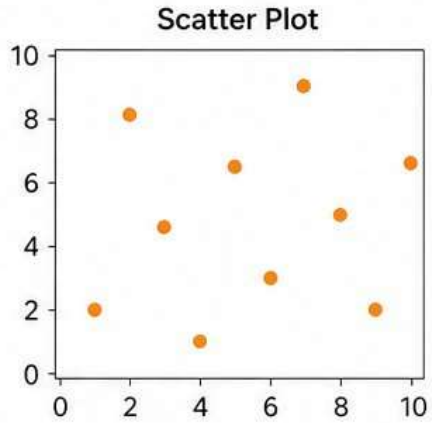


Scatter Plot

```
F: plt.subplot(2, 2, 4)
OO: axes[1, 1]
```

What it means

- Row: 1 (second row)
- Column: 1 (second column)



Quick Summary

- `plt.subplot(nrows, ncols, index)` → index starts at 1 and goes row-wise
- `axes[row, col]` → row and col start at 0

1	2
3	4

Indices (1-based)

Layout (row-wise)

Seaborn

Built On	Matplotlib
Purpose	Statistical data visualization
Syntax	High-level, simple
Data Input	Works directly with DataFrames
Default Look	Better (styled plots)
Code Size	Less code
Customization	Easy + readable
Best For	Data analysis, dashboards

Statistics: Basic Concepts

Individuals and Variables

Term	Meaning	Example
Individual	Item or entity in a dataset	Person
Variable	Property or characteristic of an individual	Hours of Study
Independent Variable (Feature)	Input used for prediction	Hours of Study
Dependent Variable (Label)	Output we want to predict	Grade

Example dataset

Person	Hours of Study	Grade
Alice	100	B
Bob	130	A
Charlie	60	E
Dave	90	C

Variable Types: Qualitative and Quantitative

Type	Characteristics	Example
Qualitative (Categorical)	Non-numeric Represents categories or labels	Grade (A, B, C, etc.)
Quantitative (Numeric)	Numeric Has measurable values	Hours of Study

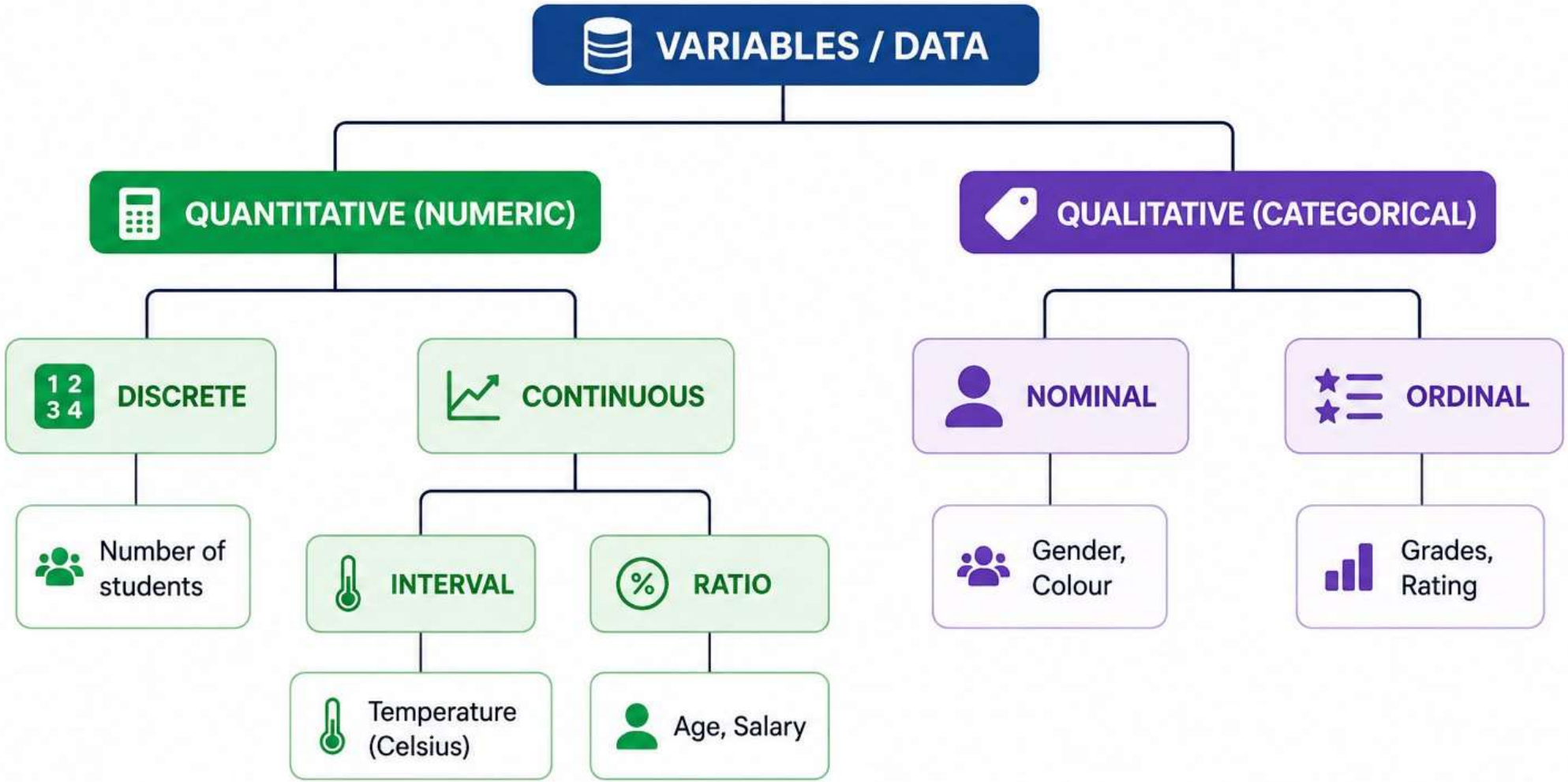
Example dataset

Person	Hours of Study (Quantitative)	Grade (Qualitative)
Alice	100	B
Bob	130	A
Charlie	60	E
Dave	90	C

Quantitative Variable Sub-types

Variable Type	Definition	Key Property	Example	Notes
Discrete	Countable values that take specific, separate numbers	Only whole numbers (no fractions)	Number of products sold (1, 2, 3, …) Number of students = 23	Cannot take values like 1.5 or 3.8
Continuous	Measurable values that can take any value within a range	Includes fractions and decimals	Height of a person (e.g., 5 ft 5 in or 2 cm) Height = 181.73 cm	Precision can go infinitely finer (mm, µm, etc.)

VARIABLE CLASSIFICATION



QUANTITATIVE (NUMERIC)
Numeric values that can be counted or measured.

QUALITATIVE (CATEGORICAL)
Non-numeric values that represent categories or labels.

Levels / Scales of Measurement of Data

Level	Type	Key Feature	Example
Nominal	Categorical	No order or ranking	Gender, Colour, Cuisine
Ordinal	Categorical	Has a meaningful order or ranking	Education level, Customer satisfaction
Interval	Numeric	Equal intervals, no true zero	Temperature (°C, °F), Student marks
Ratio	Numeric	Equal intervals, true zero	Height, Weight, Income, Distance

True zero point: does the value 0 indicate the start?

- Ratio: 0 means absence
- Interval: 0 is arbitrary

Descriptive Statistics

Measures of Location (Central Tendency)

- A single value that represents the “center” of a set of data
- Example: marks of 11 students (ascending order): 45, 56, 61, 65, 68, 71, 73, 79, 82, 88, 91

Measure	Meaning	Example	When to use	Data type
Mean	Average of all values	70.82	When data is normally distributed and has no outliers	Interval, Ratio
Median	Middle value	71	When data is skewed or has extreme values (outliers)	Ordinal, Interval, Ratio
Mode	Most common value	None	When identifying the most frequent value	Categorical (best), also numeric

Mean

- **Mean**: Average
- Sample Marks: 45,56,61,65,68,71,73,79,82,88,91
- Mean = $\frac{45+56+61+65+68+71+73+79+82+88+91}{11}$
- Mean = $\frac{779}{11}$
- Mean ≈ 70.82
- In statistical terms: $\boldsymbol{\mu} = \frac{\sum_{i=1}^{n} x_i}{n}$
- Extreme values (**outliers**) can impact mean
  - Test scores: 70, 80, 90, 85, and 75 … Mean: 80
  - Test scores: 70, 80, 90, 85, 75, and 2 … Mean: 65.4

Median

- **Median**: The *middle* value when values are in ascending/descending order
- Divides the dataset into two equal halves
  - Odd number of values in the dataset: Median is the middle value
  - Even number of values in the dataset: Median is the average of the two middle values
- Scores: 62, 78, 84, 89, 91, 95, 97 ---> Median = 89
- Scores: 62, 78, 84, 89, 91, 95 ---> Median = $\frac{84+89}{2} = 86.5$
- Outliers do not impact median as much as they impact mean
  - Mean of 56, 78, 45, 49, 55, 62 = 57.5
  - Mean of 56, 180, 45, 49, 55, 62 = 74.5
  - Corresponding medians = 55.5 and again = 55.5

Mode

- **Mode**: The value that occurs most frequently in a dataset
- Data: 62, 78, 84, 89, 91, 95, 97, 89, 91, 89
- Frequency: 62: 1, 78: 1, 84: 1, 89: 3, 91: 2, 95: 1, 97: 1
- Mode = 89
- What if there are multiple values with the same highest frequency?: **Multimodal** data
  - If we have two modes: **bi-modal**
  - If we have three modes: **tri-modal**
- Not used much in practice

Mean, Mode, Median Example

Day	Temperature (°C)
1	30
2	31
3	29
4	30
5	32
6	31
7	30
8	29
9	30
10	31
11	30
12	32
13	31
14	30

Measure	Value
Mean	30.43
Median	30
Mode	30

Measure	Value
Mean	26.64
Median	21
Mode	20

Day	Products Sold
1	20
2	22
3	21
4	20
5	23
6	21
7	20
8	22
9	21
10	20
11	22
12	21
13	20
14	100

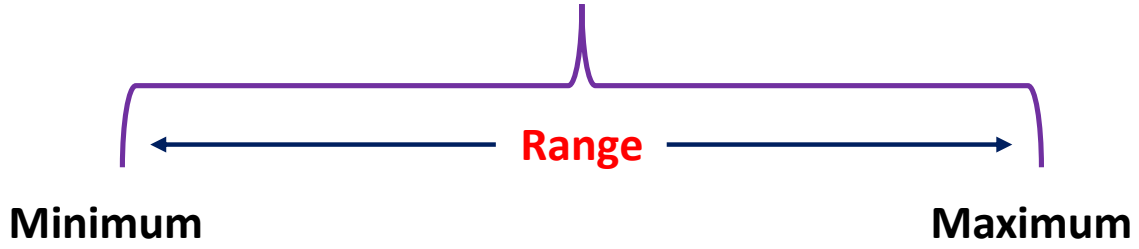
Measures of Dispersion

- Measures of Dispersion (Scatter): how and by how much our data set is spread out around its center

Measure	Meaning
Range	Difference between maximum and minimum values
Variance	Average of the squared deviations from the mean
Standard Deviation	Square root of variance; measured in the same units as the data

Range

- **Range**: Difference between the maximum value and the minimum value in the data set
- Affected by outliers



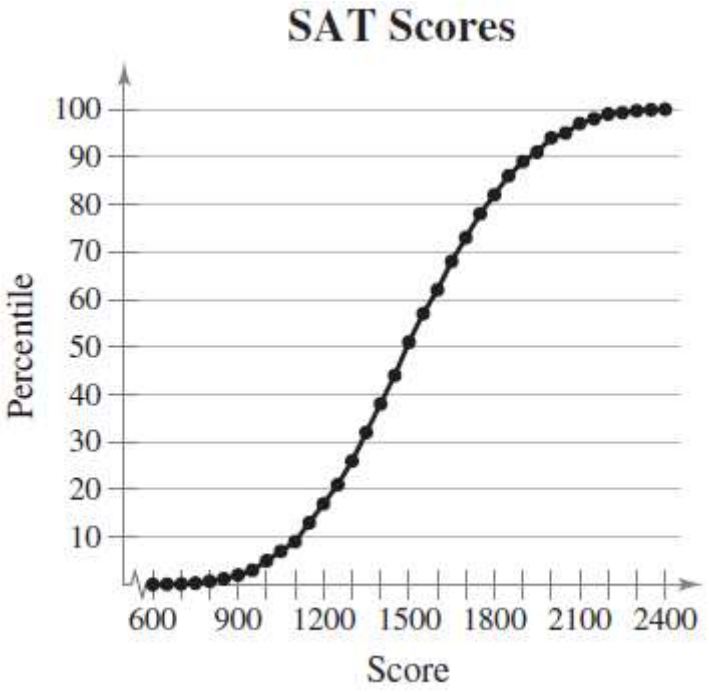
- Example: 8, 11, 5, 9, 7, 6, 2500
- Range = Max – Min = 2500 – 5 = 2495, which is quite meaningless
- Solution: **Inter Quartile Range (IQR)**
- But first, we need to understand **percentile** and **quartile**

Percentile

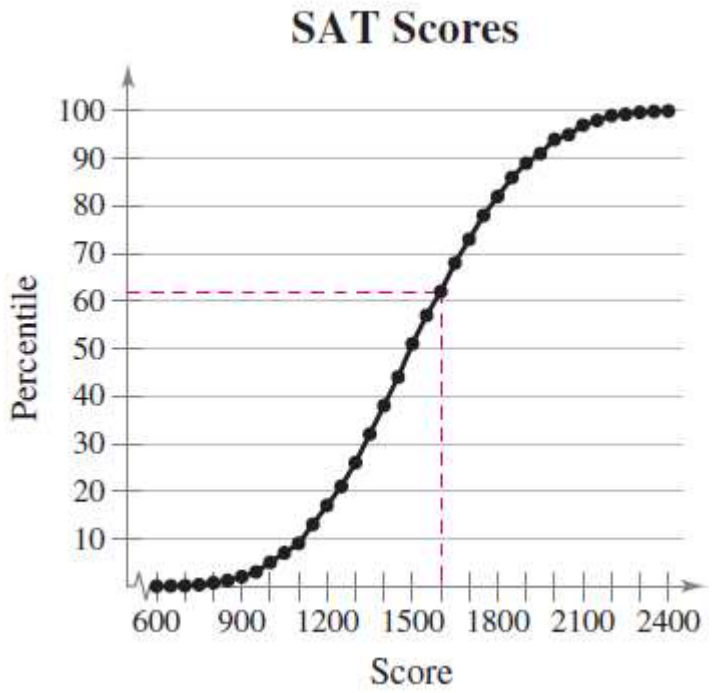
- **Percentile** (Relative): ≠ Percentage (Absolute)
- Percentile: A value below which certain percentage of observations lie
- Slices percentage data into two parts: Below a certain cut off, Above the same cut off
- k^{th} percentile = k% data is below it, and rest is above it
- Examples:
  - If you are in the 90th percentile in an examination, 90% students are below you and 10% students are above you
  - If a patient's blood pressure is in the 60th percentile, 60% patients have a blood pressure less than this patient, and 40% patients have higher blood pressure than this patient
- Median = 50th percentile

Percentile Example

- General graph



Score at the 62nd percentile



Percentile Example

In some references, we might see *Number of observations*, rather than *Number of observations + 1* … Generally does not make a big difference

- Sorted Marks of 20 students: 65,72,78,80,81,83,85,87,88,90,91,92,93,94,95,96,97,98,99,100
- 10th percentile position = $\frac{Desired\ percentile}{100}\ x\ Number\ of\ observations + 1 = \frac{10}{100}\ x\ 21 = 2.10$
- Marks at 10th percentile = Marks at the second position ~ $\frac{(72 + 78)}{2}$ = 75
- Interpretation: 10% students lie below 75 marks, 90% are above 75 marks

- 90th percentile position = $\frac{Desired\ percentile}{100}\ x\ Number\ of\ observations\ + 1\ = \frac{90}{100}\ x\ 21 = 18.9 \sim \frac{(98+99)}{2}$ = 98.5
- Marks at 90th percentile = Marks at the 18th position = 98.5
- Interpretation: 90% students lie below 98.5 marks, 10% are above 98.5 marks

Percentile Example

- Sorted Marks of 20 students: 65,72,78,80,81,83,85,87,88,90,91,92,93,94,95,96,97,98,99,100

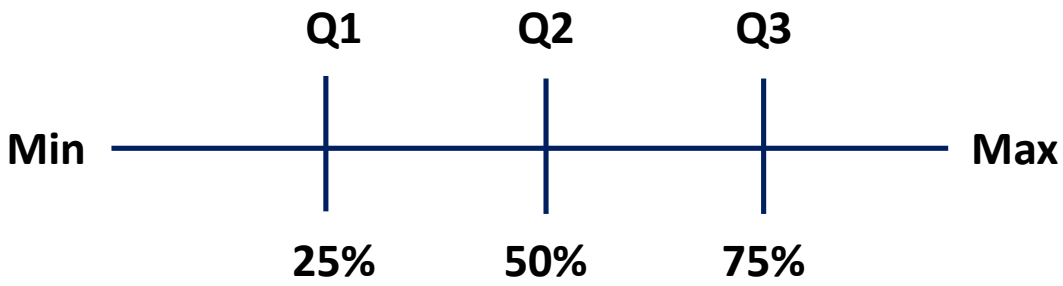
- Another type of question: At what percentile are 85 marks in the above list?
- Percentile for 85 marks = $\frac{Number\ of\ observations\ below\ the\ given\ value}{Number\ of\ observations}\ x\ 100 = \frac{6}{20}\ x\ 100$ = 30
- 85 marks are at the 30th percentile
- Interpretation: 30% students lie below 85 marks, 70% are above 85 marks
- Note: Here we use *Number of observations*, and not *Number of observations + 1*

US Household Net Worth and Percentile (Source: https://finance.yahoo.com/news/wealthy-net-worth-considered-poor-190014440.html)

Category	Percentile	Net Worth
Poor	20th	$10,000
Middle class	50th	$281,000
Wealthy	90th	$1.9 million

Quartile

- Percentile: Divides data into 100 equal parts
- **Quartile**: Divides data into 4 equal parts
- First Quartile (Q1) = 25th percentile
- Second Quartile (Q2) = 50th percentile = Median
- Third Quartile (Q3) = 75th percentile



- Sorted Marks of 20 students: 65,72,78,80,81,83,85,87,88,90,91,92,93,94,95,96,97,98,99,100
- Position of Q1 = $\frac{Percentile}{100}\ x\ (n+1) = \frac{25}{100}\ x\ 21 = \frac{525}{100} = 5.25$, So Q1 = Average of 81 and 83 = 82
- Position of Q2 = $\frac{Percentile}{100}\ x\ (n+1) = \frac{50}{100}\ x\ 21 = \frac{1050}{100} = 10.5$, So Q2 = Average of 90 and 91 = 90.5
- Position of Q3 = $\frac{Percentile}{100}\ x\ (n+1) = \frac{75}{100}\ x\ 21 = \frac{1575}{100} = 15.75$, So Q3 = Average of 95 and 96 = 95.5

Inter Quartile Range (IQR)

- **Inter Quartile Range (IQR)** = Q3 – Q1 = Middle 50% of the data
- In the given example: IQR = Q3 – Q1 = 95.5 – 82 = 13.5
- Handles outliers better than range, since the extreme values at both the ends are ignored in IQR
- Since it uses percentiles rather than actual values, it is less affected by **skewed data** (See **Skewness**)
- **Outliers**: Data points that are significantly outside of the typical range of values
  - Lower bound: Q1 – (1.5 * IQR) = 82 – (1.5 * 13.5) = 61.75
  - Upper bound: Q3 + (1.5 * IQR) = 95.5 + (1.5 * 13.5) = 102.25
  - Points below the *lower bound* or above the *upper bound* are *outliers*
  - In our example, there are no such points, so we do not have any outliers

Outlier Example

- Commute times for 14 randomly selected adults in minutes: 16, 8, 35, 17, 13, 15, 15, 5, 16, 25, 20, 20, 12, 10
- Find outliers and draw a box plot
- Solution: First sort them: 5, 8, 10, 12, 13, 15, 15, 16, 16, 17, 20, 20, 25, 35
- Create a 5-number summary: Minimum, Q1, Q2, Q3, Maximum = 5, 12, 15.5, 20, and 35
- Outlier
  - First calculate 1.5 * IQR = 1.5 x (20 – 12) = 1.5 x 8 = 12
  - Outlier calculation: Q1 – 12 = 12 – 12 = 0 and Q3 + 12 = 20 + 12 = 32
  - So, outliers = Commute time < 0 or > 32
- Boxplot: Draw a vertical line between 5 and 35; Draw a box with 12 and 20; Draw a median line at 15.5, Show outlier points (See next slide)

Percentile in Python

1. Given percentile → find absolute value

- Find the value corresponding to a percentile (e.g., 90th percentile value)

```
import numpy as np
data = [10, 20, 30, 40, 50]
p90 = np.percentile(data, 90)
print(p90)   # Output: 46.0
```

- The 90th percentile value is 46.0

2. Given absolute value → find percentile

- Find the percentile rank of a value (e.g., what percentile is the value 40?)
- Using SciPy:

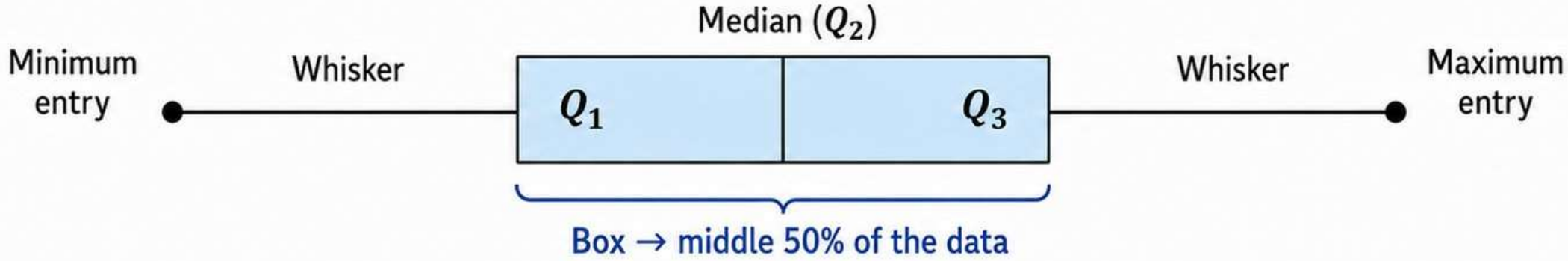
```
from scipy.stats import percentileofscore
percentileofscore([10, 20, 30, 40, 50], 40)   # 80.0
```

- Without SciPy:

```
data = np.array([10, 20, 30, 40, 50])
(data <= 40).mean() * 100   # 80.0
```

- The value 40 is at the 80th percentile

Box Plot Explained



Key Points

- Box → shows the middle 50% of the data
- Left edge of box = Q1 (first quartile)
- Middle line = Median (Q2)
- Right edge of box = Q3 (third quartile)
- Whiskers → show the smallest and largest values that are not outliers
- **Minimum entry:** The higher of True minimum and Lower bound
- **Maximum entry:** The lower of True Maximum and Upper Bound

Example

Data points: 5, 7, 8, 10, 11, 12, 14, 15, 30

- Q1 = 8
- Median (Q2) = 11
- Q3 = 14
- IQR = Q3 − Q1 = 14 − 8 = 6
- Lower bound = Q1 − 1.5 × IQR = 8 − (1.5 × 6) = −1
- Upper bound = Q3 + 1.5 × IQR = 14 + (1.5 × 6) = 23

Minimum entry: Higher of 5 and −1 = 5

Maximum entry: Lesser of 30 and 23 = 23

Outlier = 30 (outside the upper bound)

Handling Missing Values in ‘Age’ Column (Imputation)

- We fill the missing values (NaN) in the ‘age’ column using different methods – this is called Imputation
- Why? Machine learning models cannot work with missing values, so we fill them before analysis or model building

Methods 5–8 use sklearn SimpleImputer with strategy = mean, median, most_frequent, constant (fill_value = 0)

age	1. Mean	2. Median	3. Mode	4. Const 0	5. SI mean	6. SI median	7. SI most freq.	8. SI const 0
22	22	22	22	22	22	22	22	22
35	35	35	35	35	35	35	35	35
NaN	28.6	28	18	0	28.6	28	18	0
28	28	28	28	28	28	28	28	28
NaN	28.6	28	18	0	28.6	28	18	0
40	40	40	40	40	40	40	40	40
18	18	18	18	18	18	18	18	18

- Note: all known ages are unique, so there is no single mode; pandas mode()[0] and SimpleImputer(most_frequent) both pick the smallest value, 18

Outlier Detection Methods

Method	How It Works	Typical Rule	Notes
IQR Method	Uses Q1, Q3, and IQR to find extreme values	Values **< Q1 – 1.5 × IQR** or **> Q3 + 1.5 × IQR**	Very common for skewed data
Z-Score Method	Measures how many standard deviations a value is from the mean	Z-score **< –3** or **> +3**	Best for normally distributed data

Variance

Concept/Step	Explanation
Variance	Measures how far data points are from their mean
High Variance	Data points are more scattered from the mean
Low Variance	Data points are less scattered from the mean
Calculation Steps	1. Find the mean of the data set 2. Subtract the mean from each value and square the result 3. Take the average of these squared values → This is the variance

Variance Formula

- **Population variance** (Whole data)
- $\sigma^2 = \sum_{i=1}^{N} \frac{(x_i - \mu)^2}{N}$
- x_i = Data points
- μ = Population mean
- N = Number of observations (Population size)

- **Sample variance** (Selected data)
- $S^2 = \sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{n-1}$
- x_i = Data points
- $\bar{x}$ = Sample mean
- n = Number of observations (Sample size)

Variance Example

- Consider data values 5, 7, 10, 12, and 15
- Step 1: Mean (μ) : $\frac{(5 + 7 + 10 + 12 + 15)}{5}$ = 9.8
- Step 2: Subtract mean from each number and square the result
  - $(5 - 9.8)^2 = 20.24$
  - $(7 - 9.8)^2 = 6.76$
  - $(10 - 9.8)^2 = 0.04$
  - $(12 - 9.8)^2 = 5.44$
  - $(15 - 9.8)^2 = 27.04$
- Step 3: Add up the squared differences and divide by the total number of numbers: $\frac{(20.24 + 6.76 + 0.04 + 5.44 + 27.04)}{5}$ = 11.12 = Variance (σ^2)

Why are the Differences Squared?

- If we do not do this, then the positive and negative differences may cancel each other and we may not be able to find how the data points are dispersed from the mean
- Example: Consider marks of 5 students: 55, 60, 65, 54, 64
- Mean: 59.6, Variance: 20.24
- But if we do not square the differences, what will happen?

Marks	Marks - Mean	(Marks - Mean) Squared
55	-4.6	21.16
60	0.4	0.16
65	5.4	29.16
54	-5.6	31.36
64	4.4	19.36
Mean	**(Marks - Mean) / 5**	**((Marks - Mean) Squared)) / 5**
59.6	0.00	20.24

Variance

Limitation of Variance: How to Quantify?

- Quantifying/measuring variance is tough
- Example: Consider marks of 10 students: 32, 39, 44, 56, 56, 63, 71, 72, 81, 89
- Mean: 60.3, Variance: 306.81
- Variance *seems to be high*, but it does not convey any meaningful information beyond this
- This is a limitation of variance

Standard Deviation

Concept/Aspect	Explanation
Standard Deviation (σ)	Square root of variance
Scale	Same as the original data, unlike variance which is squared
Interpretation	Variance: Difficult to interpret Standard Deviation: Easy to interpret
Formula	$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$
Purpose	Measures the average distance of data points from the mean in the same units as the data

Standard Deviation Example

- Marks of 10 students: 76, 89, 53, 75, 80, 67, 61, 59, 71, 60
- **Mean (μ)**: $\frac{76 + 89 + 53 + 75 + 80 + 67 + 61 + 59 + 71 + 60}{10} = \frac{691}{10} = 69.10$

Deviation	Deviation Squared
76 – 69.10 = 6.9	47.61
89 – 69.10 = 19.9	396.01
53 – 69.10 = -16.1	259.21
75 – 69.10 = 5.9	34.81
80 – 69.10 = 10.9	118.81
67 – 69.10 = -2.1	4.41
61 – 69.10 = -8.1	65.61
59 – 69.10 = -10.1	102.01
71 – 69.10 = 1.9	3.61
60 – 69.10 = -9.1	82.81
Total:	1114.90
Variance (σ^2) =	**1114.90 / 10 = 111.49**

- **Standard deviation (σ)** = $\sqrt{111.49} = 10.56$

Range, IQR, Variance, Standard Deviation Example

Day	Temperature (°C)
1	30
2	31
3	29
4	30
5	32
6	31
7	30
8	29
9	30
10	31
11	30
12	32
13	31
14	30

Measure	Value
Range	3
IQR	1
Variance	1.07
Std Dev	1.03

Measure	Value
Range	80
IQR	2
Variance	469.02
Std Dev	21.66

Day	Products Sold
1	20
2	22
3	21
4	20
5	23
6	21
7	20
8	22
9	21
10	20
11	22
12	21
13	20
14	**100**

Exercise

- Interpret the data below
- Group A and Group B are two groups of people and we have data about their blood sugar level

- A = 125, 118, 121, 123, 119, 122, 117, 121, 124, 116, 120, 122, 119, 124, 118, 120, 123, 122, 118, 121

- B = 150, 90, 115, 145, 118, 112, 135, 121, 117, 122, 119, 128, 115, 145, 120, 90, 155, 135, 122, 115

Standard Deviation: True Indicator of Homogeneous/Heterogeneous Data

- Consider blood sugar levels of two groups of people:

- A = 125, 118, 121, 123, 119, 122, 117, 121, 124, 116, 120, 122, 119, 124, 118, 120, 123, 122, 118, 121
- Mean = 120 mg/dL, SD = 2.35 mg/dL
- Most close to the mean (Fit)

- B = 150, 90, 115, 145, 118, 112, 135, 121, 117, 122, 119, 128, 115, 145, 120, 90, 155, 135, 122, 115
- Mean = 120 mg/dL, SD = 17.31 mg/dL
- Some close to the mean (Fit), some away from the mean (Unfit)

Interpreting Standard Deviation

- Again consider A = 125, 118, 121, 123, 119, 122, 117, 121, 124, 116, 120, 122, 119, 124, 118, 120, 123, 122, 118, 121
- Mean = 120 mg/dL, SD = 2.35 mg/dL
- How to calculate *how many standard deviations away* is an observation from the mean?
- Formula: $\frac{Our\ data\ point - Mean}{Standard\ Deviation}$
- Person 1: $(\frac{125 - 120}{2.35})$ = +2.12 standard deviation away from the mean
- Person 2 $(\frac{118 - 120}{2.35})$ = -0.85 standard deviations away from the mean

Coefficient of Variation (CV)

Concept/Aspect	Explanation
Coefficient of Variation (CV)	Standard deviation divided by the mean and then multiplied by 100 (because we want percentage)
Scale	Expressed in percentage; independent of the units of data (relative measure)
Comparison with SD	SD: Absolute measure, expressed in original units CV: Relative measure, expressed in % (unitless)
High CV	Indicates large variability relative to the mean CV = 100% → SD = Mean
Formula	CV = $\frac{SD}{Mean} \times 100\%$
Example	Mean delivery time = 20 min, SD = 5 min → CV = (5 / 20) × 100% = 25%

CV Example

- A pizza delivery outlet delivers pizza on an average in 20 minutes with a standard deviation of 5 minutes

- Mean delivery time = 20 min, SD = 5 min

- CV = $\frac{SD}{Mean}\ x\ 100\ \% = \frac{5}{20}\ x\ 100\ \%$ = 25%

Coefficient of Variation (CV)

- Now suppose a second pizza restaurant has a mean delivery time of 15 minutes and the standard deviation is 10 minutes. Calculate the coefficient of variation and compare with the first one.
- $CV = \frac{SD}{Mean}\ x\ 100\ \% = \frac{10}{15}\ x\ 100\ \%$ = 66%

- Calculation for the first pizza restaurant was:
- $CV = \frac{SD}{Mean}\ x\ 100\ \% = \frac{5}{20}\ x\ 100\ \%$ = 25%

- What will be our interpretation?

Problem

- Two students A and B have appeared for five tests with the following results:
  - A: Mean score = 80, SD = 8
  - B: Mean score = 75, SD = 15
- Who is more consistent?
- CV for A = $\frac{8}{80}$ x 100 = 10%
- CV for B = $\frac{15}{75}$ x 100 = 20%
- A is more consistent

Making Investment Decisions

- Option A: Average return: 20%, SD: 15%
- Option B: Average return: 6%, SD: 2%

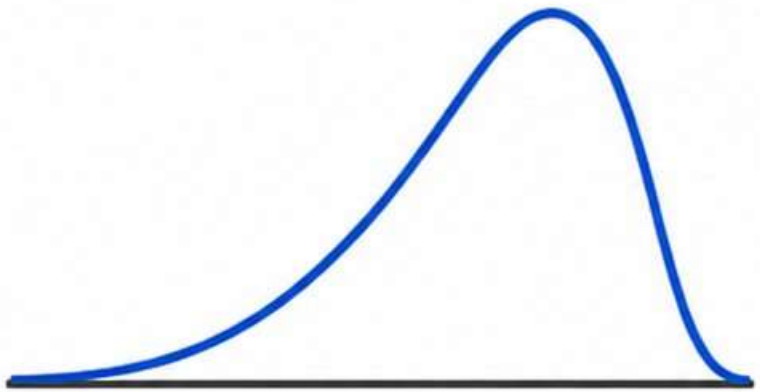
- Option A: $CV = \frac{SD}{Mean}\ x\ 100\ \% = \frac{15}{20}\ x\ 100\ \%$ = 75%
- Option B: $CV = \frac{SD}{Mean}\ x\ 100\ \% = \frac{2}{6}\ x\ 100\ \%$ = 33%

- Conclusion: Is Option B better?

Skewness

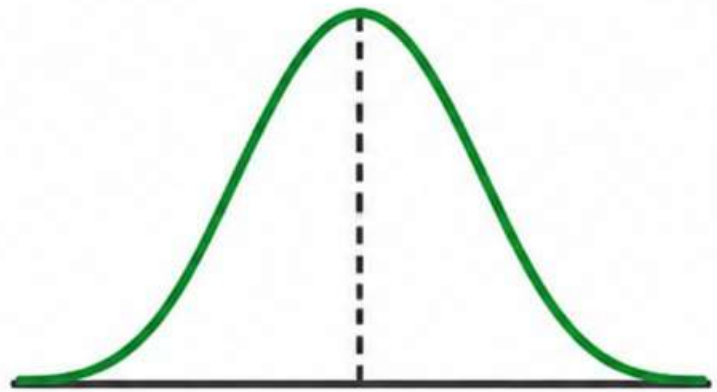
Skewness

Left Skewed



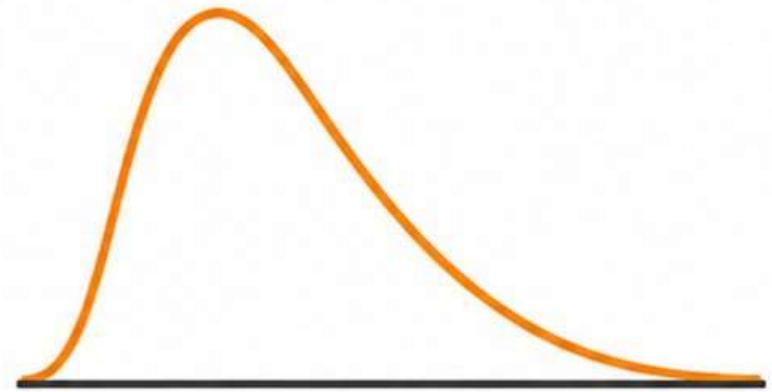
Tail on the left

Symmetric



No skew

Right Skewed



Tail on the right

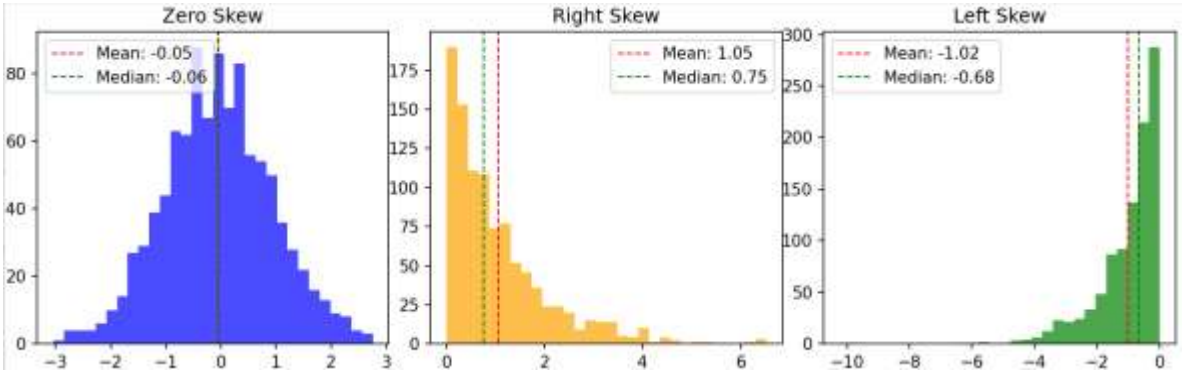
Skewness can be Confusing!

- **Skew**: Represents the asymmetry of a distribution around its mean
- **Skew = Where is the *skew*, not where is the *data***

Skew	Tail	Most Data	Interpretation
Right (Positive)	On the Right	On the Left	Most values are low
Left (Negative)	On the Left	On the Right	Most values are high

- Students appearing in an Examination
- **No-skew**: Student marks are evenly distributed
- **Right-skew**: Most students get low marks => Long tail to the right (higher values)
- **Left-skew**: Most students get high marks => Long tail to the left (lower values)

Skew



C:\code\Data Analytics\skew-visualize.py

- **Zero-skew (No-skew)**: Mean = Median

Characteristic	Right (Positive) Skew	Left (Negative) Skew
Key feature	Most values are on the lower side, a few on are the higher side	Most values are on the higher side, a few are on the lower side
Statistics	Mean > Median	Mean < Median
Mean	Skewed to the right	Skewed to the left
Tail	Comes later - On the right (Represents high values)	Comes first - On the left (Represents low values)
More data points	On the left	On the right
Outliers	On the right	On the left
Ratings Example	Most people give ratings of 1 or 2 out of 5	Most people give ratings of 4 or 5 out of 5
Example	Incomes of people	Ages at the time of death
Why?	Most people have low incomes, Very few have very high income	Most people would die in older age, Very few children die

Skewness Formulae

- **Pearson's First Coefficient of Skewness** (Mode Skewness): (Mean - Median) / Standard Deviation
- **Another variation: 3 (Mean - Median) / Standard Deviation**
- **Fisher-Pearson Coefficient of Skewness** (Moment Skewness): (Σ(xi - Mean)^3 / n) / [(Σ(xi - Mean)^2 / n)^(3/2)]
- **Quantile-Based Skewness**: (Q3 + Q1 - 2 * Median) / (Q3 - Q1)

Pearson's Coefficient of Skewness

- Formula: $Skewness = \frac{3\,(Mean - Median)}{Standard\ Deviation}$
- Consider daily commute time for a set of people in minutes: 20,22,25,30,35,40,45,50,60,70
- Mean = 397/10 = 39.70
- Median: 37.5
- SD = 15.80
- $Skewness = \frac{3\,(Mean - Median)}{Standard\ Deviation} = \frac{3\,(39.70 - 37.5)}{15.80} = 0.42$
- Interpretation: Right/positive skew … Most commute times are less than the mean, but a few outliers have increased the mean value

Kurtosis

- Kurtosis: describes the shape of the probability distribution of a real-valued random variable – it measures tail heaviness and peak sharpness

Kurtosis Type	Description	Kurtosis Value	Implications
Leptokurtic	Heavy tails and a sharp peak	> 3	More outliers / extreme values
Mesokurtic	Similar tails and peak as a normal distribution	= 3	Similar to normal distribution
Platykurtic	Lighter tails and a flatter peak	< 3	Fewer outliers / extreme values

Note: pandas and SciPy report excess kurtosis = kurtosis – 3

- So a normal distribution has value 0 (instead of 3)
- Add 3 to the value returned by pandas or SciPy to compare with the table above

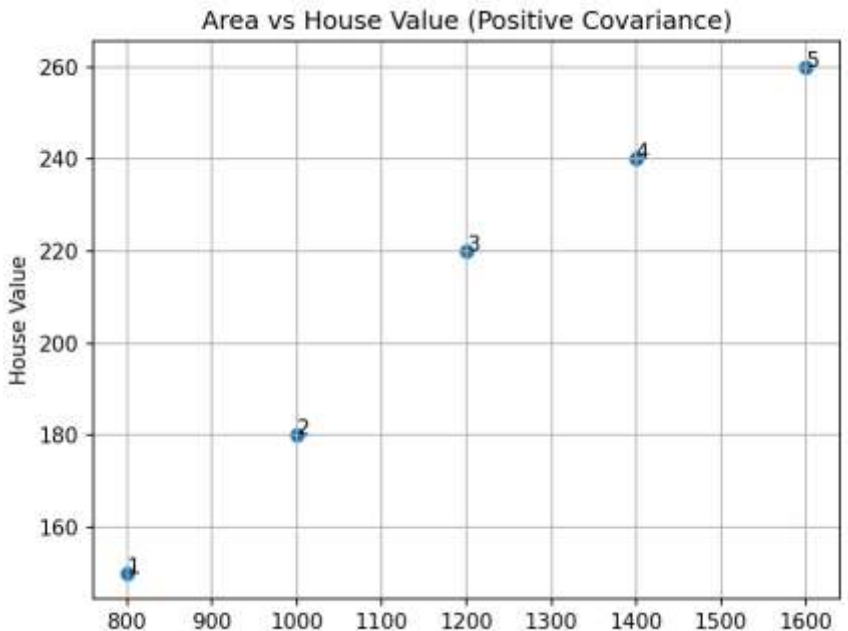
Covariance

Variance and Covariance

Measure	Definition	Example / Interpretation
Variance	Measures how far a set of numbers (a single variable) is spread out from its mean	Example: Variance in Marketing Expenditure, Variance in Revenue
Covariance	Measures the directional relationship between two variables	Example: Marketing Spend (x) and Revenue (y)
Positive Covariance	Variables tend to move in the same direction	As Marketing Spend increases, Revenue tends to increase
Negative Covariance	Variables tend to move in opposite directions	As Marketing Spend increases, Revenue tends to decrease
Near-Zero Covariance	Little to no linear relationship between the variables	Changes in Marketing Spend do not significantly affect Revenue

Covariance

House No	Area (sq ft)	House Value
1	800	150
2	1000	180
3	1200	220
4	1400	240
5	1600	260



Higher area houses tend to have higher prices
Both variables move together
Area ↑ -> House Value ↑
Positive covariance

Covariance: Measures whether two variables move together

But, if we had
Area ↑ → Price ↓
Negative covariance

Covariance Example

Height	Weight
65	68
67	69
68	70
66	69
64	65

- Cov (x,y) = $\sum_{i=1}^{n} \frac{(x_i-\bar{x})(y_i-\bar{y})}{n-1}$
- Mean height ($\bar{x}$) = $\frac{65+67+68+66+64}{5}$ = 66
- Mean weight ($\bar{y}$) = $\frac{68+69+70+69+65}{5}$ = 68.2
- For Heights: $x_i - \bar{x}$ = [65−66,67−66,68−66,66−66,64−66]=[−1,1,2,0,−2]
- For Weights: $y_i - \bar{y}$ = [68−68.2,69−68.2,70−68.2,69−68.2,65−68.2]=[−0.2,0.8,1.8,0.8,−3.2]
- $(x_i - \bar{x})(y_i - \bar{y})$ = (-1x-0.2) + (1x0.8) + (2x1.8) + (0x0.8) + (-2x-3.2) = 11
- Cov (x,y) = $\sum_{i=1}^{n} \frac{(x_i-\bar{x})(y_i-\bar{y})}{n-1} = \frac{11}{4}$ = 2.75

Covariance and Variance

- As we know, Cov (x,y) = $\sum_{i=1}^{n} \frac{(x_i - \bar{x})(y_i - \bar{y})}{n-1}$

- What if we use x instead of y?
- Cov (x,x) = $\sum_{i=1}^{n} \frac{(x_i - \bar{x})(x_i - \bar{x})}{n-1}$

- Remember: Variance (x) = $\sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{n-1} = \sum_{i=1}^{n} \frac{(x_i - \bar{x})(x_i - \bar{x})}{n-1}$
- So: Var (x) = Cov (x,x)

Covariance/Correlation Matrix Interpretation

```
Overall Covariance between Height and Weight:
            Height       Weight
Height   14.803473   114.242656
Weight  114.242656  1030.951855
```

```
Overall Correlation between Height and Weight:
            Height    Weight
Height    1.000000  0.924756
Weight    0.924756  1.000000
```

Aspect	Value / Example	Interpretation
Diagonal elements	Height variance = 14 Weight variance = 1030	Variability of each variable individually
Off-diagonal elements	Covariance (Height, Weight) = 114	Positive → as Height increases, Weight tends to increase
Covariance meaning	114	Shows direction of relationship, but magnitude is hard to interpret because it depends on units
Correlation (r)	0.92	Standardized measure: easy to interpret, close to 1 → strong positive linear relationship

Correlation

Correlation

Concept	Definition / Formula	Example	Notes
Correlation	Measures whether two numeric variables are related	Exercise & Health, Study & Marks	Captures direction and strength of relationship
Correlation Formula	$\frac{Cov\ (x, y)}{\sigma_x \sigma_y}$	—	Standardized measure of covariance
Pearson Correlation Coefficient	Measures linear relationship between continuous variables	Experience & Salary	Sensitive to linear relationships, ranges from -1 to +1
Spearman Rank Correlation Coefficient	Measures monotonic relationships, not necessarily linear	Rank of Students vs Rank of Marks	Works for ordinal data or non-linear but monotonic relationships

Pearson Correlation Coefficient Interpretation

Correlation Type	Range	Description	Example
Positive correlation	0 to +1	Variables move up together	Correlation of 0.80 between Hours spent studying and test scores
Negative correlation	-1 to 0	As one variable moves up, the other moves down	Correlation of -0.70 between Hours spent watching TV and physical fitness
Zero correlation	0	Variables are unrelated	Correlation of 0.02 between Shoe size and IQ score

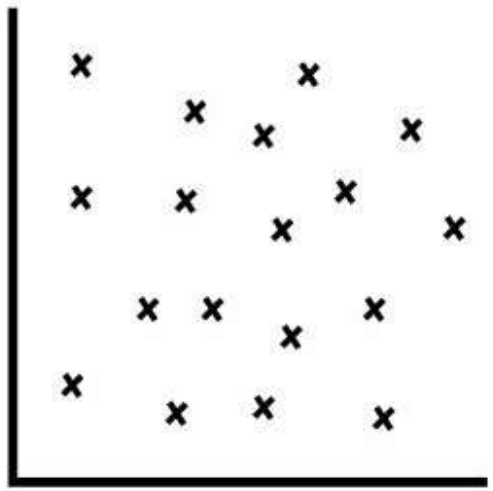
Positive, Negative, No Correlation



Positive Correlation



Negative Correlation



No Correlation

Pearson Correlation Coefficient for Earlier Example

- Corr(x,y) = $\frac{Cov\ (x,y)}{\sigma_x \sigma_y}$
- Corr(x,y) = $\frac{2.75}{1.58\ x\ 1.92}$
- Corr(x,y) = $\frac{2.75}{1.58\ x\ 1.92}$ = 0.90

Height	Weight
65	68
67	69
68	70
66	69
64	65

Pearson Versus Spearman Method

Feature	Pearson Correlation	Spearman Correlation
Definition	Measures linear relationship between two continuous variables	Measures monotonic relationship (whether linear or not) based on ranks
Data Type	Continuous, normally distributed	Ordinal, continuous, or not normally distributed
Calculation	Covariance of variables divided by the product of their standard deviations	Pearson correlation of the rank values of the variables
Range	-1 to +1	-1 to +1
Assumptions	Linear relationship, normality	Monotonic relationship, no strict normality requirement
Use Case Example	Height vs Weight, Study hours vs Exam score	Rank of students vs Rank of exam results, Likert scale survey data (Strongly agree…Disagree)

Spearman Rank Correlation Formula

Aspect	**Details**
Formula (covariance-based)	$r_s = \frac{Cov\ (R(x),R(y))}{\sigma(R(x))\ \sigma(R(y))}$
Explanation	R(x) and R(y) = ranks of x and y Cov = covariance of the ranks σ = standard deviation of the ranks
Formula (difference-based, popular)	$r_s = 1 - \frac{6\ \Sigma\ d^2}{(n^2-1)}$
Explanation	d^2 = square of difference between ranks of each pair n = number of data points
Range	-1 (perfect decreasing) to +1 (perfect increasing)

Spearman Rank Correlation

- Use when data does not use precise intervals but represents an ordered sequence
- Examples:
  - Class rank (1st, 2nd, 3rd place) vs. hours spent studying
  - Customer satisfaction ratings (Unhappy, Neutral, Happy) vs. money spent

Spearman Rank Correlation

- Formula: $r_s = 1 - \frac{6\,\Sigma\,d^2}{(n^2-1)}$

Height	Weight
65	68
67	69
68	70
66	69
64	65

- d^2 = Square of differences in the ranks of the two coordinates for each point (x,y)
- n = Number of data points
- $r_s = 1 - \frac{6\,\Sigma\,d^2}{(n^2-1)} = 1 - \frac{6\,x\,0.50}{(5^2-1)}$
- $r_s = 1 - \frac{3}{24} = 1 - 0.125 = 0.87$

x_i	R_x	y_i	R_y	$d_i = R_x - R_y$	d_i^2
65	2	68	2	0	0
67	4	69	3.5	0.5	0.25
68	5	70	5	0	0
66	3	69	3.5	-0.5	0.25
64	1	65	1	0	0

Spearman Rank Correlation Example (correlation_spearman.py)

How Spearman Rank Correlation is Calculated (Example)

Between Age and Health_Level_Indicator

1 Original Data (Two Variables)

Person	Age (X)	Health_Level_Indicator (Y)
1	56	6
2	69	3
3	46	7
4	32	7
5	60	4

2 Rank the Age (X)

Person	Age (X)	Rank of Age (R_x)
1	56	3
2	69	5
3	46	2
4	32	1
5	60	4

3 Rank the Health_Level_Indicator (Y)

Person	Health_Level_Indicator (Y)	Rank of Y (R_y)
1	6	3
2	3	1
3	7	4.5
4	7	4.5
5	4	2

4 Compute the Differences of Ranks (d) and d^2

Person	R_x	R_y	$d = R_x - R_y$	d^2
1	3	3	0	0
2	5	1	4	16
3	2	4.5	-2.5	6.25
4	1	4.5	-3.5	12.25
5	4	2	2	4

$\sum d^2 = 38.5$

5 Apply Spearman Rank Correlation Formula

$$\rho = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

where,

ρ = Spearman rank correlation coefficient

$\sum d^2$ = Sum of squared differences

n = Number of pairs

6 Calculate

$n = 5$

$\sum d^2 = 38.5$

$$\rho = 1 - \frac{6 \times 38.5}{5(5^2 - 1)}$$

$$\rho = 1 - \frac{231}{5(24)}$$

$$\rho = 1 - \frac{231}{120}$$

$\rho = 1 - 1.925 = -0.925$

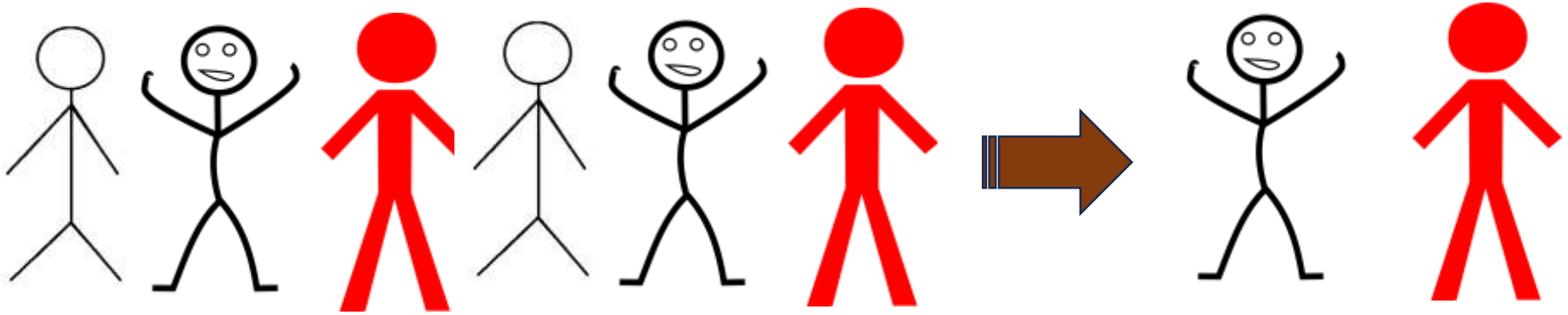
Interpretation:

The Spearman rank correlation between Age and Health_Level_Indicator is −0.925, indicating a strong negative monotonic relationship.

Population and Sample, Probability Theory

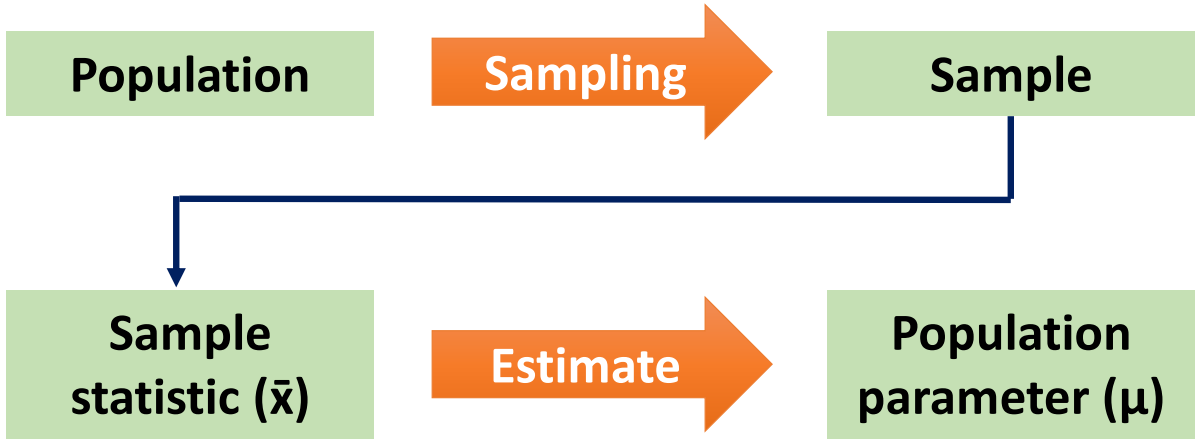
Sampling and Population

- **Population (**N) = Complete data **Sample** (n) = Subset of data



- Calculate sample mean $\bar{x}$ -> **Estimation** of -> Population mean u
- **Point estimate**: Exact estimate (e.g. average age = 25)
- **Interval estimate**: A range of estimates (e.g. average age = 23-27)

Population and Sample



- 10000 students in a university
- Question: What is the average student height?
  - Difficult: Measure the actual height of the **population**, i.e. all 10000 students to calculate the average (**parameter**)
  - Easier: Take a **sample** of 500 students, calculate the average and use it to estimate the population average (**statistic**)

Population Parameter	**Symbol**	**Sample Statistic**	**Symbol**
Mean	**μ** (Mu)	Sample Mean	**$\bar{x}$** (x bar)
Standard Deviation	**σ** (Sigma)	Sample Standard Deviation	**s**
Proportion	**π** (Pi)	Sample Proportion	**$\hat{p}$** (p-hat)

Sampling Types

Univariate Sampling	Bivariate Sampling
Analysis of a single variable	Analysis of two variables to find relationship between them
Analyzing the ages of individuals in a population Examining the heights of a sample of students Studying the test scores of students in a class	Amount of time spent studying (variable 1) and the Test scores achieved (variable 2) Education (variable 1) and the Income level (variable 2)
Descriptive statistics such as mean, mode, median, variance, standard deviation	Correlation, Regression
Histogram, Box plot	Scatter plot, Heatmap

Sampling Techniques

- **Non-probability sampling**: Non-random selection
  - Not every individual has a chance of being included
  - Easier and cheaper to access, has a higher risk of **sampling bias**
  - Used in qualitative research
- **Probability sampling**: Random selection
  - Every member of the population has a chance of being selected
  - Most appropriate
  - Mainly used in quantitative research

Non-Probability Sampling

Type	Brief description	Method	Example
Convenience	Just pick sample data as per one's convenience	No systematic method	To research opinions about student support services in our university, we ask our fellow students to complete a survey on the topic
Voluntary-response	People voluntarily come forward	Start with the people who are interested – May be biased	Ask employees about problems faced through a third-party survey – unhappy employees will come forward on their own
Purposive	Done for a specific purpose, also called **judgment sampling**	Reach the right people and then do a detailed study	We want to know the opinions of foreign students at C-DAC, so we specifically talk to them about a variety of topics
Snowball	Recruit more participants via existing participants	Create a **snowball** of participants	Homeless people survey – Initially difficult to find such people to speak – But they may help us speak with others soon

Probability Sampling

Type	Brief description	Method	Example
Simple	Every member of the population has an equal chance	Random number generator	Just pick n random samples from 1 million credit card transactions
Systematic	Similar but easier	Choose at regular intervals	Pick order number # 7, 17, 27, 37, 47, …
Stratified	More representative, if we have sub-categories within data	Divide population into strata (sub-groups) and then pick proportionate samples from each	From 5 different income groups, select proportionate samples
Cluster	Similar, but use entire sub-groups, rather than samples within sub-groups	Divide population into clusters and pick entire clusters as samples	Select 5 different branches of a bank out of say 25 and consider all the customers from these 5 branches

Sample Space and Events, Probability

Sample Space and Events

- **Sample space (S)**: All the possible outcomes of an experiment
- **Event (E)**: A subset of the sample space



Sample space (S)
Tossing a Coin

Event (E)
Heads

Event (E)
Tails



Sample space (S)
Throwing a Dice

Event (E)
1

Event (E)
3

Event (E)
5

Event (E)
2

Event (E)
4

Event (E)
6

Probability Definition

- Probability: how likely is it that something will occur?
- Written as a number or fraction (e.g. 0.5 or 50%)
  - Impossible = 0 (0%) • Equal chance = 0.5 (50%) • Certain = 1 (100%)
- Example: what is the probability that it will rain today?
  - Sample space S = {R, NR}; P(R) = 1/2 = 0.5 or 50%; P(NR) = 1/2 = 0.5 or 50% → Theoretical probability
  - But what about the rainy season? It should be P(R) = 100% and P(NR) = 0% → Empirical probability
- Theoretical probability: what will happen in theory?
- Empirical / Practical probability: what actually happened?

A Team Winning Consecutive Matches

- Symbols: Win = 1, Loss = 0

Matches	Possible Cases	Theoretical Probability (all wins)	Percentage
1	0, 1	1/2	50%
2	00, 01, 10, 11	1/4	25%
3	000, 001, 010, 011, 100, 101, 110, 111	1/8	12.5%
4	0000, 0001, 0010, 0011, 0100, 0101, 0110, 0111, 1000, 1001, 1010, 1011, 1100, 1101, 1110, 1111	1/16	6.25%

- Imagine the team actually winning four consecutive matches
  - Theoretical probability would diminish to 6.25%
  - But the team has actually won! So, the empirical probability will be 100%
- Theoretical probability: what could happen in theory. Empirical probability: what actually happened.

Marginal, Joint and Conditional Probability

	Marginal Probability	Joint Probability	Conditional Probability
Meaning	The probability of a single event occurring, irrespective of any other event	The probability that two (or more) events occur together	The probability of an event occurring, given that another event has already occurred
Formula	$P(A) = $ (Number of outcomes where A occurs) / (Total number of outcomes)	$P(A \cap B) = $ (Number of outcomes where A and B both occur) / (Total number of outcomes)	$P(A \mid B) = P(A \cap B) / P(B)$, $P(B) > 0$
Example	A die is rolled. What is the probability of getting a 3? $P(3) = 1/6$	Two dice are rolled. What is the probability that the first die shows 3 and the second die shows 5? $P(3 \text{ and } 5) = 1/36$	In a class, 60% of students study, and 80% of those who study pass the exam. What is the probability of passing given that a student studies? $P(\text{Pass} \mid \text{Study}) = 0.80$
Focus	One event only (A)	Both events together (A and B)	Event A, given that B has occurred

Sample Dataset - Student

PRN	studied	passed
101	1	1
102	1	1
103	1	0
104	0	0
105	0	1

Marginal Probability

Probability of a single event, ignoring everything else

Examples

What is the probability that a student passed?

P(passed) = $\frac{3}{5}$ = 0.60

What is the probability that a student did not study?
P(did not study) = 0.40

Joint Probability

Looks at the entire dataset and looks for an overlap

Example

What is the probability that a student studied and passed?

P(studied and passed) = $\frac{2}{5}$
= 0.40

Conditional Probability

Looks a subset and then asks for an outcome within the subset

Example

Of the students who studied, what is the probability that they passed?

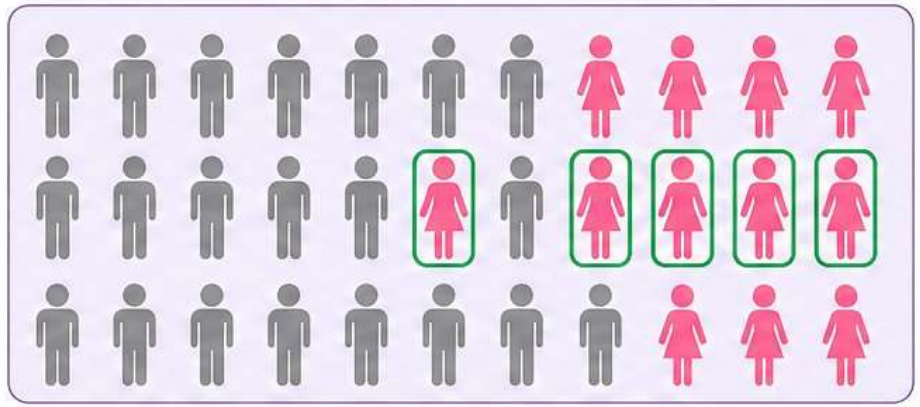
P(studied and passed) = $\frac{2}{3}$
= 0.66

JOINT PROBABILITY vs CONDITIONAL PROBABILITY

A Joint Probability:
P(Female and Survived)

The probability that a passenger was **female AND survived.**

All passengers



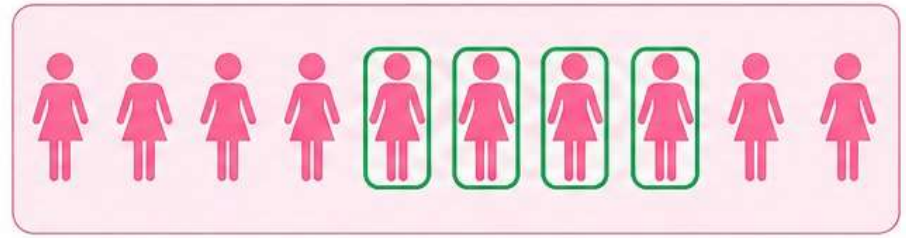
We look at **everyone** on the ship.
We ask: what fraction are both **female and survived**?

$$P(\text{Female and Survived}) = \frac{\#\text{ of female passengers who survived}}{\text{Total \# of passengers}}$$

B Conditional Probability:
P(Survived | Female)

The probability that a passenger survived **given that she was a female.**

Only female passengers



We look only at female passengers.
Among them, what fraction survived?

$$P(\text{Survived} \mid \text{Female}) = \frac{\#\text{ of female passengers who survived}}{\text{Total \# of female passengers}}$$

Key Difference: Joint probability looks at the entire population and asks for the overlap. Conditional probability focuses on a subset (females) and asks for the outcome within that subset.

Bayes' Theorem

Step		Formula
1	Example	P(survived | female) = P(survived and female) / P(female)
2	Conditional probability	P(A | B) = P(A, B) / P(B)
3	Reverse the roles	P(B | A) = P(A, B) / P(A)
4	Use the multiplication rule	P(A, B) = P(A | B) × P(B)
5	Substitute and simplify	P(B | A) = P(A | B) × P(B) / P(A)

Bayes' Theorem: Names of the Terms

Step		Formula
1	Example	P(survived | female) = P(survived and female) / P(female)
2	Conditional probability	P(A | B) = P(A, B) / P(B) ← P(A, B) is the joint probability
3	Reverse the roles	P(B | A) = P(A, B) / P(A)
4	Use the multiplication rule	P(A, B) = P(A | B) × P(B)
5	Substitute and simplify	P(B | A) = P(A | B) × P(B) / P(A)

In the final formula:

- P(B | A) = Posterior – updated belief in B after seeing A
- P(A | B) = Likelihood – how likely the evidence A is if B is true
- P(B) = Prior – belief in B before seeing the evidence
- P(A) = Evidence (marginal likelihood) – overall probability of A

TITANIC DATA

WE KNOW:

$P(\text{Survived} \mid \text{Female}) = 74\%$

	Survived	Did Not Survive	Total
Female	233	81	314
Male	109	468	577
Total	342	549	891

CAN WE FIND:

$P(\text{Female} \mid \text{Survived})$?



STEP 1 — Compute what we know

$$P(\text{Survived} \mid \text{Female}) = \frac{233}{314} = 74\%$$

STEP 2 — Reverse the question



Among survivors, how many were female?

STEP 3 — Use Bayes' Theorem

$$P(\text{Female} \mid \text{Survived}) = \frac{P(\text{Survived} \mid \text{Female}) \times P(\text{Female})}{P(\text{Survived})}$$

STEP 4 — Plug in the numbers

- $P(\text{Survived} \mid \text{Female}) = 233/314$
- $P(\text{Female}) = 314/891$
- $P(\text{Survived}) = 342/891$

RESULT →

$$P(\text{Female} \mid \text{Survived}) = \frac{233}{342} = 68\%$$

Among all 342 survivors, 233 were female.





$P(\text{Survived} \mid \text{Female})$

74%

Females who survived

≠



$P(\text{Female} \mid \text{Survived})$

68%

Survivors who are female

BAYES = REVERSING THE CONDITION

Bayes' Theorem: Why a Positive Test Doesn't Mean You Have the Disease

- Let A = you have the disease, B = test is positive

$P(A \mid B) = P(B \mid A) \times P(A) / P(B)$

- Suppose:
  - 1 in 1,000 people has the disease → P(A) = 0.001
  - The test is positive for 99% of people who actually have the disease → P(B | A) = 0.99
  - The test comes back positive 5% of the time overall → P(B) = 0.05

$P(A \mid B) = (0.99 \times 0.001) / 0.05 = 0.00099 / 0.05 = 0.0198$

Even with a positive test, the chance you have the disease is only about 2%

Bayes' Theorem: Loan Example

- Given a new customer with good credit, what's the chance she gets a loan?

1. From historical data		2. Convert to probabilities	3. Apply Bayes' theorem
Total applicants	1,000	P(A) = 300 / 1000 = 0.300	P(B | A) = P(A | B) × P(B) / P(A)
Good credit (A)	300	P(B) = 200 / 1000 = 0.200	= (0.80 × 0.200) / 0.300
Granted loans (B)	200	P(A | B) = P(A, B) / P(B) = 160 / 200 = 0.80	= 0.5333 (53.33%)
Good credit & granted loan P(A, B)	160		

Answer: there is a 53.33% chance she gets a loan given she has good credit

Bayes' Theorem Example: Liver Disease

Question: if a patient is alcoholic, what's the probability they have liver disease?

- 10% of patients entering a clinic have liver disease; 5% are alcoholic; 40% of patients who have liver disease are alcoholics
- Let A = liver disease, B = alcoholic

Quantity	Meaning	Value
P(A)	Probability patient has liver disease	0.10
P(B)	Probability patient is alcoholic	0.05
P(B | A)	Probability patient is alcoholic given liver disease	0.40
P(A | B)	Probability patient has liver disease given alcoholic	?

$P(A \mid B) = P(B \mid A) \times P(A) / P(B) = (0.40 \times 0.10) / 0.05 = 0.80$

- If a patient is alcoholic, there is an 80% chance they have liver disease

Bayes' Theorem Example

- Rain prediction
- Overall historical probability of rain: P(R) = 0.30
- Sky condition
  - P(Overcast|Rain) = 0.8
  - P(Clear-sky|Rain) = 0.2
  - P(Overcast) = 0.6
- Find P(Rain|Overcast), because today it is overcast
- Suppose A = Rain, B = Overcast sky condition

Quantity	**Meaning**	**Value**
P(A)	Probability of **Rain**	**0.30**
P(B)	Probability of **Overcast sky**	**0.60**
P(B | A)	Probability of overcast sky **given it rains**	**0.80**
Goal: P(A | B)	Probability of rain **given overcast sky**	**?**

Bayes' Theorem Example

- $P(Rain|Overcast) = \frac{P(Overcast|Rain)\ x\ P(Rain)}{P(Overcast)}$
- i.e. $P(A|B) = \frac{P(B|A)\ x\ P(A)}{P(B)}$
- $P(Rain|Overcast) = \frac{0.8\ x\ 0.3}{0.6}$
- $P(Rain|Overcast) = 0.4$ or 40%

Bayes' Theorem

- Dangerous fires are rare (1%)
- Smoke is quite common (10%)
- 90% dangerous fires cause smoke
- What is the probability that we have a dangerous fire when there is smoke?

Quantity	Meaning	Value
P(A)	Probability of a **dangerous fire**	**0.01**
P(B)	Probability of **smoke**	**0.10**
P(B | A)	Probability of smoke **given a dangerous fire**	**0.90**
Goal: P(A | B)	Probability of a dangerous fire **given that we see smoke**	**?**

Bayes' Theorem

- Dangerous fires are rare (1%), Smoke is quite common (10%), 90% dangerous fires cause smoke
- What is the probability that we have a dangerous fire when there is smoke? Suppose A = Fire, B = Smoke
- $P(A \mid B) = \frac{P(B|A)\ x\ P(A)}{P(B)}$
- $P(Fire \mid Smoke) = \frac{P(Smoke \mid Fire)\ x\ P(Fire)}{P(Smoke)}$
- $P(Fire \mid Smoke) = \frac{90\%\ x\ 1\%}{10\%}$
- $P(Fire \mid Smoke) = 9\%$

Random Variables

Random Variable: X = number of heads in two tosses

	1st Toss	2nd Toss	
X = 2 (2 heads)			→ X = 2
X = 1 (1 head)			→ X = 1
X = 0 (0 heads)			→ X = 0

X is a random variable that assigns a number **(0, 1, or 2)** to each outcome.

Random Variable: 3 Credit Card Transactions

All possible outcomes (N = Not Fraud, F = Fraud)

Txn 1	Txn 2	Txn 3	X (# of frauds)
N	N	N	0
N	N	F	1
N	F	N	1
N	F	F	2
F	N	N	1
F	N	F	2
F	F	N	2
F	F	F	3

Probability distribution of X

X (# of frauds)	Probability (fraction)	Probability (decimal)
0	1/8	0.125 (12.5%)
1	3/8	0.375 (37.5%)
2	3/8	0.375 (37.5%)
3	1/8	0.125 (12.5%)
Total	8/8	1.000 (100%)

Random Variables: Example

- A life insurance agent has two elderly clients with policy value of Rs 1 crore each, paid upon death
- Y = younger one dies, P(Y) = 0.05; O = older one dies, P(O) = 0.10; Y and O are independent
- X = amount of money paid out (in crores) based on who dies; possible values of X: 0, 1, 2 crore

Value of X (crores)	Meaning	Probability	Calculation	Result
0	Nobody dies	$P(Y^c\ O^c)$	0.95×0.90	0.855 or 85.5%
1	One of them dies	$P(Y^c\ O) + P(Y\ O^c)$	$(0.95 \times 0.10) + (0.05 \times 0.90) = 0.095 + 0.045$	0.140 or 14%
2	Both die	$P(Y\ O)$	0.05×0.10	0.005 or 0.5%

Another Example

- A restaurant has two high-demand dishes: Dish-A and Dish-B (Only one sell of each dish)
- Chef prepares ingredients every morning, but leftovers must be thrown away, causing a loss
- Estimates for a loss on any given day
  - P(Dish-A is not ordered) = 0.30
  - P(Dish-B is not ordered) = 0.20
- If a dish is not ordered, the restaurant's loss is Rs 500, regardless of the ingredients
- Events are independent

- Random Variable (X) = Total loss because of the leftovers in rupees
- Possible values
  - 0 -> No loss
  - 500 -> One dish left unsold
  - 1000 -> Both dishes left unsold

Random Variable X (Loss)	**Event**	**Probability**	**Calculation**	**Result**
0 (Both dishes sold)	$P^c B^c$	$P(P^cB^c)$	(0.70 × 0.80)	**0.56 → 56%**
500 (Exactly one unsold)	$P\ B^c$ OR $P^c\ B$	$P(P\ B^c) + P(P^c\ B)$	(0.30×0.80) + (0.70×0.20)	**0.38 → 38%**
1000 (Both unsold)	P B	P(PB)	(0.30 × 0.20)	**0.06 → 6%**

Random Variable Types

Type of Random Variable	Meaning	Examples
Discrete	Can only take certain values (Usually integers)	Number of orders placed on a shopping site (500), Number of students in a class (60)
Continuous	Any value in a range (Allows for an infinite range)	Market share of a company (45.23784126…) or Temperature today (23.3802918435…)

Random Variables: Titanic

Gender	Count
male	577
female	314

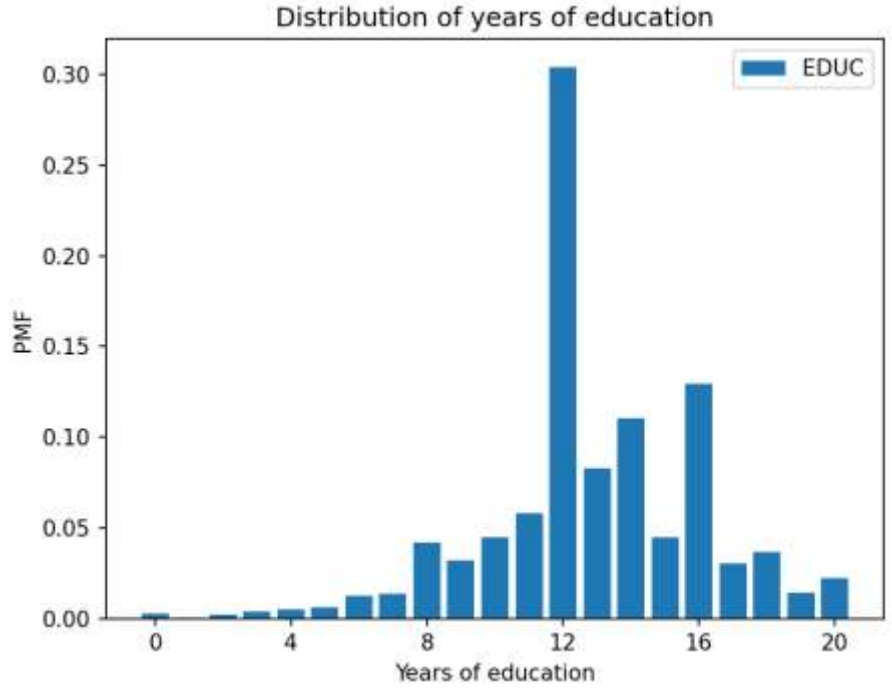
X = gender	P(X)
P(X = male)	577 / 891 = 0.64
P(X = female)	314 / 891 = 0.35

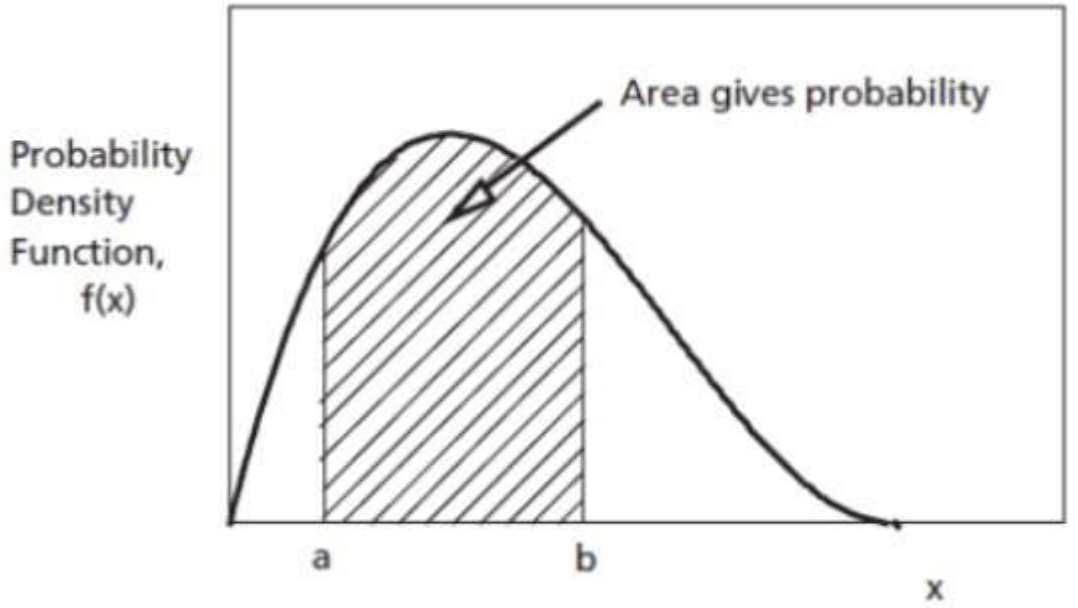
Passenger Class	Count
1	216
2	184
3	491

X = Passenger Class	P(X)
P(X = 1)	216 / 891 = 0.24
P(X = 2)	184 / 891 = 0.20
P(X = 3)	491 / 891 = 0.55

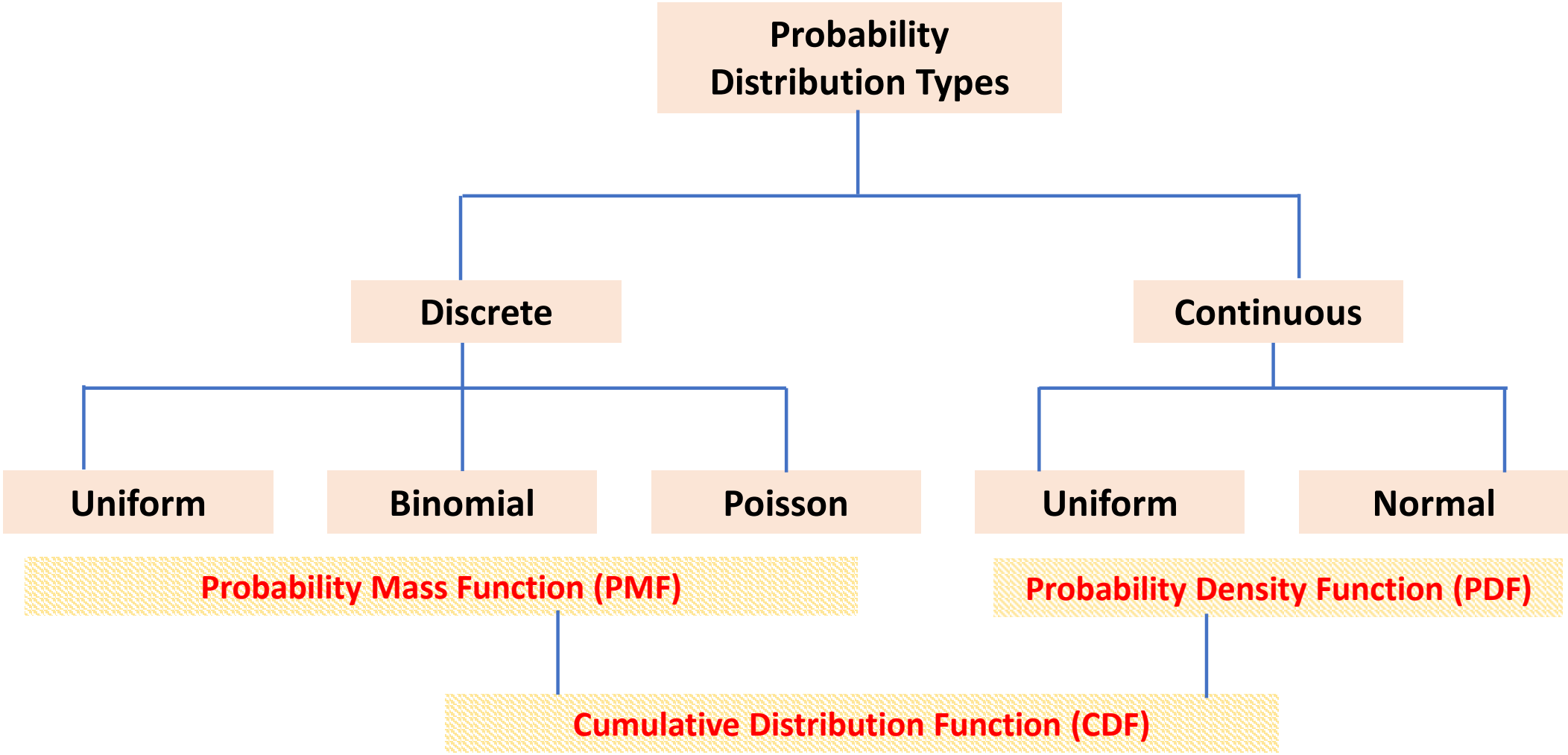
Probability Distribution

- **Distribution**: Describes all the probable outcomes of a **variable**
- **Discrete distribution**: Sum of all individual probabilities = 1
- **Continuous distribution**: Total area under the **probability curve** (**density**) = 1

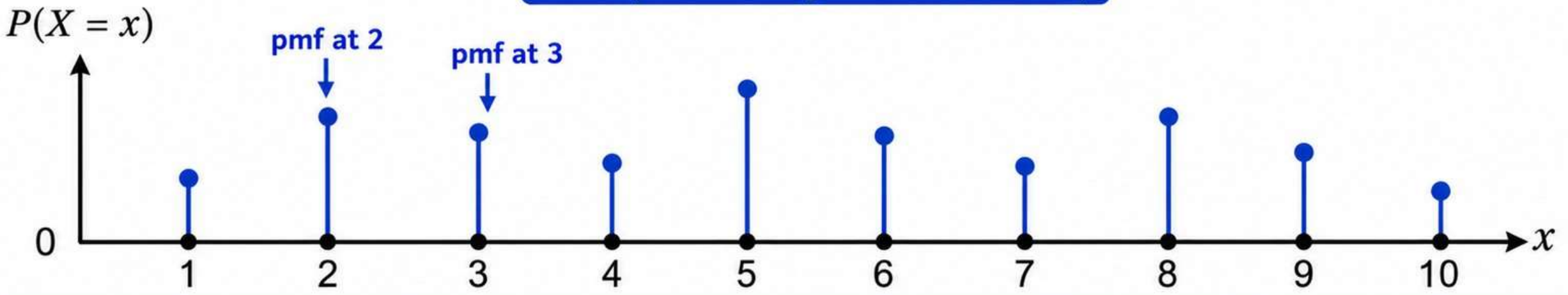




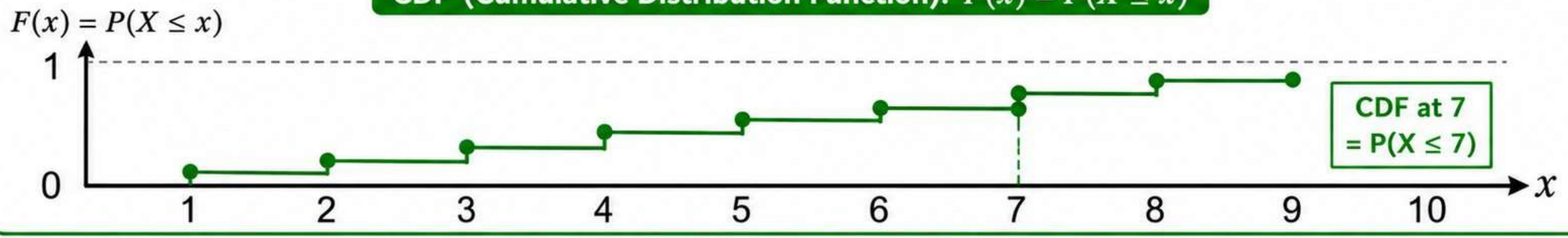
Probability Distribution Types and Estimations



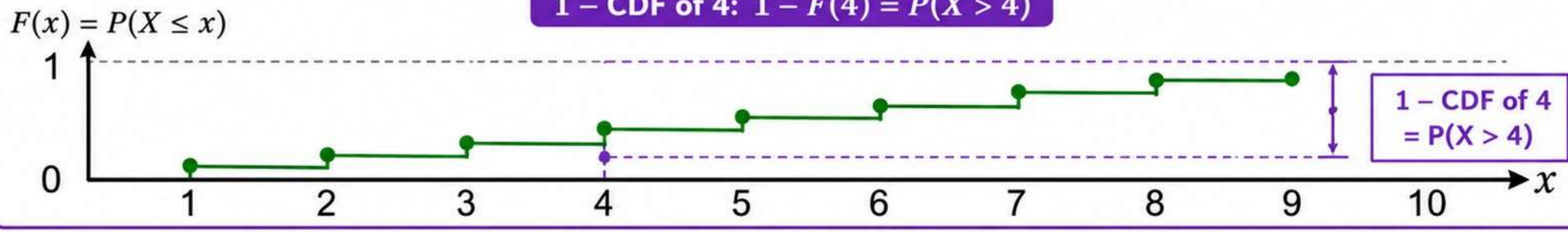
PMF (Probability Mass Function)



CDF (Cumulative Distribution Function): $F(x) = P(X \leq x)$



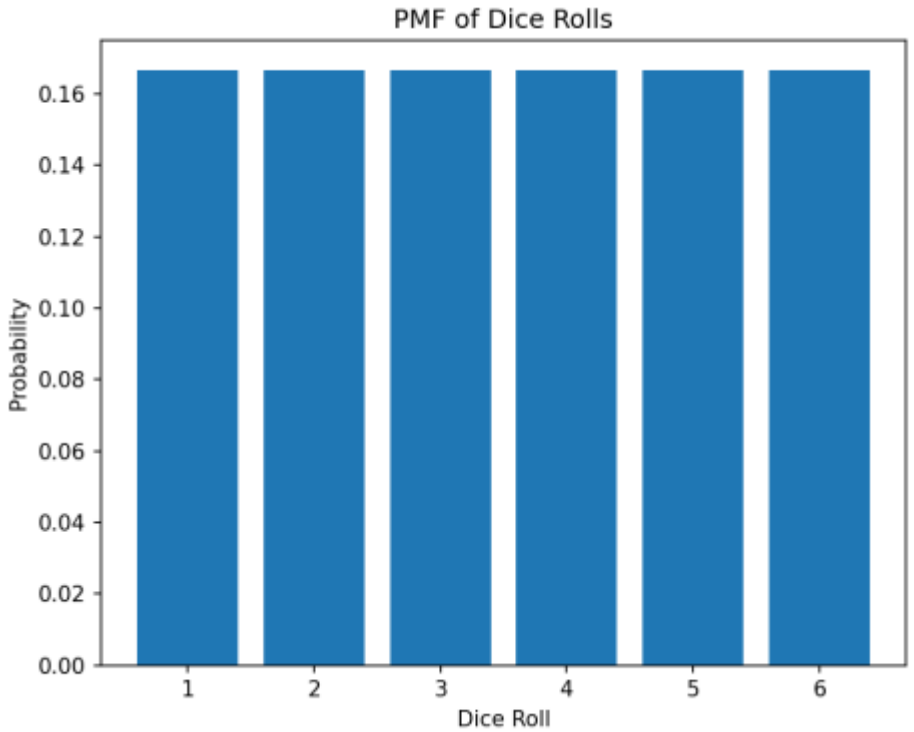
1 – CDF of 4: $1 - F(4) = P(X > 4)$

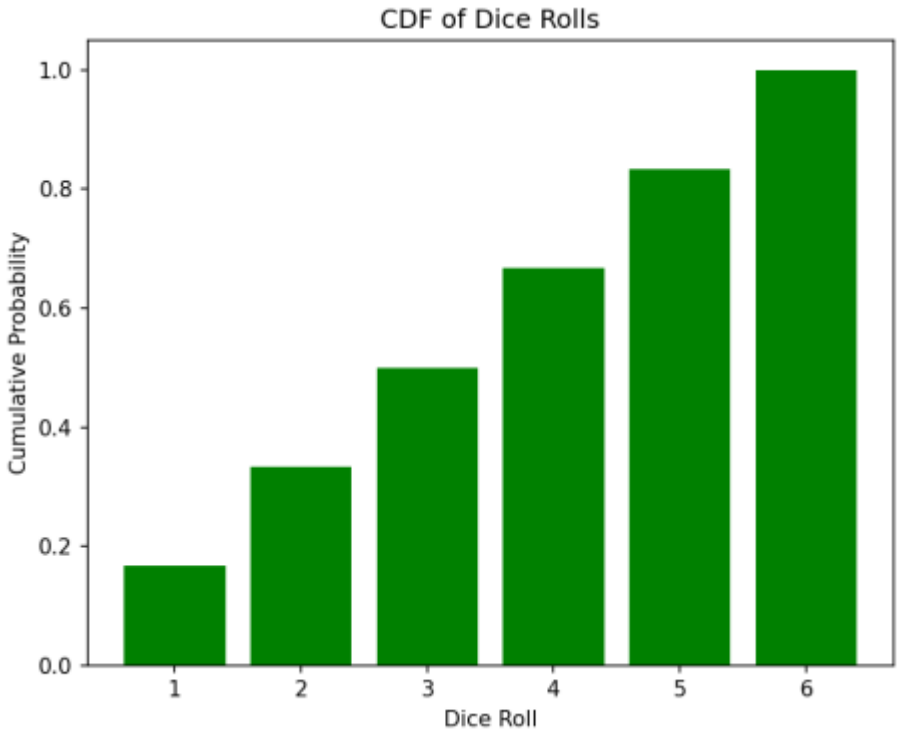


Uniform Distribution



- Rolling a fair die: 6 discrete, equally probable outcomes
- We can roll a 1 or 2, but not 1.5
- Code: C:\code\Data Analytics\uniform_dist.py





Binomial Distribution

- **Bernoulli Trial**: An experiment with only two possible outcomes – success or failure
- Example: Heads or Tails, On or Off, Sick or Healthy
- Many such Bernoulli trials = **Binomial distribution**
- So, tossing a coin once = Bernoulli trial, tossing it say 10 times = Binomial distribution
- Binomial distribution: The two outcomes do not always have to be equally probable
  - Example: In a fruit basket, there are 7 apples and 3 oranges
  - P(A) = $\frac{7}{10}$, P(O) = $\frac{3}{10}$

Binomial Distribution – Key Characteristics

- Fixed number of trials : n
- Each trial has two outcomes: success or failure
- The probability of success: p is constant across the trials
- Trials are independent
- Example: You flip a coin 10 times. What's the probability of getting exactly 7 heads?

Binomial Distribution Formula

Problem: Calculate the probability that *3 out of 5 students like Python*, when in general 66% students like Python

Number of students liking Python, e.g. 3

Probability that someone likes Python, e.g. 66%

$$p(x \mid n, p) = \left(\frac{n!}{x!(n-x)!}\right) p^x (1-p)^{n-x}$$

Probability

Number of students we asked, e.g. 5

Binomial Distribution Example (C:\code\Data Analytics\binom_student.py)

- Problem: Calculate the probability that 3 out of 5 students like Python, when in general 66% students like Python
- **p(x | n, p) =** $\left(\frac{n!}{x!(n-x)!}\right) p^x\ (1-p)^{n\ -\ x}$
- **p(x = 3 | n = 5, 0.66) =** $\left(\frac{5!}{3!(5-3)!}\right) 0.663\ (1-0.66)^{5\ -\ 3}$
- **p(x = 3 | n = 5, 0.66) =** $\left(\frac{5x4\cancel{x3x2x1}}{\cancel{3x2x1}(2x1)}\right) (0.66\ x\ 0.66\ x\ 0.66)\ (0.33)^2$
- **p(x = 3 | n = 5, 0.66) =** $\left(\frac{5\ x\ 4}{2}\right) 0.287\ x\ 0.115$
- **p(x = 3 | n = 5, 0.66) =** $10\ x\ 0.287\ x\ 0.115$
- **p(x = 3 | n = 5, 0.66) = 0.33** Result: **Binomial Probability Mass Function (PMF)**

Exercise (Solve + Code)

- Calculate the probability that 2 out of 7 students prefer online learning over in-person classes when, in general, 55% of students prefer online learning.
- **p(x | n, p) =** $\left(\frac{n!}{x!(n-x)!}\right) p^x\ (1-p)^{n\ -\ x}$
- **p(x = 2 | n = 7, 0.55) =** $\left(\frac{7!}{2!(7-2)!}\right) 0.55^2\ (1-0.55)^{7\ -\ 2}$
- **p(x = 2 | n = 7, 0.55) =** $\left(\frac{7x6x\cancel{5x4x3x2x1}}{2x1(\cancel{5x4x3x2x1})}\right) (0.55\ x\ 0.55)\ (0.45)^5$
- **p(x = 2 | n = 7, 0.55) =** $\left(\frac{42}{2}\right) 0.3025\ x\ 0.0184$
- **p(x = 2 | n = 7, 0.55) = 21 x** $0.3025\ x\ 0.0184$
- **p(x = 2 | n = 7, 0.55) = 0.1168**

Poisson Distribution

- Binomial distribution: Perform *n* trials and determine success/failure
- **Poisson distribution**: Considers the number of successes *per unit of time* (or another continuous unit, such as distance)
- Example: On an average, we read 10 pages of a book in an hour – what are the chances of reading 8 pages in the next hour?
- P(x) = $\frac{\lambda^x e^{-\lambda}}{x!} = \frac{10^8 e^{-10}}{8!} = \frac{10^8 x\ 2.718^{-10}}{8x7x6x5x4x3x2x1}$ = 0.113 or 11%

- Note: e = 2.718, Euler's constant

Poisson Distribution: Practical Usage

- Generally 5 accidents happen at a dangerous road corner in a month and the police department wants to calculate the probability that exactly 0, 1, 2, 3, or 4 accidents will happen in a month
- Poisson distribution formula

$$P(x) = \frac{\lambda^x \times e^{-\lambda}}{x!}$$

Number of accidents	Probability formula	Probability
0	P(0) = (5)^0 x (e^-5) / 0!	0.00674
1	P(1) = (5)^1 x (e^-5) / 1!	0.03370
2	P(2) = (5)^2 x (e^-5) / 2!	0.08425
3	P(3) = (5)^3 x (e^-5) / 3!	0.14042
4	P(4) = (5)^4 x (e^-5) / 4!	0.17552

Sample steps for one case:
P(0) = (5)^0 x (e^-5) / 0!
P(0) = (1) x (0.00674) / 1
P(0) = 0.00674

Poisson Distribution: Practical Usage

Number of accidents	Probability formula	Probability
0	P(0) = (5)^0 x (e^-5) / 0!	0.00674
1	P(1) = (5)^1 x (e^-5) / 1!	0.03370
2	P(2) = (5)^2 x (e^-5) / 2!	0.08425
3	P(3) = (5)^3 x (e^-5) / 3!	0.14042
4	P(4) = (5)^4 x (e^-5) / 4!	0.17552

- Now we can answer many questions
- Probability of 0, 1, or 2 accidents in a month = P(0 or 1 or 2) 0.12469
- Probability of more than 3 accidents = 1 – [P(0) + P(1) + P(2) + P(3)] = 1 - 0.26511 = 0.73489

Poisson Distribution

- Example: Usually, 15 cars pass a toll plaza every hour. We want to know the probability that 13 cars will pass through in the next hour.

- P(x) = $\frac{\lambda^x e^{-\lambda}}{x!} = \frac{15^{13}\ 2.718^{-15}}{13!} = \frac{10^8\, x\ 2.718^{-10}}{8x7x6x5x4x3x2x1}$ = 0.0956 or 9.56%

Geometric Distribution

- How many Bernoulli trials (yes/no events) do we need until we get the first success?
- Type: discrete probability distribution
- Used to model the number of trials till we get the first success
- Example: toss a coin and find the number of tosses till we get the first head
  - P(Success) = p = 0.5
  - P(Failure) = q = 0.5
  - X = number of trials to reach first success

PMF: P(X = k) = (1 – p)^(k–1) × p, for k = 1, 2, 3, … and 0 < p ≤ 1

(Probability that the first success occurs on the kth trial)

Geometric Distribution: Example

- Example: Toss a coin and find the number of tosses till we get the first head
  - P(Success) = p = 0.5
  - P(Failure) = q = 0.5
  - X = Number of trials to reach first success
- PMF: $\mathbf{P(X = k) = (1 - p)^{k - 1} x p}$ … Probability of success at the kth trial
- Probability of getting the first head on the third trial = PMF (X = 3) =

  $(1 - 0.5)^{3-1}$ x 0.5 = $(0.5)^2$ x 0.5 = 0.25 x 0.5 = 0.125

Geometric Distribution: Job Applications Example

PMF: P(X = k) = (1 − p)^(k−1) × p – probability that the first success occurs on the kth trial

- Success = receiving the first job offer; each application results in Offer / No Offer
- Let p = 0.2 (probability that one application leads to an offer)
- Find the probability that the first offer arrives from the 4th application

Calculation

- P(X = 4) = (1 − 0.2)^(4−1) × 0.2
- = $(0.8)^3$ × 0.2
- = 0.512 × 0.2

= 0.1024

Interpretation

- Application 1 → Rejected
- Application 2 → Rejected
- Application 3 → Rejected
- Application 4 → Offer received

Key idea: the geometric distribution models the number of Bernoulli trials needed until the first success occurs

Continuous Distribution: Exponential Distribution

Exponential Distribution

- Continuous probability distribution used to model time between events in a Poisson process
- Example: On an average, a customer arrives every 4 minutes in a supermarket
- **Exponential distribution** will describe the time between the arrival of two customers
- Events occur continuously and independently at a constant average rate
- PDF: f(x; λ) = $\begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$
- CDF: f(x; λ) = $\begin{cases} 1 - e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$
- λ = Rate parameter (Average number of events per unit time)
- Mean (μ) = Average time between events = $\frac{1}{\lambda}$
- e = 2.718

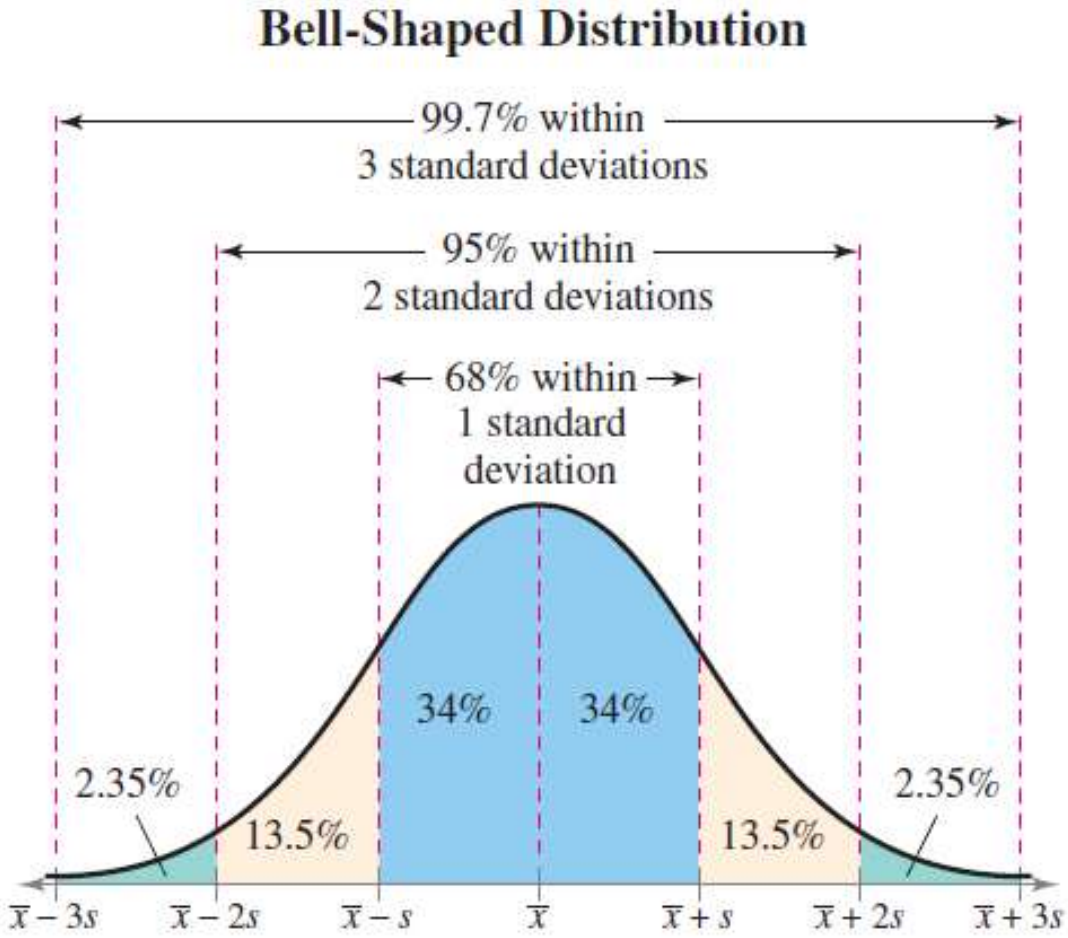
Exponential Distribution Example

- Let X = amount of time (in minutes) at which a new customer arrives at a supermarket
- μ = 4 minutes (Average time between arrivals)
- $\lambda = \frac{1}{\mu} = \frac{1}{4} = 0.25$ (Rate parameter)
- Suppose we have x = 5, then f(x; λ) = $\lambda e^{-\lambda x}$ = 0.25 x 2.718 ^ (-0.25 x 5)
- So f(5; 0.25) = 0.072 = Relative likelihood (Density) that the next customer arrives exactly after 5 minutes
- Note: We cannot say exact probability, since this is continuous distribution
- Code: exponential_distribution_pizza_orders.py

Normal Distribution

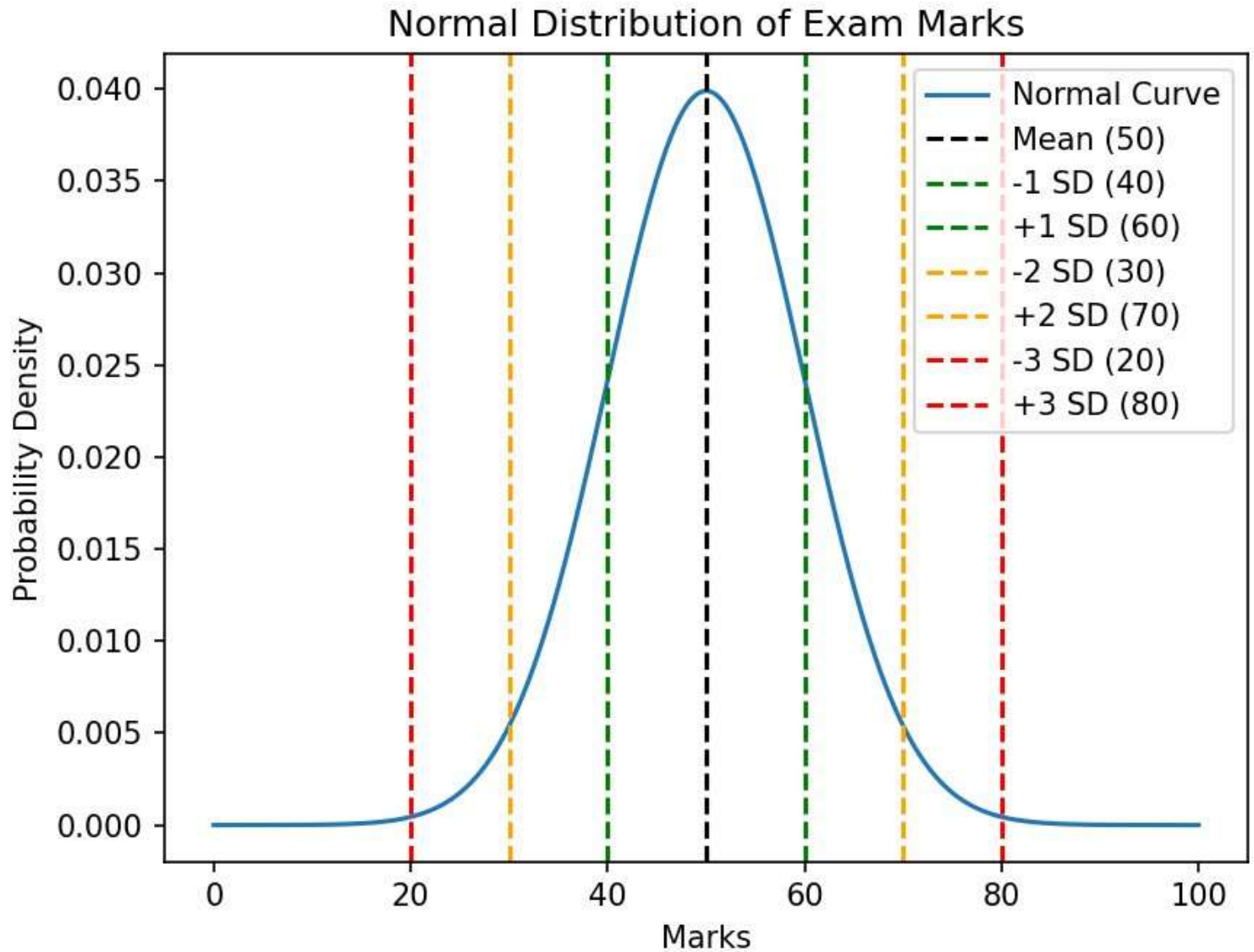
- **Normal Distribution/Gaussian Distribution**
- Example: Random Blood Glucose Test
  - Estimated population mean: 100 mg/dL
  - Estimated standard deviation: 15 mg/dL

Blood Glucose	Within	Why?
85 to 115	68%	Mean ± 1 SD
70 to 130	95%	Mean ± 2 SD
55 to 145	99.7%	Mean ± 3 SD



Empirical Rule

Normal Distribution

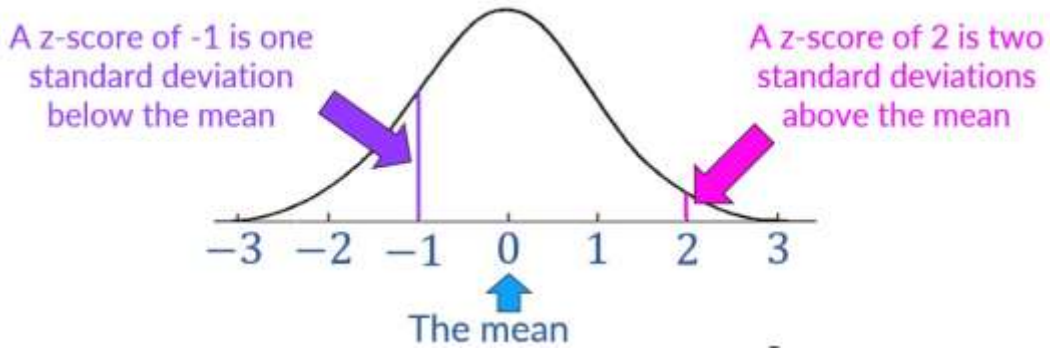


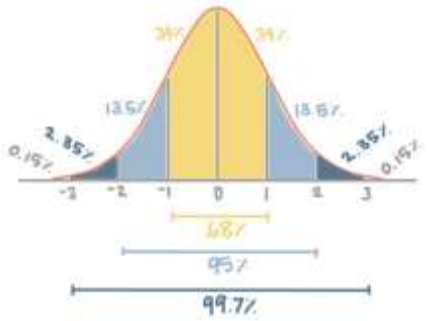
Z-Score

Z-score
Describes *how many standard deviations away* a data point is *from the mean*

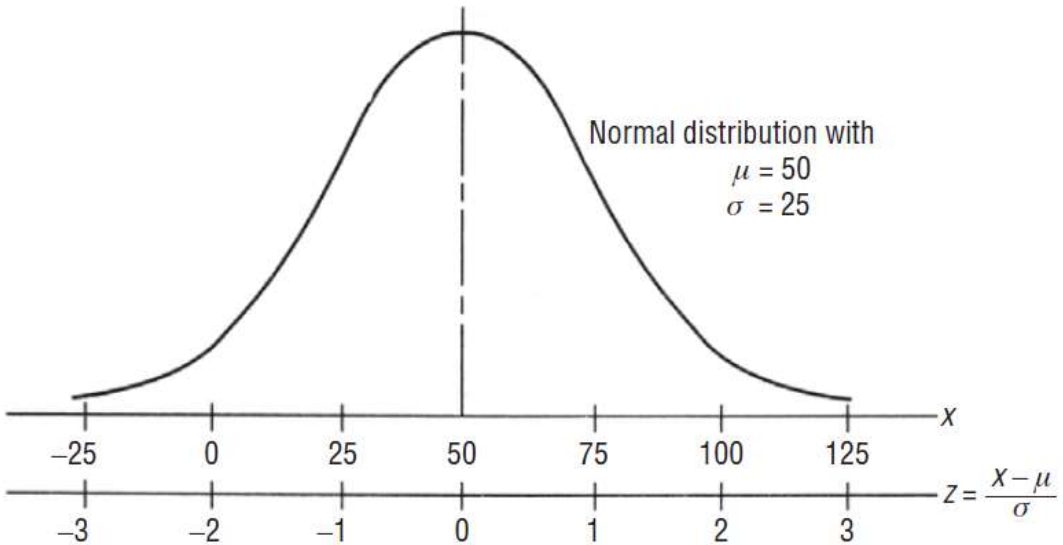
X = Our data point
μ = Hypothesized population mean
σ = Hypothesized standard deviation

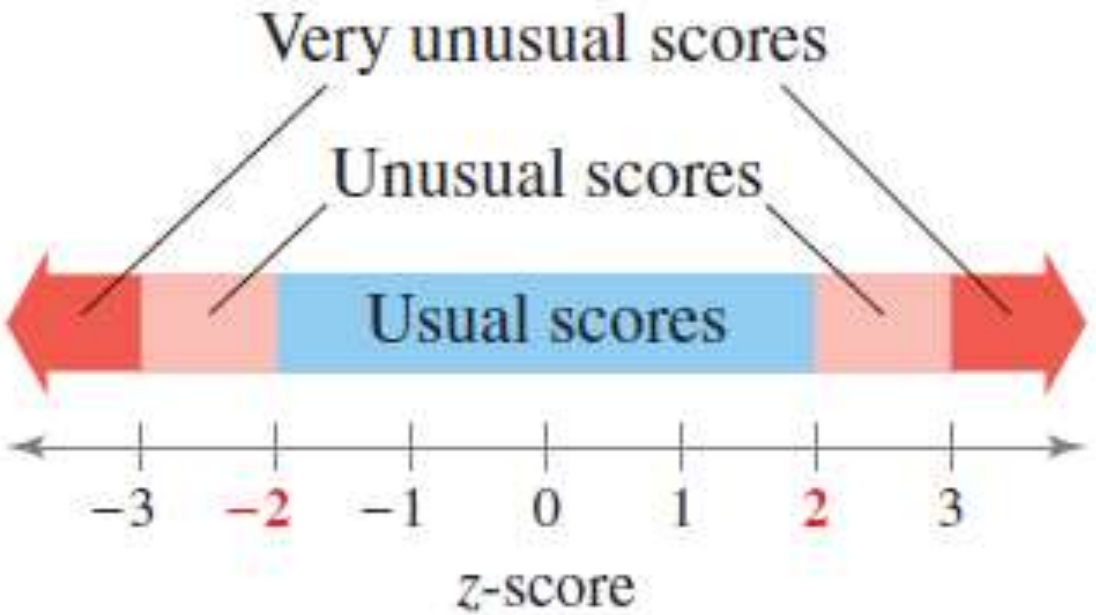
- **Z-score (Standard score)**: **Relative position** of an observation = $\frac{X - \mu}{\sigma}$
- A patient has blood glucose level = 115
- Z = $\frac{X - \mu}{\sigma}$ = $\frac{115 - 100}{15}$ = 1
- From Z-table,
  - Cumulative probability at Z-score of 1 = 0.8413
  - Meaning: 84.13% data lies below +1
  - Cumulative probability at Z-score of -1 = 0.1587
  - Meaning: 15.87% data lies below -1
- Conclusion: Probability/Area under the curve between -1 and +1
  - 0.8413 – 0.1587 = 0.6826 ≈ 68%





Z-Score Interpretation





Normal Distribution: Example

- In the US, a baby is just born with weight of 2.91 kg (2910 grams)
- The mother is told that the baby's weight is *below average* and hence she is concerned
- Is the weight unusually low?
- Mean: 3480 grams, Standard deviation: 462 grams

- $Z - Score = \frac{X - \mu}{\sigma}$
  - x: individual data value
  - μ: population mean
  - σ: population standard deviation

- $Z = \frac{2910 - 3480}{462} = -1.23$
- Interpretation: The baby's weight is 1.2 SD below the mean (below, because it is negative)

Using Z-Score

- We know the Z-score for the current baby: -1.23
- Go to Z-table (https://www.z-table.com/)
- Look for -1.2 in row and 3 in column
- Interpretation: ~11% babies have lesser weight
- So, this baby is in the 11th percentile

z	.00	.01	.02	.03
–3.4	.0003	.0003	.0003	.0003
–3.3	.0005	.0005	.0005	.0004
–3.2	.0007	.0007	.0006	.0006
–3.1	.0010	.0009	.0009	.0009
–3.0	.0013	.0013	.0013	.0012
–2.9	.0019	.0018	.0018	.0017
–2.8	.0026	.0025	.0024	.0023
–2.7	.0035	.0034	.0033	.0032
–2.6	.0047	.0045	.0044	.0043
–2.5	.0062	.0060	.0059	.0057
–2.4	.0082	.0080	.0078	.0075
–2.3	.0107	.0104	.0102	.0099
–2.2	.0139	.0136	.0132	.0129
–2.1	.0179	.0174	.0170	.0166
–2.0	.0228	.0222	.0217	.0212
–1.9	.0287	.0281	.0274	.0268
–1.8	.0359	.0351	.0344	.0336
–1.7	.0446	.0436	.0427	.0418
–1.6	.0548	.0537	.0526	.0516
–1.5	.0668	.0655	.0643	.0630
–1.4	.0808	.0793	.0778	.0764
–1.3	.0968	.0951	.0934	.0918
–1.2	.1151	.1131	.1112	.1093

Z-Score: Problem

- A runner participated in a 200m race and a 500m race
- Consider the following, calculate Z-scores and determine where she did better

Race	Average time	Standard deviation	Runner's time
200m	31s	1.5s	28s
500m	125s	8.2s	132s

Z-Score Example

- A runner participated in a 200m race and a 500m race
- Consider the following and determine where she did better

Race	Average time	Standard deviation	Runner's time
200m	31s	1.5s	28s
500m	125s	8.2s	132s

200m

$$Z = \frac{X - \mu}{\sigma} = \frac{28 - 31}{1.5} = -2$$

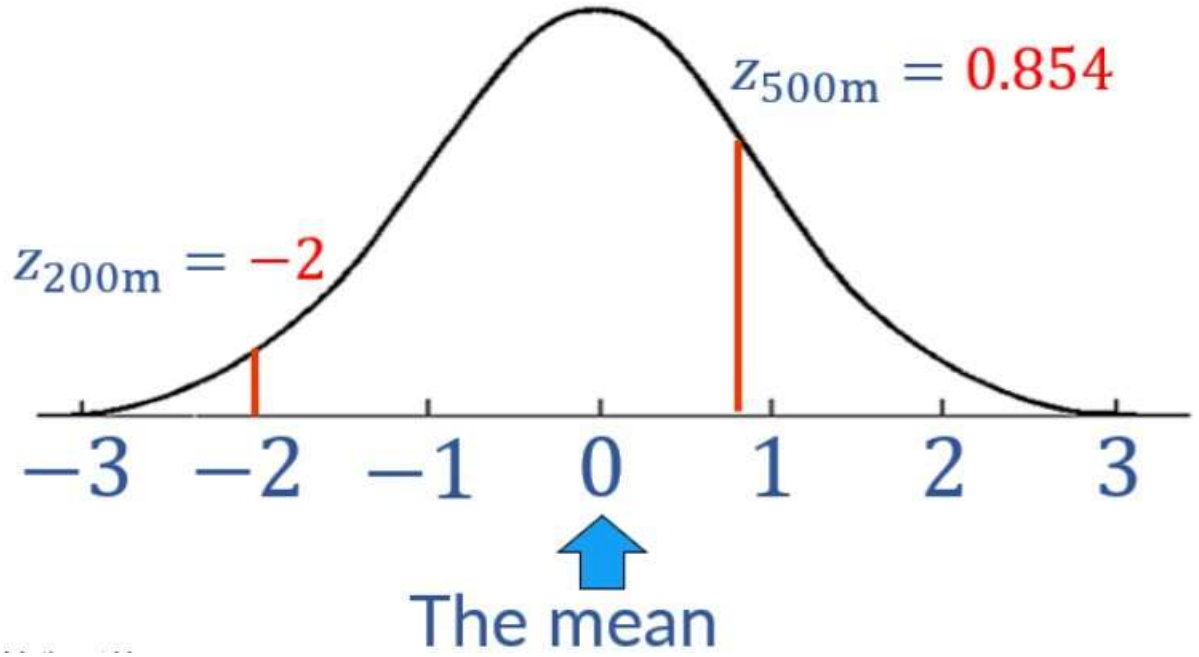
500m

$$Z = \frac{X - \mu}{\sigma} = \frac{132 - 125}{8.2.} = 0.854$$

Visualizing Z-Scores

In other examples, positive/higher Z-score will be better, e.g. marks obtained by a student – Because, here the student would want to be *above* average

- In this example, a lower time would be preferable when completing a race and so, the lower z-score would be better



The 200m time was below the mean and was therefore the quicker than the average runner

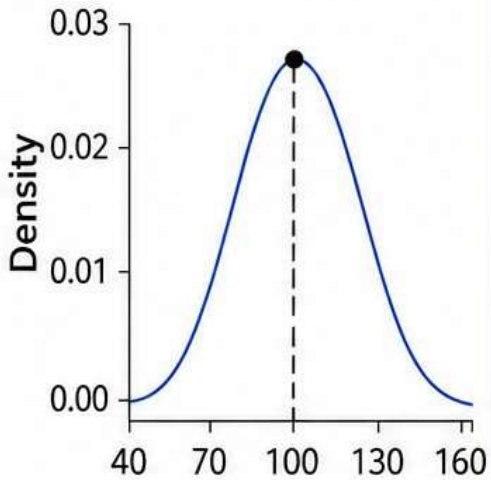
PDF

(Probability Density)

pdf(x)

Density at a value x.

```
x = 100
norm.pdf(x, mu, sigma)
```



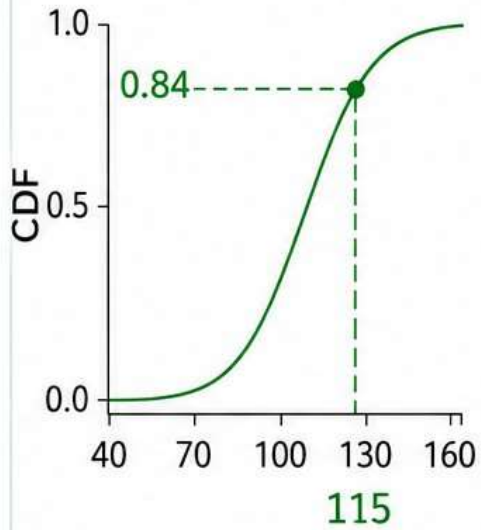
CDF

(Cumulative Probability)

cdf(x)

$P(X \leq x)$.

```
x = 115
norm.cdf(x, mu, sigma)
```



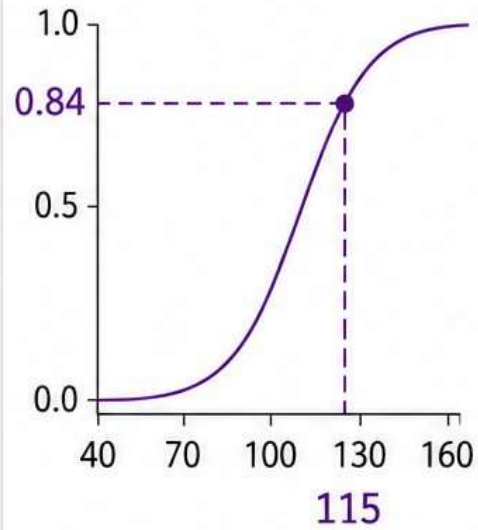
PPF

(Inverse CDF)
(Percentile)

ppf(q)

Value x such that $P(X \leq x) = q$.

```
q = 0.84
norm.ppf(q, mu, sigma)
```



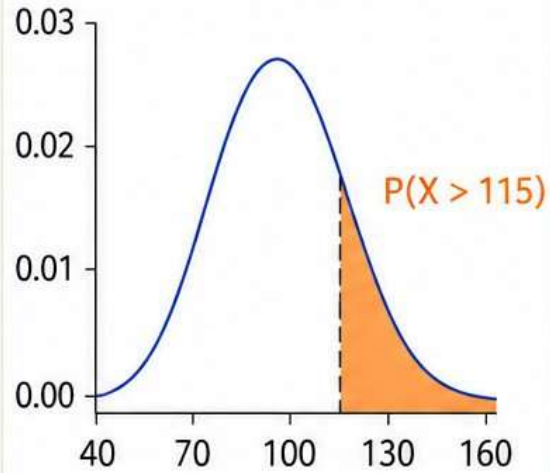
SF

(Survival Function)

sf(x)

$P(X > x) = 1 - CDF(x)$.

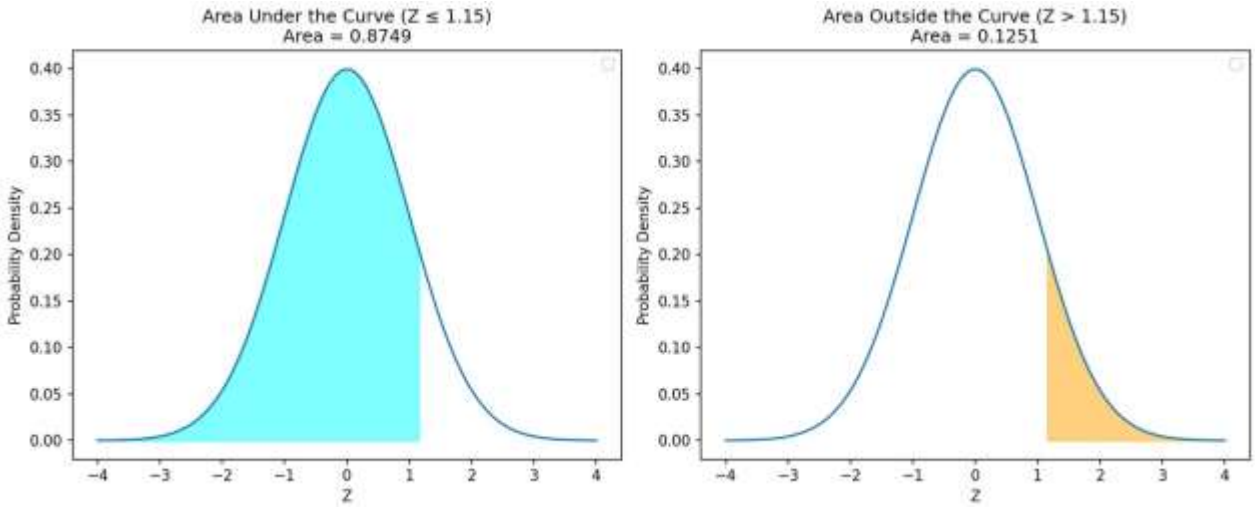
```
x = 115
norm.sf(x, mu, sigma)
```



Understanding Z-Score and Area Under the Curve (Probability)

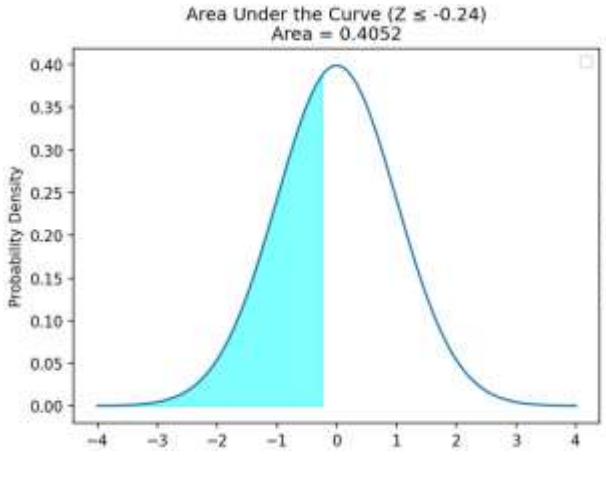
- Suppose Z-Score = 1.15

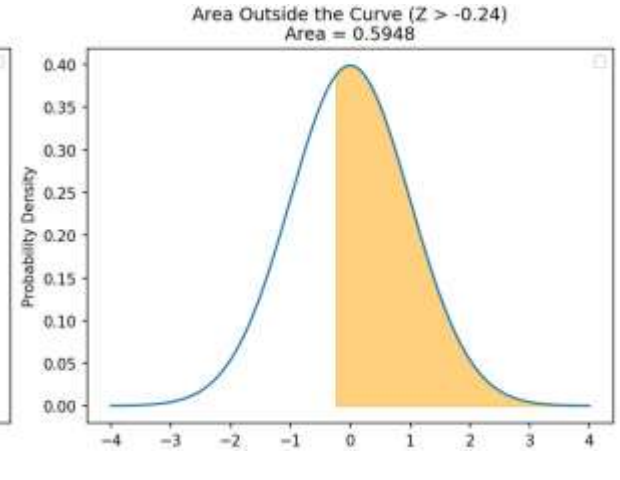
z	.00	.01	.02	.03	.04	.05	.06
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279



- Suppose Z-Score = -0.24

z	.09	.08	.07	.06	.05	.04	.03
−3.4	.0002	.0003	.0003	.0003	.0003	.0003	.0003
−3.3	.0003	.0004	.0004	.0004	.0004	.0004	.0004
−3.2	.0005	.0005	.0005	.0006	.0006	.0006	.0006
−0.5	.2776	.2810	.2843	.2877	.2912	.2946	.2981
−0.4	.3121	.3156	.3192	.3228	.3264	.3300	.3336
−0.3	.3483	.3520	.3557	.3594	.3632	.3669	.3707
−0.2	.3859	.3897	.3936	.3974	.4013	.4052	.4090
−0.1	.4247	.4286	.4325	.4364	.4404	.4443	.4483
−0.0	.4641	.4681	.4721	.4761	.4801	.4840	.4880





Student Example: Z-Scores, Probabilities, Percentiles

- Boy with a height of 62 inches
- Z–Score $= \frac{62 - 67.16}{2.88} = -1.79$

- From negative scores Z-table (Row = -1.7, Column = 0.09), Probability = 0.0367
- This student is in the 3.67th percentile
  - He is taller than 3.67% boys
  - 96.33% boys are taller than him

- Girl with a height of 63 inches
- Z–Score $= \frac{63 - 61.98}{2.11} = 0.48$

- From positive scores Z-table (Row = 0.4, Column = 0.08), Probability = 0.6844
- This student is in the 68.44th percentile
  - She is taller than 68.44% girls
  - Only 31.56% girls are taller than her

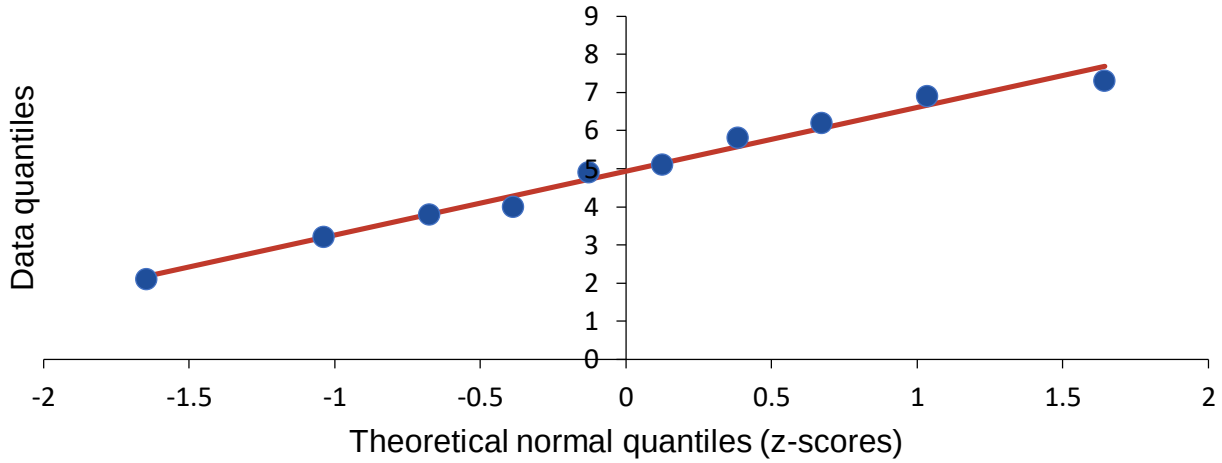
Checking Normality: Shapiro Test and QQ Plot

Test / Plot	What It Is	When to Use	How to Interpret
Shapiro–Wilk Test	Statistical test: checks if data come from a normal distribution	≤ 5000 rows	H0: data looks normal p > 0.05 → fail to reject H0 p < 0.05 → not normal
Anderson Test	Goodness-of-fit test: checks how well data follow a normal distribution. Focuses on tails.	> 5000 rows	Statistic < critical value → fits normal Statistic > critical value → not normal
QQ Plot (Quantile–Quantile Plot)	Graphical tool: compares data quantiles to a normal distribution	All datasets	Straight line → normal Curved → skewness S-shape → heavy/light tails Extreme deviations → outliers

From Data to Q–Q Plot

1. Sample dataset (n = 10), sorted
2. Compute theoretical normal quantiles (mean = 0, sd = 1) for the same n, using p = (i − 0.5) / n
3. Plot data quantiles (y-axis) against theoretical normal quantiles (x-axis)

Rank (i)	Data (y-axis)	Z-score (x-axis)
1	2.1	-1.6449
2	3.2	-1.0364
3	3.8	-0.6745
4	4.0	-0.3853
5	4.9	-0.1257
6	5.1	0.1257
7	5.8	0.3853
8	6.2	0.6745
9	6.9	1.0364
10	7.3	1.6449

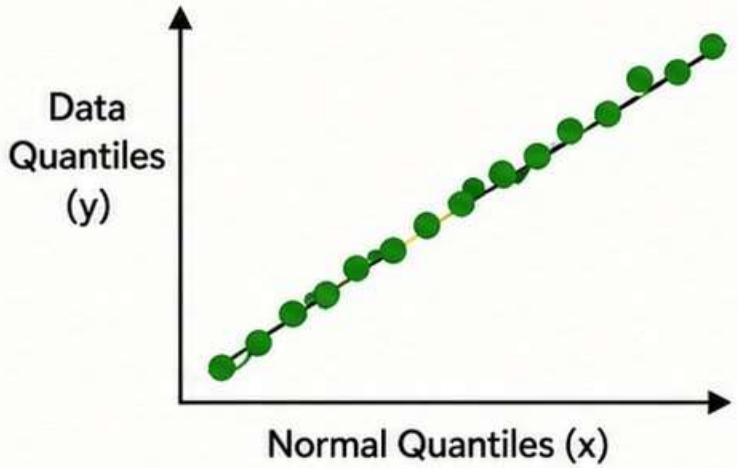


- If the data are approximately normal, the points fall close to the straight line
- Here the points are close to the line ⇒ the data look approximately normal

Q–Q Plot: Normality at a Glance

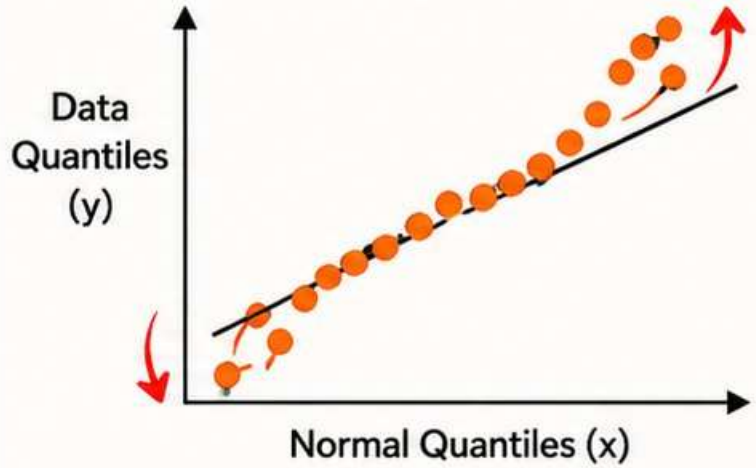
Compare your data (y-axis) to Normal quantiles (x-axis).

Normal



Points fall on a **straight line** → **Normal**

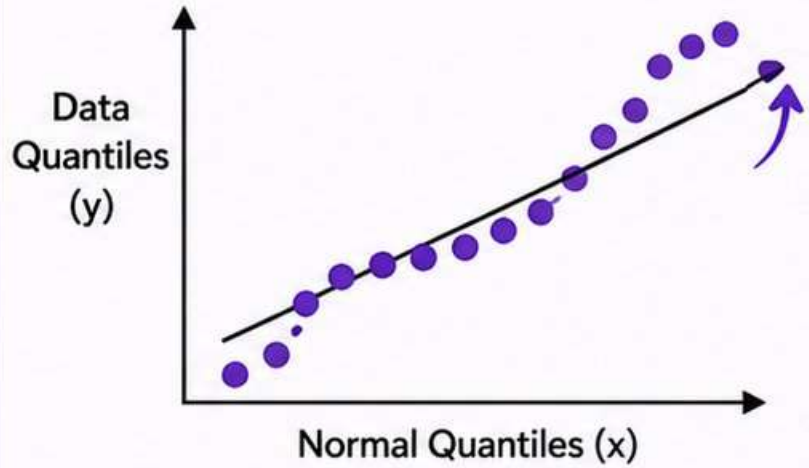
Heavy Tails



Curves away at both ends

→ **Heavy tails** (more extremes)

Skewed



Curves away on one side

→ **Skewed**

Closer to the straight line → More Normal

Shapiro-Wilk Test in Python

- The Shapiro-Wilk test checks if your data is normally distributed

1. Perform Shapiro-Wilk test

```
from scipy.stats import shapiro
stat, p_value = shapiro(data)
```

2. Interpret the result

```
alpha = 0.05
if p_value > alpha:
    print("Data is normally distributed (Fail to
reject H0)")
else:
    print("Data is not normally distributed (Reject
H0)")
```

What it does

- Calculates a test statistic (W) and a p-value
- Null hypothesis (H0): data is normally distributed
- If p-value > 0.05 → fail to reject H0 (data is normal)
- If p-value ≤ 0.05 → reject H0 (data is not normal)

Example output

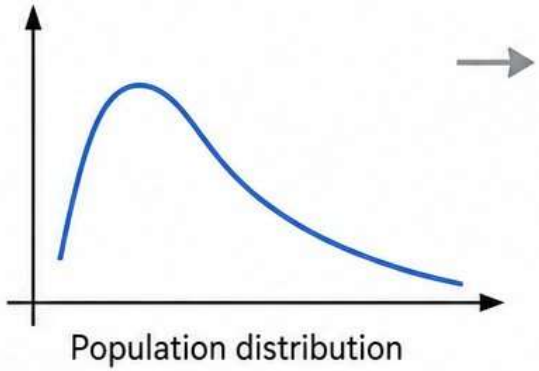
```
W statistic: 0.9715
p-value: 0.8987
Data is normally distributed (Fail to reject H0)
```

Central Limit Theorem (CLT)

No matter what the population looks like,
the distribution of sample means is approximately normal (if $n \geq 30$).

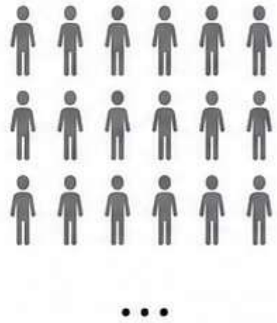
1 Population

May not be normal



2 Take Many Samples

e.g., 500 samples,
each of size n
($n \geq 30$)



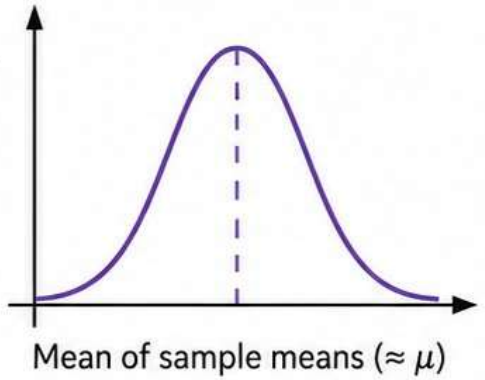
3 Calculate Sample Mean

Find $\bar{x}$ for each sample
Collect all $\bar{x}$'s

$\bar{x}_1$
$\bar{x}_2$
$\bar{x}_3$
$\vdots$
$\bar{x}_{500}$

4 Distribution of Sample Means

Approximately normal



Key Takeaways

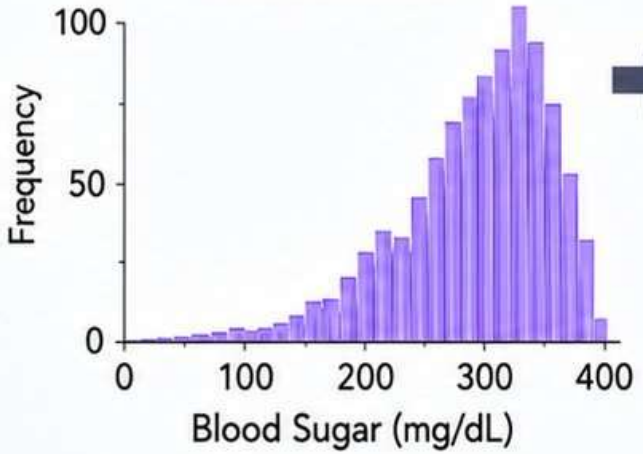
- If sample size $n \geq 30$, the sampling distribution of the mean is approximately normal.
- This is true **even if the population is not normal**.
- More samples give a smoother (more reliable) normal shape.

Central Limit Theorem (CLT) in Action

1 Population (Blood Sugar)

All 768 patients (mg/dL)

Left-skewed (Not normal)



2 Take Many Samples

Draw 500 random samples, each of size **n = 30**

Sample 1 ···

Sample 2 ···

Sample 3 ···

⋮

Sample 500 ···

3 Compute Sample Mean

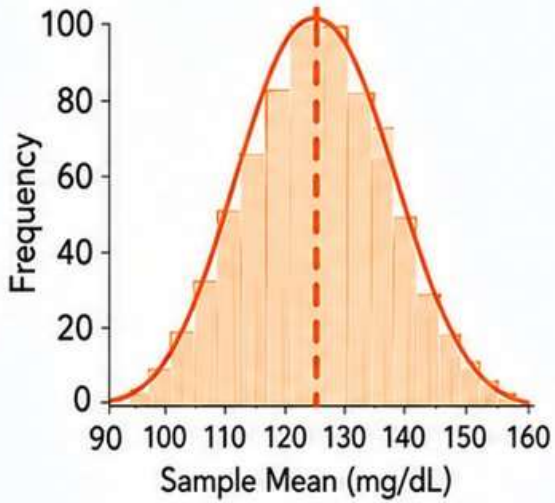
Calculate mean blood sugar for each sample

Sample 1	→	127.6
Sample 2	→	130.2
Sample 3	→	124.8
⋮		⋮
Sample 500	→	129.1

500 sample means

4 Distribution of Sample Means

Plot the 500 sample means



Approximately normal!

Key Takeaway: Even though the blood sugar values are left-skewed (not normal) with a high average, the sample means **(n ≥ 30)** form an approximately normal distribution.

Hypothesis: The Foundation of Statistical Inference

Concept	Explanation	Notes
Hypothesis = Assumption	A hypothesis is simply an assumption about a population parameter (e.g., mean, proportion)	Examples: Mean (μ), Proportion (p), Variance (σ^2)
Why Needed?	Testing the entire population is impossible or impractical → so we use a sample and draw conclusions about the population	Sample data helps us make informed decisions
Example	Average cola in a bottle is 500 ml. So, μ = 500 ml	We test whether data supports this claim
Purpose of Hypothesis Testing	To evaluate how well a sample supports an assumption about the population	Helps in decision-making under uncertainty

Hypothesis Concept

Concept	Explanation
Null Hypothesis (H0)	A neutral or status-quo statement. It assumes no change, no difference, or no effect – the baseline we start with.
Alternative Hypothesis (H1 or Ha)	What we actually want to test or prove – a claim that there is a change, difference, or effect.
Decision Rule	We never prove a hypothesis – we either reject H0 (if data give strong evidence) or fail to reject H0 (if evidence is not strong enough). Note: modern usage sometimes ignores this.
Key Takeaway	We use data to decide whether there is enough evidence to reject the null hypothesis.

Hypothesis Example

- Example: average cola in a bottle is 500 ml ($\mu = 500$ ml)

Hypothesis	Statement	Symbolic Form	When to Use	Rejection Region
Null Hypothesis (H0)	Average cola in a bottle is 500 ml	$\mu = 500$, $\mu \geq 500$ or $\mu \leq 500$	Always includes =, ≥, or ≤ because it defines the reference point	—
Alternative Hypothesis (H1)	There is a significant difference (checking if there's any change)	$\mu \neq 500$	Two-tailed test (checks both sides)	Both tails
	Reject if the quantity is less than 500 ml	$\mu < 500$	Left-tailed test (checks only left side)	Left tail
	Reject if the quantity is more than 500 ml	$\mu > 500$	Right-tailed test (checks only right side)	Right tail

- Note: choose the alternative hypothesis based on the direction of the effect you want to detect

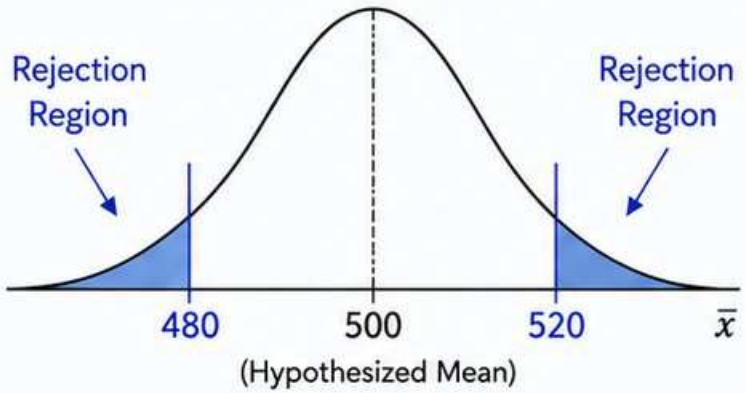
REJECTION REGIONS FOR HYPOTHESIS TESTING

TWO-TAILED TEST

H_0: $\mu = 500$ ml vs H_1: $\mu \neq 500$ ml

Rejection regions exist on both sides.
It is **harder to reject** H_0 and **easier to fail to reject.**

Rejection Regions: $\bar{x} < 480$ or $\bar{x} > 520$



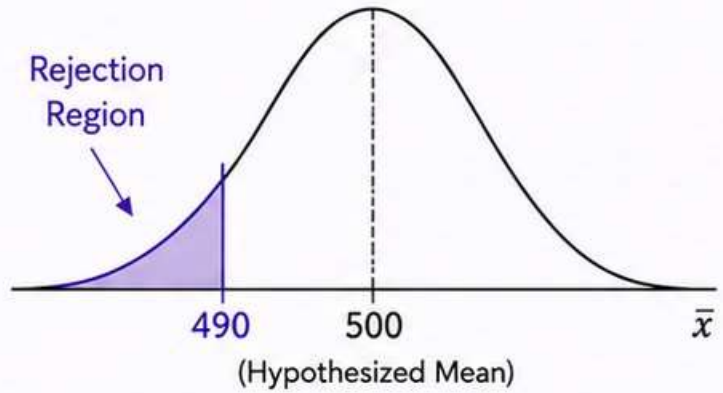
Reject H_0 if $\bar{x} < 480$ or $\bar{x} > 520$
(evidence that the mean is **different** from 500 ml)

ONE-TAILED TEST (LEFT)

H_0: $\mu \geq 500$ ml vs H_1: $\mu < 500$ ml

The entire rejection region is on the **left** side only.
So, it is **easier to reject** H_0 when the mean is small, but **harder to fail to reject.**

Rejection Region: $\bar{x} < 490$



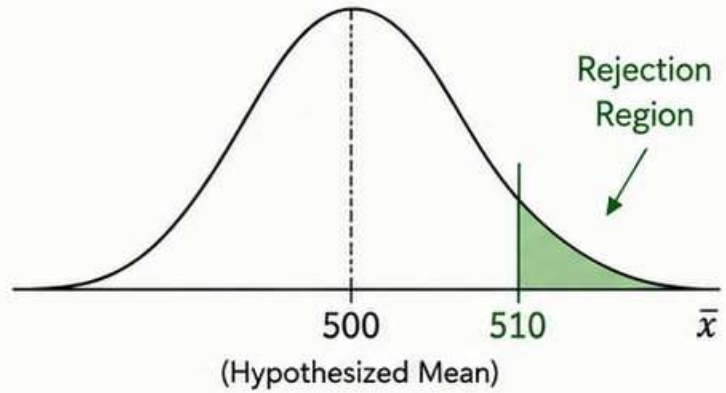
Reject H_0 if $\bar{x} < 490$
(evidence that the mean is **less than** 500 ml)

ONE-TAILED TEST (RIGHT)

H_0: $\mu \leq 500$ ml vs H_1: $\mu > 500$ ml

The entire rejection region is on the **right** side only.
So, it is **easier to reject** H_0 when the mean is **large**, but **harder to fail to reject.**

Rejection Region: $\bar{x} > 510$



Reject H_0 if $\bar{x} > 510$
(evidence that the mean is **greater than** 500 ml)

Note: The equality is always included in H_0.

Framing Hypothesis

Categorical Scenarios	Null Hypothesis (H0)	Alternative Hypothesis (H1)
Email spam detection	Email is NOT spam	Email IS spam
Medical diagnosis	Patient does NOT have the disease	Patient HAS the disease
Credit card fraud	Transaction is NOT fraudulent	Transaction is fraudulent
Manufacturing inspection	Product is NOT defective	Product IS defective
Cybersecurity login check	Login attempt is NOT an attack	Login attempt is an attack

Numerical Scenarios	Null Hypothesis (H0)	Alternative Hypothesis (H1)
Average height of students	$\mu = 165$ cm	$\mu \neq 165$ cm
Mean delivery time	$\mu \leq 2$ days	$\mu > 2$ days
Mean weight of packaged chips	$\mu = 50$ g	$\mu \neq 50$ g
Average monthly electricity bill	$\mu = ₹3,000$	$\mu \neq ₹3,000$
Mean blood pressure after medication	$\mu1 = \mu2$	$\mu1 \neq \mu2$

Hypothesis Testing

Step	What to do
1. State the assumptions	Null hypothesis (H0) → the default assumption Alternate hypothesis (H1) → what we want to test/prove
2. Choose significance level (α)	α = probability of making a Type I error (rejecting a true H0) Common values: 0.05, 0.01, 0.10
3. Calculate test statistic	Measures how far the sample statistic (e.g., $\bar{x}$) is from the hypothesized population value (e.g., μ) under H0
4. Convert to probability (p-value)	p-value = probability of observing a result at least as extreme as our sample result, assuming H0 is true
5. Make a decision	Compare p-value with α: p-value < α → reject H0 (assumption was wrong) p-value ≥ α → fail to reject H0 (assumption stands)
Alternative method	Compare test statistic with critical value: |Test statistic| > critical value → reject H0 Otherwise → fail to reject H0

Hypothesis Testing: Example

1. Null Hypothesis (H0)	Average cola in a bottle (μ) = 500 ml
2. Alternate Hypothesis (H1)	Average cola in a bottle (μ) ≠ 500 ml
3. Data Collection	We collect 100 bottles from 100 different shops, measure cola in each bottle, and calculate the average
4. Our Finding	Average cola in 100 sample bottles ($\bar{x}$) = 498 ml, with a standard deviation (s) = 5 ml
5. Decision	Should we reject or not reject H0?

Hypothesis Testing Example: Is the Average Cola Different from 500 ml?

Step	What we do	Result
1	Null Hypothesis (H0): assume average cola = 500 ml	$\mu = 500$ ml
2	Alternate Hypothesis (H1): assume average cola ≠ 500 ml (two-tailed test)	$\mu \neq 500$ ml
3	Sample size: number of bottles	$n = 100$
4	Sample mean: observed average	$\bar{x} = 498$ ml
5	Sample standard deviation: sample variability	$s = 5$ ml
6	Standard error: $SE = s / \sqrt{n}$	0.5 ml
7	Z-score: $Z = (\bar{x} - \mu) / (s / \sqrt{n})$	$Z = -4$
8	Critical Z ($\alpha = 0.05$, two-tailed)	±1.96: reject H0 if $Z < -1.96$ or $Z > 1.96$
9	P-value: probability of seeing this result by chance	0.00006 (extremely tiny)
10	Decision: compare Z with critical values	Reject H0 ($Z = -4 < -1.96$)
11	Conclusion	Average cola ≠ 500 ml

There is strong evidence that the true average cola in a bottle is not 500 ml

Significance Level: Alpha (α)

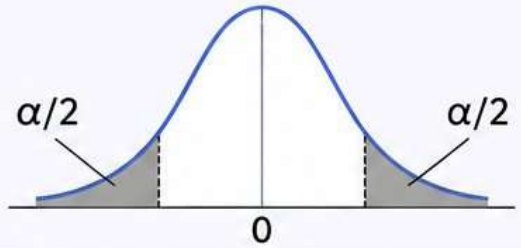
- Goal: test the claim that the average cola in a bottle is 500 ml ($\mu = 500$)
- We collect sample data, calculate the sample mean ($\bar{x}$) and compare it with the population mean ($\mu = 500$)
- Question: is the sample mean "close enough" to 500?

Sample Mean ($\bar{x}$)	Observation	Human Thinking (Unscientific)
495 (close to 500)	Very close to $\mu = 500$ (is this close enough?)	Do not reject H0 (uncertain)
350 (far from 500)	Far away from $\mu = 500$ (is this far enough?)	Reject H0 (uncertain)

- Summary: this is an unscientific method. Is 495 reasonably close to 500? Is 350 reasonably close to 500?
- Problem: we cannot rely on human interpretation – we need statistical proof

Solution: use the significance level, alpha (α), to make an objective decision

Significance Level: Alpha (α)



- **Alpha (α):** Area of rejection or Rejection Zone
- Probability of rejecting the null hypothesis even though it is true

Why?

Assume
$\mu = 500$

We get
$\bar{x} = 350$

Are they close?
We think they are not and reject H_0.

But rejecting H_0 could be a **mistake!**

Example: Toss a coin 5 times



H_0: Coin is fair

We get all five heads

H H H H H

Is the coin fair?

We might reject H_0, but we may be **wrong** ... the coin could still be **fair**!

Therefore, **α (alpha)** = Probability of rejecting the null hypothesis even though it is true

Significance Level: Alpha (α)

Let's recall our example

Population mean (μ) = 500. We collected a sample and got:

Sample mean ($\bar{x}$) = 350 | **Sample SD** (s) = 10 | **Sample size** (n) = 50

Compute Z-score

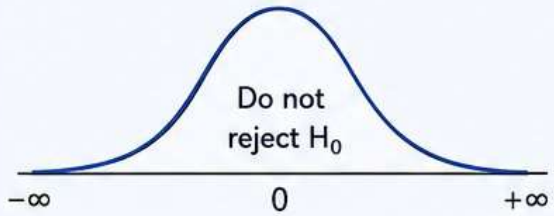
$$Z = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{350 - 500}{10/\sqrt{50}} = \frac{-150}{1.4142} = \boxed{-106.07}$$

This Z-score is way beyond extreme!
But how do we decide whether to reject or not reject H_0?

We need a Z-score cut-off.
That's where **Alpha (α)** comes in!

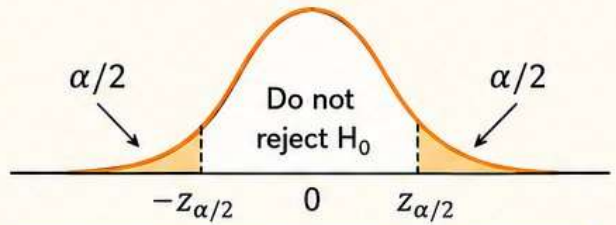
What does Alpha (α) mean?

$\alpha = 0$ (No Rejection Zone)



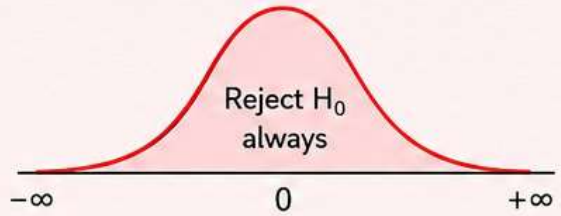
We never reject H_0, no matter what the data is.

$0 < \alpha < 1$ (Balanced Decision)



We reject H_0 only if Z is in the shaded rejection zones (in either tail).

$\alpha = 1$ (Full Rejection Zone)



We always reject H_0, no matter what the data is.

Alpha (α) (Rejection Zone Size)	Meaning
0	No rejection zone → Never reject H_0
Between 0 and 1	Some rejection zone → Reject H_0 only for extreme results
1	Full rejection zone → Always reject H_0

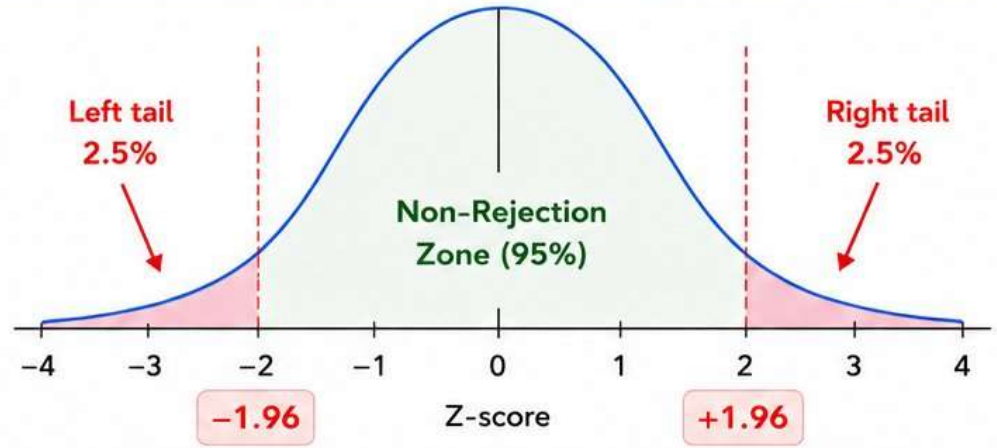
Alpha (α) controls how strict we are in rejecting H_0.

Common choices:

0.10, 0.05, 0.01

Significance Level: Alpha (α)

1. First, consider $\alpha = 0.05$ or 5%.
2. **Meaning:** We will reject H_0 if the Z-score of our sample is in the extreme 5% of Z-score values, split into two areas of 2.5% each (two-tailed test).
3. So, the middle 95% is the non-rejection zone, and the two tails (2.5% each) are the rejection zones.
4. **What is the Z-score at the start of the boundary of the rejection zone?**
5. **Why ±1.96?** Let's use the Standard Normal (Z) Table.



Reject H_0 if $Z \leq -1.96$ or $Z \geq +1.96$

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
−2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
−2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
−1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
−1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
−1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367

In the Z-table:
Look for the cumulative area closest to 0.0500 / 2 = **0.0250**.

→

In row −1.9 and column 0.06:
The area is **0.0250**.
This is the area in one tail (2.5%).

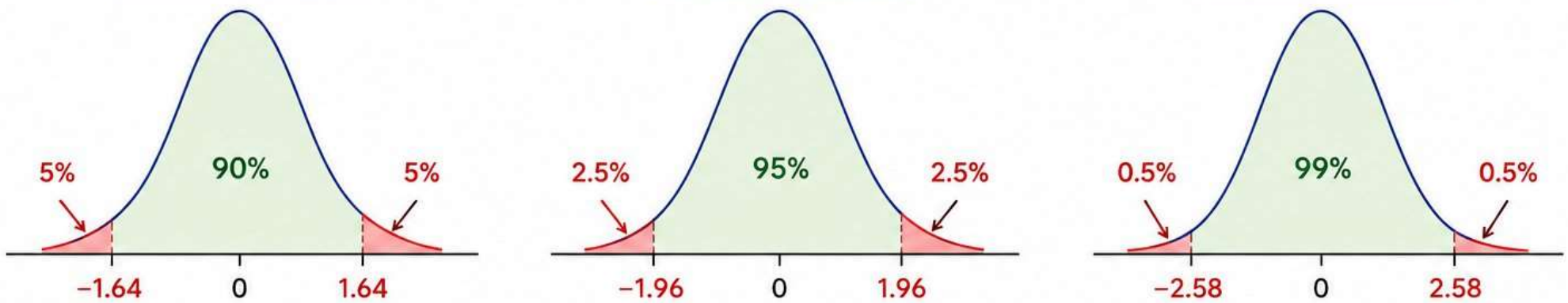
→

By symmetry, the other tail also has area 0.0250.
Therefore, the critical Z-scores are **−1.96** and **+1.96**.

Significance Level (α) and Confidence Level (1 – α)

α is the total area in the two tails (rejection region). The middle area is the confidence level (non-rejection region).

α = 0.10 (10%) → α = 0.05 (5%) → α = 0.01 (1%)



Confidence level = 90%
Significance level (α) = 10%
Critical z-value = ±1.64

Confidence level = 95%
Significance level (α) = 5%
Critical z-value = ±1.96

Confidence level = 99%
Significance level (α) = 1%
Critical z-value = ±2.58

Intuition

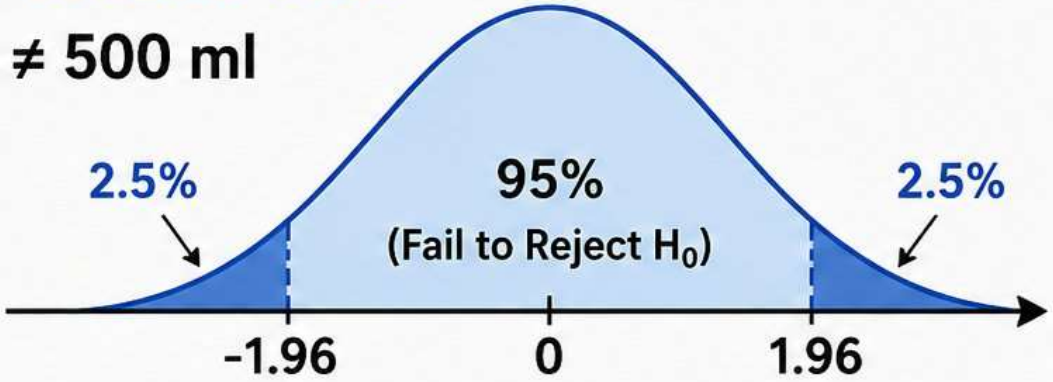
- As α decreases, the rejection regions (tails) get smaller.
- The non-rejection region (confidence level) gets larger.
- Therefore, you need more extreme Z-scores to reject H_0, so the critical value increases.

H_0: μ = 500 ml (The cola bottle contains 500 ml)

Test Statistic (z) = 1.75

TWO-TAILED TEST

H_1: μ ≠ 500 ml



Critical Values: z = -1.96 and z = 1.96

INTERPRETATION:

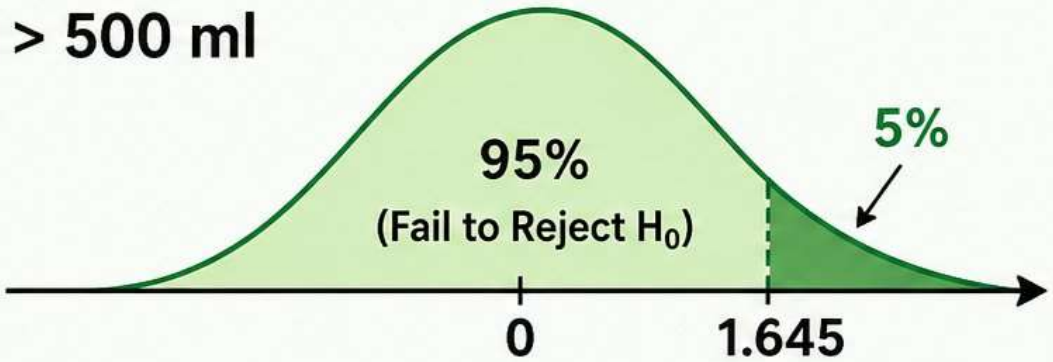
Since test statistic z = 1.75 and -1.96 < 1.75 < 1.96,

FAIL TO REJECT H_0.

There is not enough evidence to conclude that the mean is different from 500 ml.

ONE-TAILED TEST (RIGHT-TAILED)

H_1: μ > 500 ml



Critical Value: z = 1.645

INTERPRETATION:

Since test statistic z = 1.75 and 1.75 > 1.645,

REJECT H_0.

There is enough evidence to conclude that the mean is greater than 500 ml.

Hypothesis Test for Average Daily Customer Complaints

1. PROBLEM

Goal: A company has organized a training program to reduce the number of daily consumer complaints from 6.

Hypotheses

Hypothesis	Notation	Meaning
H_0 (Null Hypothesis)	$\mu = 6$	Average number of daily complaints = 6
H_1 (Alternative Hypothesis)	$\mu \neq 6$	Average number of daily complaints ≠ 6

Sample Data

Statistic	Value
Sample size (n)	100 days
Sample mean ($\bar{x}$)	7.2 complaints
Sample standard deviation (s)	3.5 complaints

2. SOLUTION

1 Test statistic (σ unknown, so use z-test)

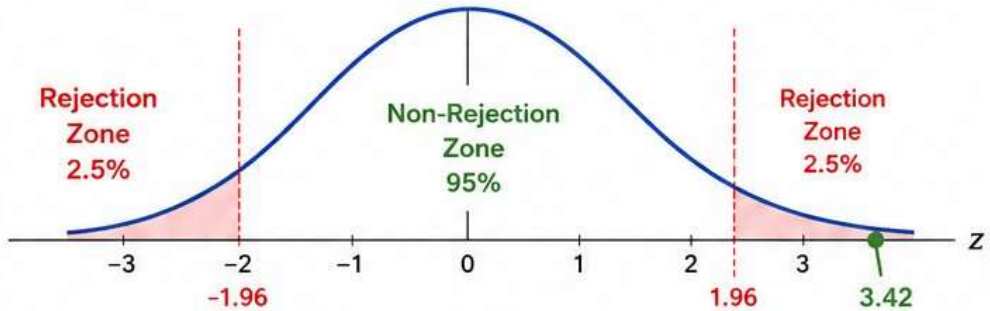
$$Z = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{7.2 - 6}{3.5/\sqrt{100}} = \frac{1.2}{0.35} = 3.42$$

2 Significance level

$\alpha = 0.05$ (5%) → Critical value (two-tailed) = ±1.96

3 Decision

$|Z| = 3.42 > 1.96$ → **Reject H_0**



4 Conclusion

There is sufficient evidence to conclude that the average number of customer complaints is significantly different from 6.

Since the sample mean (7.2) is greater than 6, training has actually made the situation **worse** — complaints have increased.

In Short

We tested if the average complaints ≠ 6. → Computed Z = 3.42. → Compared to critical value ±1.96. → Since |Z| > 1.96, we reject H_0.

Practice Problems

1. Manufacturing Quality Control

- A car manufacturer's specification states that the average lifespan of a particular tire model must be 40,000 km. The company conducts a test to ensure a recent batch of tires hasn't changed its performance (either decreased or increased).
  - H0: $\mu = 40{,}000$ km (null hypothesis)
  - H1: $\mu \neq 40{,}000$ km (alternative hypothesis)
  - $n = 100$
  - $\bar{x} = 39{,}200$ km
  - $s = 3{,}000$ km
  - $\alpha = 0.05$

2. Healthcare Wait Times

- A regional hospital system claims that the average waiting time for patients in its emergency room is 90 minutes (μ).
  - H0: $\mu = 90$ minutes (null hypothesis)
  - H1: $\mu \neq 90$ minutes (alternative hypothesis)
  - $n = 120$
  - $\bar{x} = 95.5$ minutes
  - $s = 30$ minutes
  - Try for $\alpha = 0.05$ and $\alpha = 0.01$

Solutions to Practice Problems

1 Manufacturing Quality Control

Given

- H_0: $\mu = 40{,}000$ km
- H_1: $\mu \neq 40{,}000$ km
- $n = 100$
- $\bar{x} = 39{,}200$ km
- $s = 3{,}000$ km
- $\alpha = 0.05$

Test Statistic (z-test)

$$Z = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{39{,}200 - 40{,}000}{3{,}000/\sqrt{100}} = \frac{-800}{300} = -2.67$$

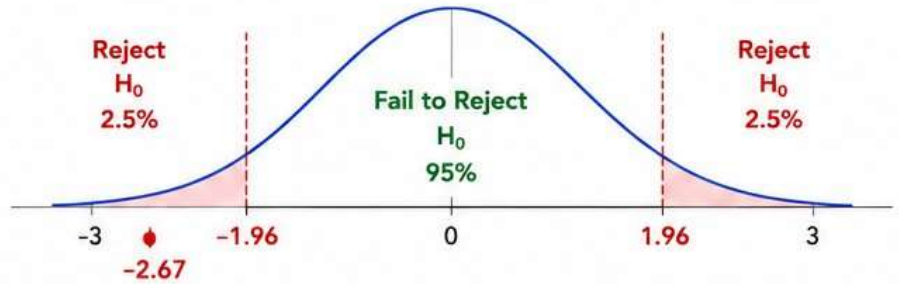
Critical Value (two-tailed, $\alpha = 0.05$)

± 1.96

Decision

Since $|Z| = 2.67 > 1.96$, **reject H_0**.

Two-Tailed Test ($\alpha = 0.05$)



Conclusion

There is sufficient evidence to conclude that the average lifespan of this tire model is different from 40,000 km.

2 Healthcare Wait Times

Given

- H_0: $\mu = 90$ minutes
- H_1: $\mu \neq 90$ minutes
- $n = 120$
- $\bar{x} = 95.5$ minutes
- $s = 30$ minutes

Test Statistic (z-test)

$$Z = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{95.5 - 90}{30/\sqrt{120}} = \frac{5.5}{2.7386} = 2.01$$

For $\alpha = 0.05$ (two-tailed)

Critical Value: ± 1.96

Decision:

Since $|Z| = 2.01 > 1.96$, **reject H_0**.

For $\alpha = 0.01$ (two-tailed)

Critical Value: ± 2.576

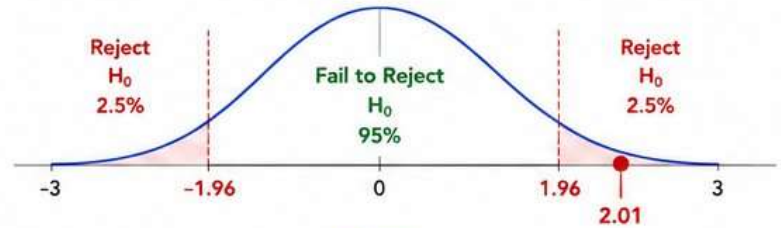
Decision:

Since $|Z| = 2.01 < 2.576$, **fail to reject H_0**.

Two-Tailed Test

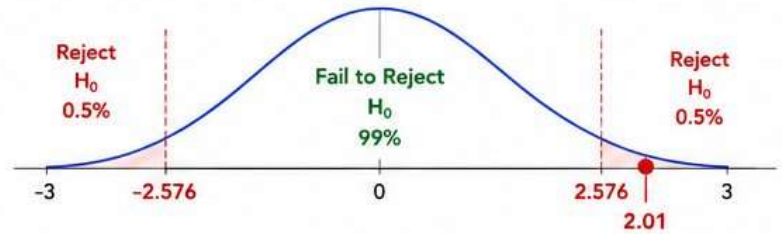
$\alpha = 0.05$

Critical Values:
± 1.96



$\alpha = 0.01$

Critical Values:
± 2.576



Conclusion

- At $\alpha = 0.05$: There is sufficient evidence to conclude that the average waiting time is different from 90 minutes.
- At $\alpha = 0.01$: There is not sufficient evidence to conclude that the average waiting time is different from 90 minutes.

Type I & Type II Errors (Alpha & Beta)

	H0 is True	H0 is False
Reject H0	Type I Error (Alpha, α) False alarm	Correct
Do not reject H0	Correct	Type II Error (Beta, β) Missed detection

- Type I error (α): rejecting H0 even though it is actually true
- Type II error (β): failing to reject H0 even though it is actually false
- Why does this happen? Because of random variation in samples (bad sample data)

Key takeaways

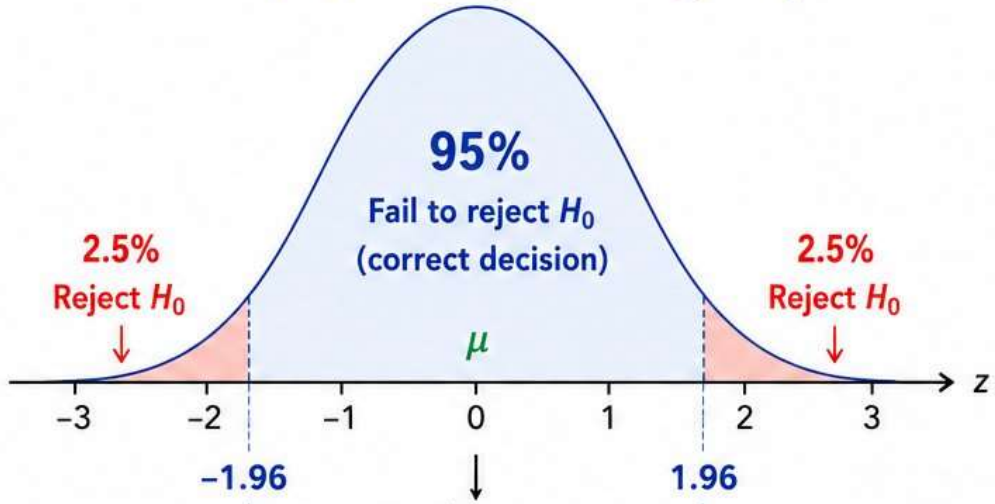
- To reduce Type I error, we set a small α (be stricter to reject H0)
- But reducing α increases Type II error (β) – the trade-off

Visualizing Type I Errors (α) – Rejecting H_0 When It Is True

If we repeat this many times...

95% of all sample means ($\bar{x}$) fall in this region when H_0 is true.

$\alpha = 0.05$



Hypotheses

$H_0: \bar{x} = \mu$

$H_a: \bar{x} \neq \mu$

(Two-tailed test)

Fail to reject H_0	$\bar{x}_1$	Correct (fail to reject H_0)
Fail to reject H_0	$\bar{x}_2$	Correct (fail to reject H_0)
Fail to reject H_0	$\bar{x}_3$	Correct (fail to reject H_0)
Fail to reject H_0	$\bar{x}_4$	Correct (fail to reject H_0)
Reject H_0	$\bar{x}_5$	Type I error (reject H_0)
Fail to reject H_0	$\bar{x}_6$	Correct (fail to reject H_0)

Type I error (α)

If we took a sample and it was by chance like $\bar{x}_5$, we would **incorrectly reject the null hypothesis.**

Key Idea: If we set $\alpha = 0.05$, we are willing to incorrectly reject a true H_0 about 5% of the time (Type I error).

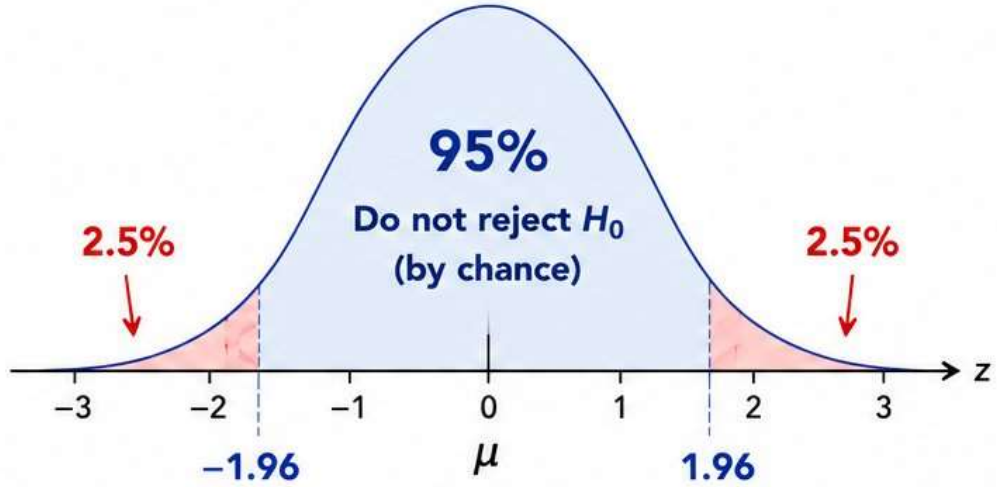
Note: We choose α before collecting data. Smaller $\alpha \Rightarrow$ fewer Type I errors, but harder to detect a real effect.

Type II Error (β) — Failing to Reject H_0 When It Is False

If we repeated this many times...

95% of all sample means ($\bar{x}$) would fall in this "**do not reject H_0**" region when H_0 is true.

Significance level: $\alpha = 0.05$



Hypotheses

$H_0: \bar{x} = \mu$

$H_a: \bar{x} \neq \mu$

(Two-tailed test)

Reject H_0 (statistically significant)	$\bar{x}_1$	In left tail (Reject H_0)
Reject H_0 (statistically significant)	$\bar{x}_2$	In left tail (Reject H_0)
Reject H_0 (statistically significant)	$\bar{x}_3$	In left tail (Reject H_0)
Fail to reject H_0 (not statistically significant)	$\bar{x}_4$	In center region (Fail to reject H_0)
Reject H_0 (statistically significant)	$\bar{x}_5$	In right tail (Reject H_0)
Reject H_0 (statistically significant)	$\bar{x}_6$	In right tail (Reject H_0)

If we took a sample and it was by chance like $\bar{x}_4$, we would **incorrectly "accept"** (fail to reject) H_0, even though H_0 is false.

β Meaning: If by chance, the only sample we tested was $\bar{x}_4$, we would commit a **Type II error**.

(Fail to reject H_0 when it is false.)

Key Idea: Reducing Type I error (α) by making the critical region smaller increases the chance of Type II error (β).

Trade-off:

Lower α → Higher β

Higher α → Lower β

Type I and Type II Errors – Example

H0: John's used car is safe to drive

	What John thinks	Reality	Outcome	Error Type
a)	His car may be safe	Car is not safe	Error	Type II (FN)
b)	His car may be safe	Car is safe	OK	–
c)	His car may not be safe	Car is not safe	OK	–
d)	His car may not be safe	Car is safe	Error	Type I (FP)

- Remember: Type I error occurs when H0 is true; Type II error occurs when H0 is false
- In this example H0 (the car is safe) – so (d) is a Type I error and (a) is a Type II error

Which error is more serious? Type II – failing to detect an unsafe car is potentially dangerous!

Type I and Type II Errors – Another Example

Hypothesis (H0): the defendant is not guilty (innocent)

	What the Jury Believes	Reality	Outcome	Error Type
a.	Defendant is guilty	Defendant is not guilty	Error	Type I (FP)
b.	Defendant is guilty	Defendant is guilty	OK	–
c.	Defendant is not guilty	Defendant is guilty	Error	Type II (FN)
d.	Defendant is not guilty	Defendant is not guilty	OK	–

- Remember: Type I error occurs when H0 is true; Type II error occurs when H0 is false
- So (a) is a Type I error (false positive) and (c) is a Type II error (false negative)

Which one is more dangerous? It depends on the situation and the crime. Many would argue a Type I error (convicting an innocent person) is more serious.

H0: A person is not pregnant



Significance Level (α) and p-value

α (significance level)

- Set before the test: the threshold or acceptable probability of rejecting the null hypothesis when it is actually true
- Usually set to 5% (α = 0.05)
- Meaning: we are okay with a 5% chance of rejecting the null hypothesis incorrectly
- Why not keep it at 0%? We would never reject H0!

p-value

- Calculated after the test: the result of the actual test (area of rejection)
- Low p-value: reject H0; high p-value: fail to reject H0

How it works

- 1. Choose α (e.g., 0.05)
- 2. Run the test and calculate the p-value
- 3. Decide: if p-value ≤ α → reject H0; if p-value > α → do not reject H0

- Example (α = 0.05): if p-value = 0.02 → reject H0; if p-value = 0.10 → do not reject H0

In short: set α before the test, use the p-value after the test. If p-value ≤ α, reject H0; otherwise, do not reject H0.

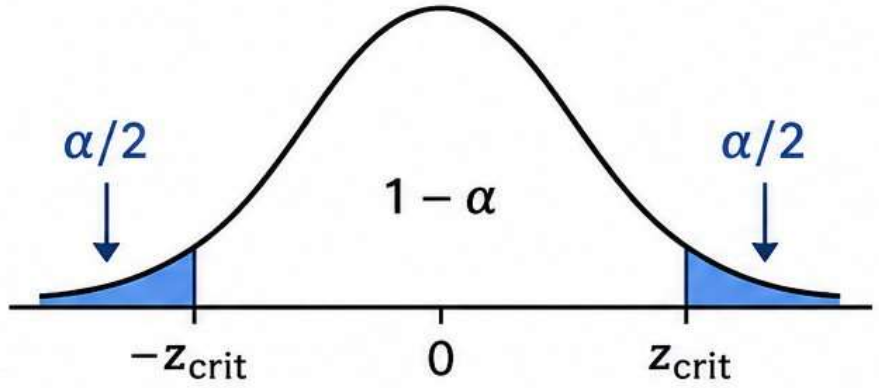
p-Value Example

- H0: The coin is fair (P(H) = 0.5)
- H1: The coin is not fair (P(H)≠ 0.5)

Toss Number	Result	Current Situation	Probability of This Exact Sequence under H_0	Two-Tailed p-value
1	H	H	1/2 = 50%	**100%**
2	H	HH	1/4 = 25%	**50%**
3	H	HHH	1/8 = 12.5%	**25%**
4	H	HHHH	1/16 = 6.25%	**12.5%**
5	H	HHHHH	1/32 = 3.125%	**6.25%**
6	**H**	**HHHHHH**	1/64 = 1.5625%	**3.125%**

- Why double the probability?
  - 6 heads = P(HHHHHH) = 1/64
  - 6 tails = P(TTTTTT) = 1/64
  - We need to consider both for a two-tailed test
  - So, p-value = 1/64 + 1/64 = 2/64 = 0.03125
- Decision at 6 heads
  - p-value = 3.125% and α = 5%
  - Since p-value < α, reject H0
  - Conclusion: Coin is not fair

TWO-TAILED Z-TEST USING Z-TABLE



In a two-tailed test:

- Total significance level = α
- Each tail gets $\alpha/2$
- Reject H_0 if $z \leq -z_{crit}$ or $z \geq z_{crit}$

APPROACH 1

Compare one-tail probability with $\alpha/2$

① Find the area in one tail from the Z-table.

② Compare with $\alpha/2$

Reject H_0 if (one-tail area) $\leq \alpha/2$

OR

APPROACH 2

Find two-tailed p-value and compare with α

① Find the area in one tail from the Z-table.

② Two-tailed p-value = 2 × (one-tail area)

③ Compare with α

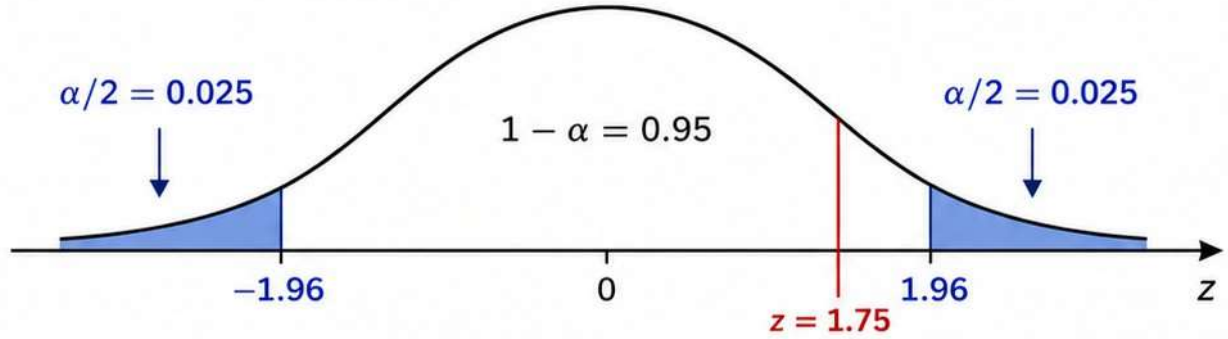
Reject H_0 if p-value $\leq \alpha$

Both approaches give the same conclusion.

One-tail area ↔ compare with $\alpha/2$ | Two-tail p-value ↔ compare with α

Two-Tailed Z-Test (Example)

Test statistic: $z = 1.75$ | **Significance level:** $\alpha = 0.05$ | **Hypotheses:** $H_0: \mu = \mu_0$ vs $H_1: \mu \neq \mu_0$



METHOD 1: Compare one-tail probability with $\alpha/2$

1. Find the area in one tail for $z = 1.75$ (from Z-table)

$$P(Z > 1.75) = 0.0401$$

2. Compare with $\alpha/2$

$$\alpha/2 = 0.025$$

Decision:
$0.0401 > 0.025 \Rightarrow$ **Fail to Reject H_0**
(at $\alpha = 0.05$ level)

METHOD 2: Find two-tail p-value and compare with α

1. Area in one tail (from Z-table)

$$P(Z > 1.75) = 0.0401$$

2. Two-tail p-value = 2 × (one-tail area)

$$\text{p-value} = 2 \times 0.0401 = 0.0802$$

3. Compare with α

$$\alpha = 0.05$$

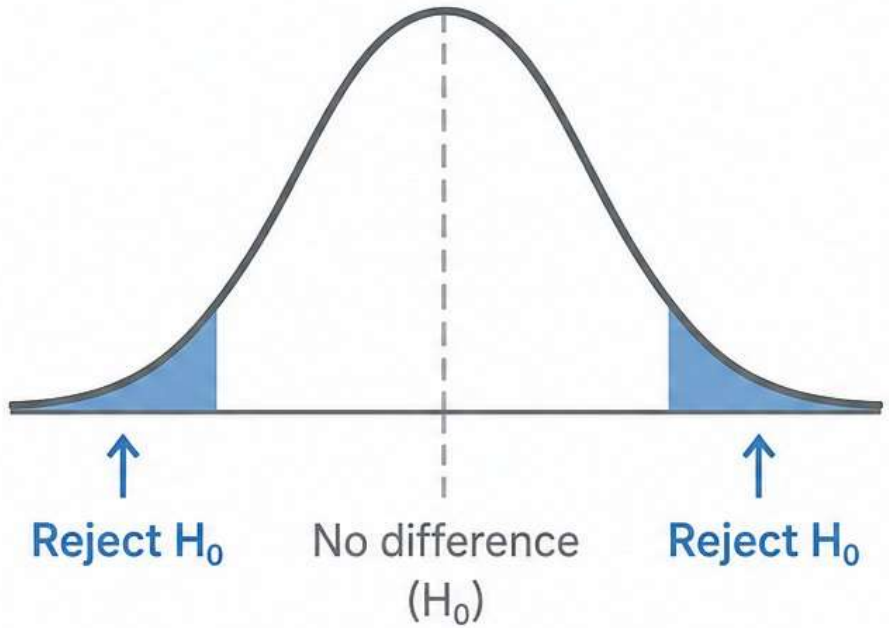
Decision:
$0.0802 > 0.05 \Rightarrow$ **Fail to Reject H_0**
(at $\alpha = 0.05$ level)

Both methods give the same conclusion: Fail to Reject H_0 at the 0.05 significance level.

(One-tail area = 0.0401 compared with $\alpha/2 = 0.025$ OR Two-tail p-value = 0.0802 compared with $\alpha = 0.05$)

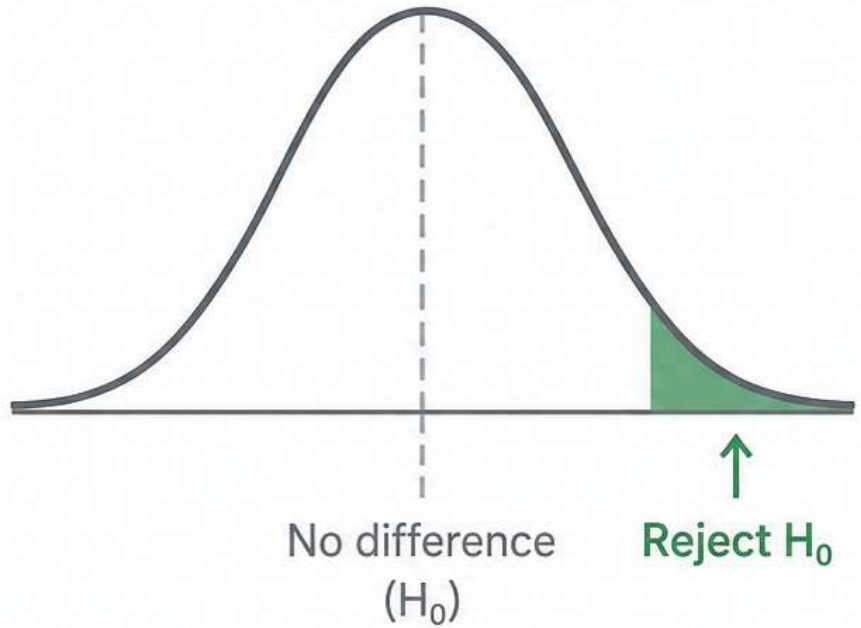
Two-tailed test

Looking for a difference
in either direction



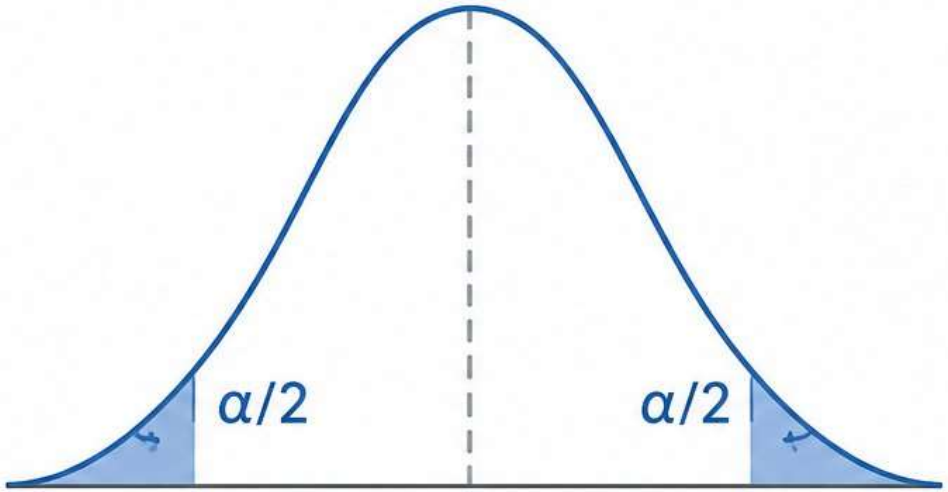
One-tailed test

Looking for a difference
in one specific direction



Two-tailed test

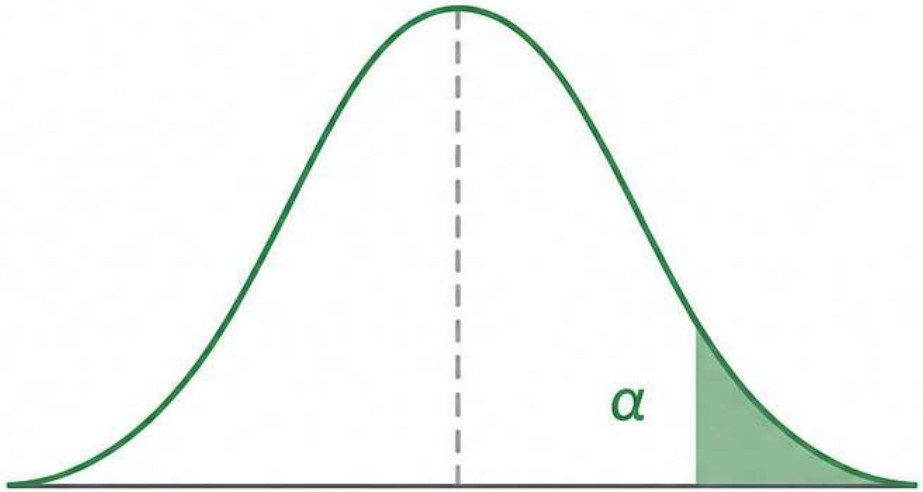
Alpha (α) is split between both tails



Total $\alpha = \frac{\alpha}{2} + \frac{\alpha}{2} = \alpha$

One-tailed test

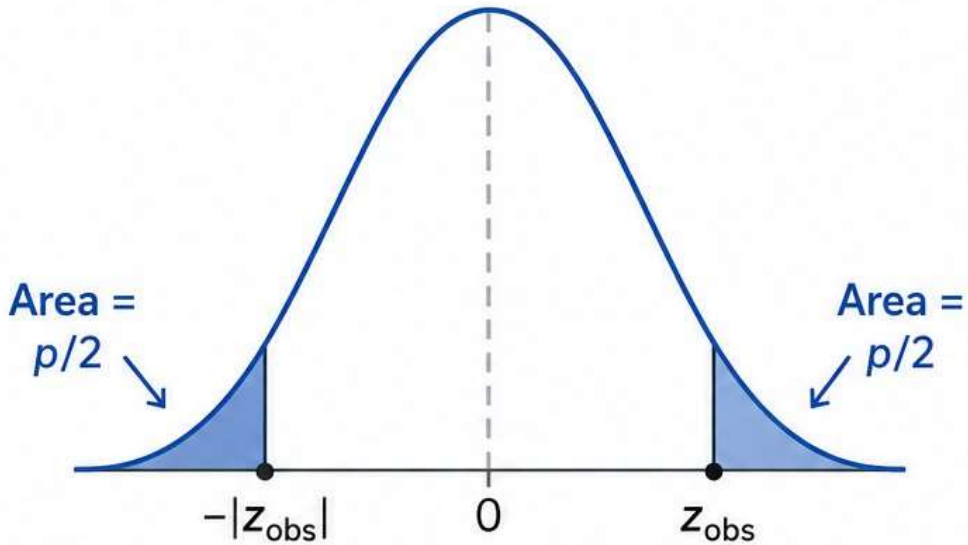
Alpha (α) is concentrated in one tail



Total $\alpha = \alpha$

Two-tailed test

P-value = area in both tails

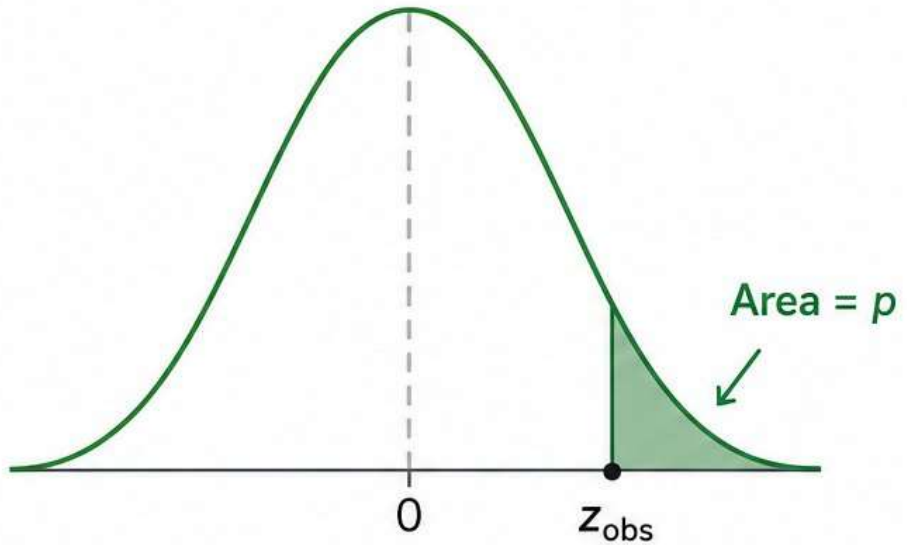


P-value = 2 × (area in one tail)

Larger p-value (for the same $|z_{obs}|$)
→ harder to reject H_0

One-tailed test

P-value = area in one tail



P-value = area in one tail

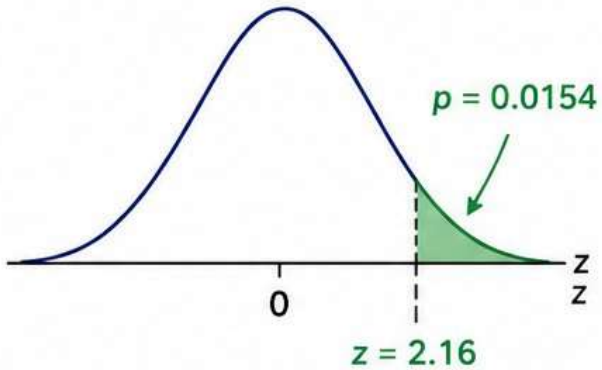
Smaller p-value (for the same z_{obs})
→ easier to reject H_0

Key takeaway: For the same test statistic, one-tailed test gives a smaller p-value (more evidence in the specified direction).

WHEN DO WE USE p AND WHEN DO WE USE $2 \times p$?

RIGHT-TAILED TEST

$H_1: \mu > \mu_0$

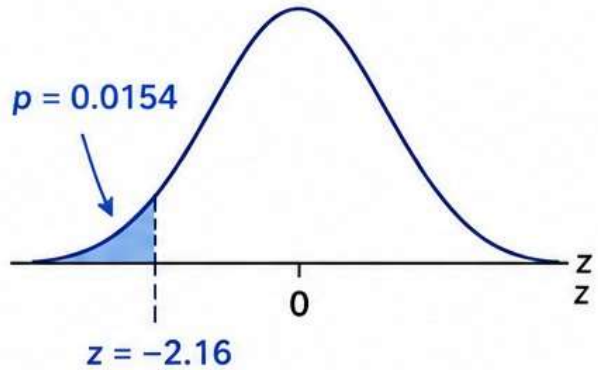


We care about values **GREATER** than 2.16

p-value = p = 0.0154
(area in one tail)

LEFT-TAILED TEST

$H_1: \mu < \mu_0$

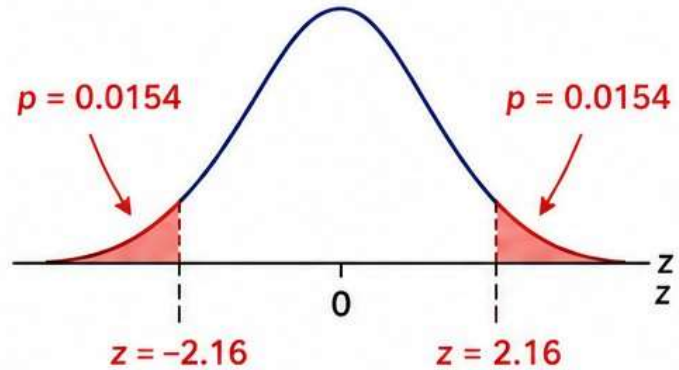


We care about values **LESS** than −2.16

p-value = p = 0.0154
(area in one tail)

TWO-TAILED TEST

$H_1: \mu \neq \mu_0$



We care about values **DIFFERENT** from 0 (in both directions)

p-value = $2 \times p = 2 \times 0.0154 = 0.0308$
(area in both tails)

THE SIMPLE RULE

Count the tails mentioned by H_1. →

- One tail (>, <) → use p
- Two tails (≠) → use $2 \times p$

REMEMBER

One direction → p

Both directions → $2 \times p$

Two-tailed test

Example:

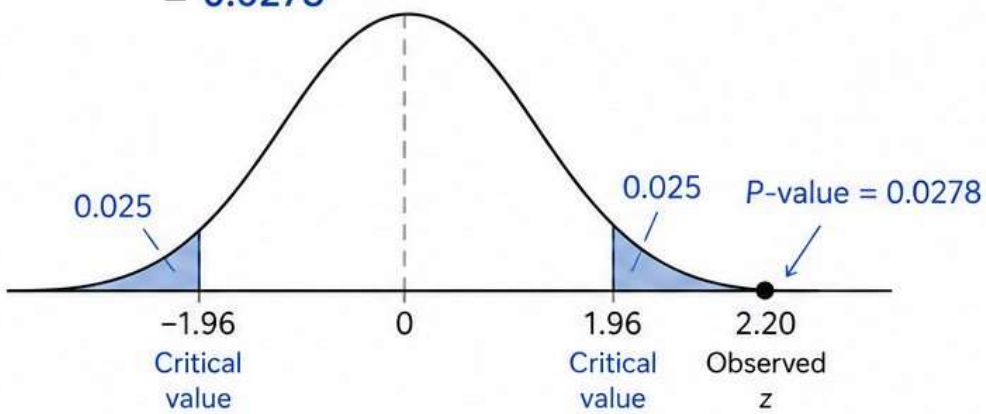
We test a new drug.

Test statistic: $z = 2.20$

Significance level: $\alpha = 0.05$

P-value calculation

$$P\text{-value} = 2 \times P(Z > |2.20|)$$
$$= 2 \times 0.0139$$
$$= \mathbf{0.0278}$$



$\alpha = 0.05$, P-value = 0.0278

Since P-value $< \alpha$, **reject H_0.**

There is evidence of a difference (in either direction).

One-tailed test (greater)

Example:

We test a new drug.

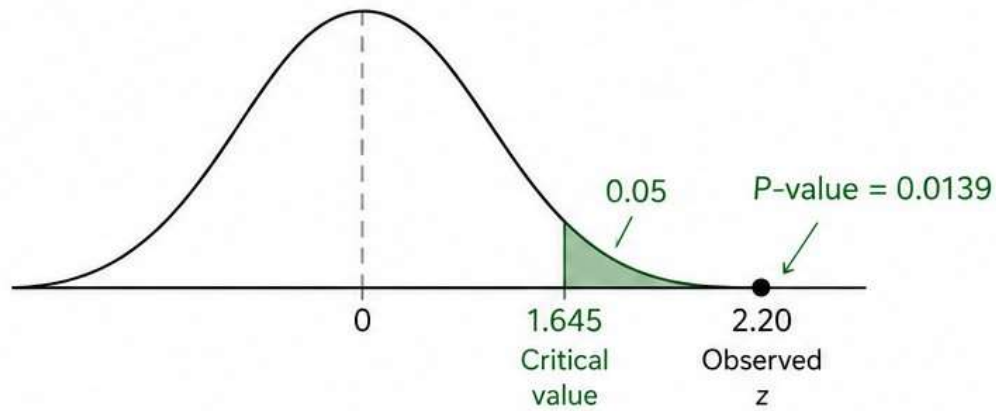
Test statistic: $z = 2.20$

Significance level: $\alpha = 0.05$

Alternative hypothesis: mean is greater

P-value calculation

$$P\text{-value} = P(Z > 2.20)$$
$$= \mathbf{0.0139}$$



$\alpha = 0.05$, P-value = 0.0139

Since P-value $< \alpha$, **reject H_0.**

There is evidence the mean is greater.

Key takeaway: For the same test statistic ($z = 2.20$), the one-tailed test has a smaller p-value (more evidence in the specified direction).

One-Sample t-Test in SciPy: Two-Tailed and One-Tailed Tests

Python (SciPy) code

```
from scipy import stats
x = np.array([82, 90, 78, 85, 88, 92, 75, 80, 84, 79])
mu0 = 80   # hypothesized population mean
# Two-tailed: H0: mu = 80, H1: mu != 80
t_two, p_two = stats.ttest_1samp(x, popmean=mu0,
alternative='two-sided')
# Left-tailed: H0: mu >= 80, H1: mu < 80
t_left, p_left = stats.ttest_1samp(x, popmean=mu0,
alternative='less')
# Right-tailed: H0: mu <= 80, H1: mu > 80
t_right, p_right = stats.ttest_1samp(x,
popmean=mu0, alternative='greater')
print("Two-tailed: t =", t_two, ", p =", p_two)
```

Test Type	H0	H1	SciPy option
Two-tailed	$\mu = \mu_0$	$\mu \neq \mu_0$	'two-sided'
Left-tailed	$\mu \geq \mu_0$	$\mu < \mu_0$	'less'
Right-tailed	$\mu \leq \mu_0$	$\mu > \mu_0$	'greater'

Interpretation

- If p-value < α (e.g., 0.05), reject H0
- The test statistic t and p-value are returned by stats.ttest_1samp()
- Use 'two-sided' for H1: $\mu \neq \mu_0$; 'less' for $\mu < \mu_0$; 'greater' for $\mu > \mu_0$

Interpreting the Output (t-statistic = 3.0)

The test statistic is positive (t = +3.0).
So the evidence is in the right tail (mean > 500).

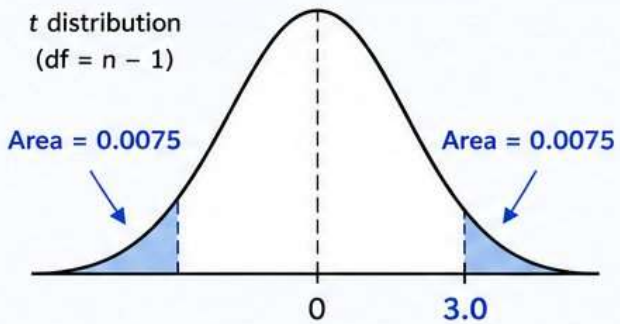
Example: One-sample t-test (μ_0 = 500)

Output you got:

- Two-tailed p-value = 0.015
- Right-tailed p-value = 0.0075
- Left-tailed p-value = 0.9925

1) TWO-TAILED TEST

H_1: $\mu \neq 500$



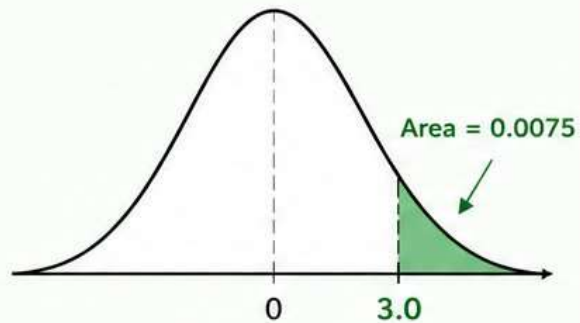
p-value (two-tailed)
= Area(left tail) + Area(right tail)
= 0.0075 + 0.0075 = 0.015

Interpretation

We look in both tails because the alternative says the mean is different from 500 (either greater or less).
Only 1.5% of the area lies in the two tails.

2) RIGHT-TAILED TEST

H_1: $\mu > 500$



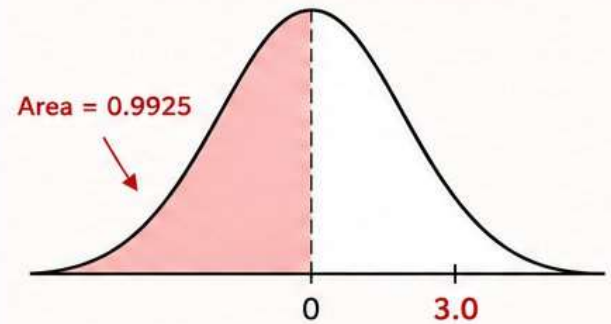
p-value (right-tailed)
= Area(right tail)
= 0.0075

Interpretation

We look only in the right tail because the alternative says the mean is greater than 500.
Only 0.75% of the area lies to the right of 3.0.

3) LEFT-TAILED TEST

H_1: $\mu < 500$



p-value (left-tailed)
= Area(left tail)
= 0.9925

Interpretation

We look only in the left tail because the alternative says the mean is less than 500.
Since the statistic is positive, almost the entire area (99.25%) is in the left tail.

Key Takeaways

- Two-tailed p-value = 2 × (one tail) when the statistic is positive.
- Right-tailed p-value is smaller because the statistic (3.0) is in the right tail.
- Left-tailed p-value is very large (close to 1) because the statistic is in the opposite direction.

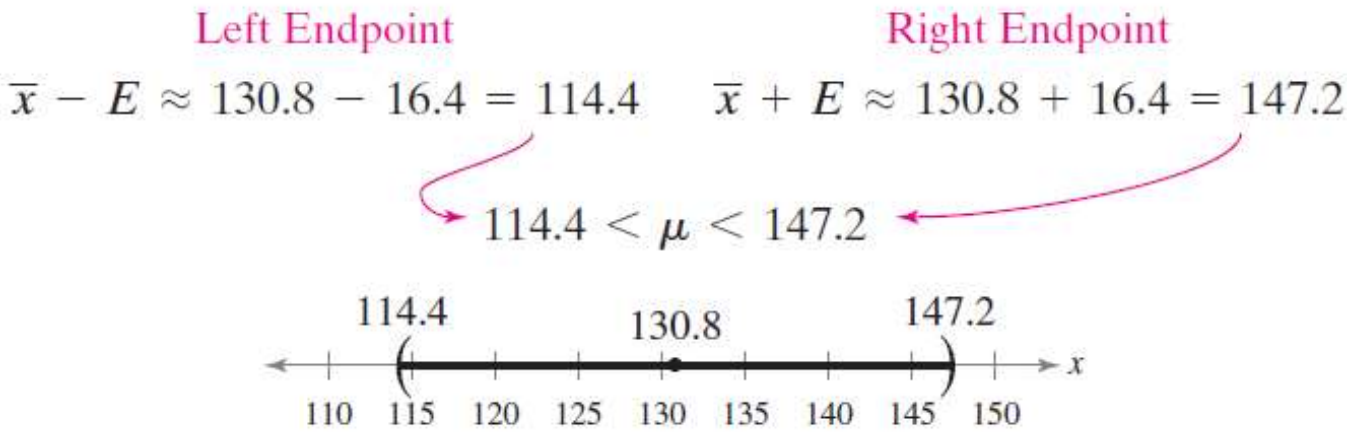
General Rule

If $t > 0$:

- Right tail ($\mu > \mu_0$) → small p-value
- Left tail ($\mu < \mu_0$) → large p-value

Margin of Error

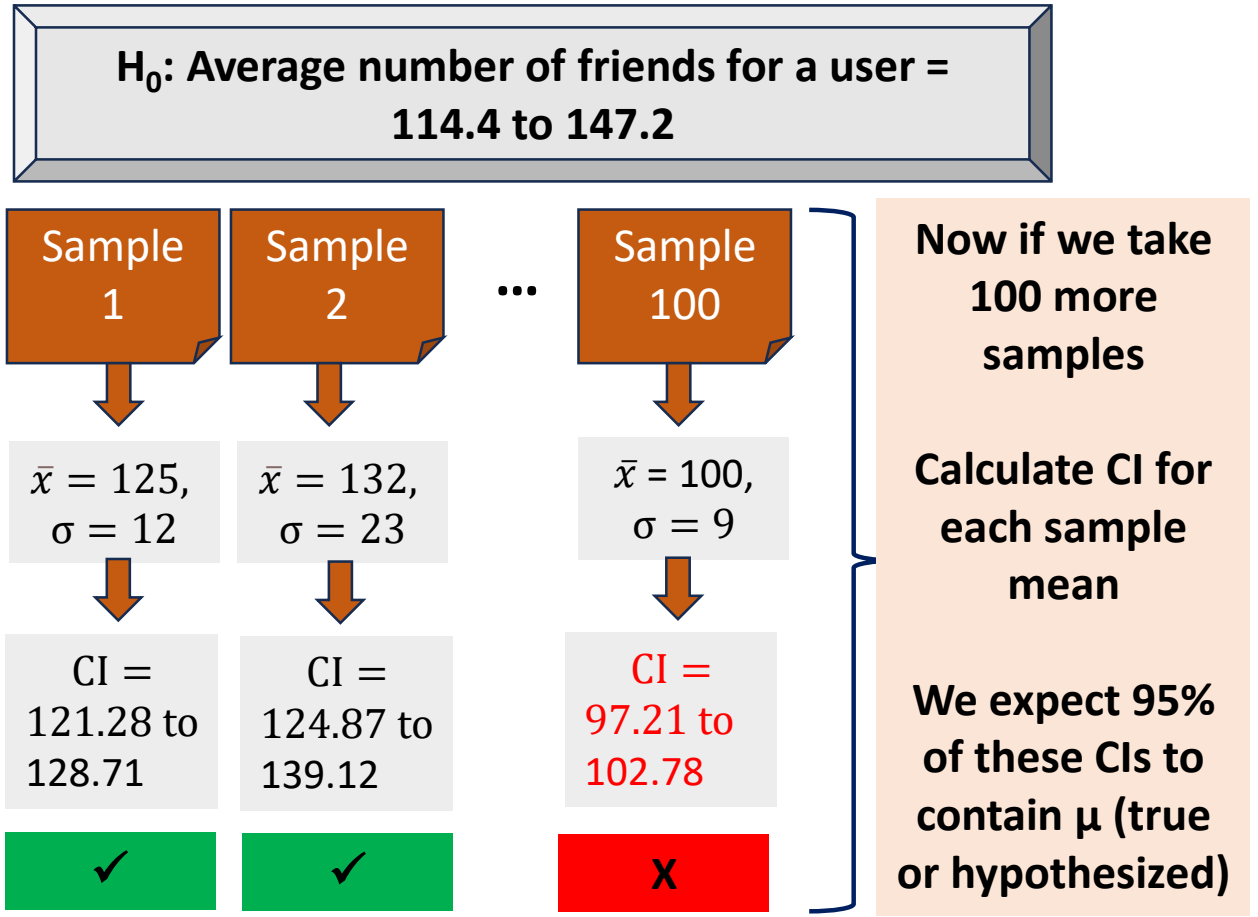
- H0: On a social networking site, a random sample of 40 users tells us that the average number of friends per user is 130.8 with a SD of 53
- **Point estimate ($\bar{x}$)**: 130.8 friends (Single value estimate for a population parameter)
- **Margin of error**: Add a range of uncertainty around the point estimate (e.g. instead of I am reaching in 10 minutes, say 5-15 minutes)
- **Interval estimate**: Point estimate + Margin of Error
- Margin of Error = Z_c x Standard Error
- As we know: Standard Error = $\frac{s}{\sqrt{n}}$
- So, Margin of Error = 1.96 x $\frac{53}{\sqrt{40}}$ = 16.4
- Then Interval estimate = 130.8 ± 16.4
- Interval estimate = 114.4 < μ <147.2
- Now we can use this in H0



Confidence Interval
Confidence Level

* Calculate MoE and CI as shown on the previous slide, for each sample

- Interval Estimate = 114.4 to 147.2
- **Confidence interval** = *Calculate* for each sample*
- **Confidence level** = Confidence about the confidence interval
- 95% confidence level means: If we take 100 samples, calculate sample mean, calculate sample CI* around the sample mean, 95% of such sample CIs would contain the true/hypothesized value of the population mean (μ)



Statistical Tests

Test	Null Hypothesis
One-sample T-Test	There is no significant difference between the sample mean and the hypothesized population mean
Independent T-Test	There is no statistically significant difference between the means of the two groups
Paired T-Test	There is no statistically significant difference between the means of the two groups BEFORE and AFTER
ANOVA	There is no statistically significant difference between the means of the various groups
U Test (Mann–Whitney U)	There is no statistically significant difference between the distributions (medians) of the two groups
Chi-Square Test	There is no association between the two categorical variables (they are independent)

Parametric Tests

What it is

- Assumes data follows a normal distribution and uses population parameters (mean, standard deviation)

Assumptions

- Normal distribution; equal variances
- Data is interval or ratio scale; observations are independent

Advantages

- More powerful, more precise, smaller sample sizes

Disadvantages

- Sensitive to outliers; invalid if assumptions are violated

Purpose	Test
Compare one mean	One-sample t-test
Compare two means	Independent t-test
Paired samples	Paired t-test
Compare >2 means	ANOVA
Relationship between variables	Pearson correlation
Prediction	Linear regression

Non-Parametric Tests

What it is

- Do not assume a specific distribution. Use ranks or medians, not means.

Characteristics

- No normality assumption; suitable for ordinal data
- Works well with small samples; robust to outliers

Advantages

- Fewer assumptions; handles skewed data well; works with ordinal or ranked data

Disadvantages

- Less powerful than parametric tests; limited modeling capabilities

Parametric Equivalent	Non-Parametric Test
One-sample t-test	Wilcoxon signed-rank
Independent t-test	Mann–Whitney U
Paired t-test	Wilcoxon signed-rank
ANOVA	Kruskal–Wallis
Pearson correlation	Spearman correlation
Repeated measures ANOVA	Friedman test

One-Sample T-Test

Purpose	Compare sample mean with hypothesized population mean
Null Hypothesis (H0)	Sample mean ($\bar{x}$) = Population mean (μ)
Formula	$t = (\bar{x} - \mu) / SE = (\bar{x} - \mu) / (s / \sqrt{n})$ = (Sample mean − Population mean) / (Sample standard deviation / √Sample size)
Symbols	$\bar{x}$ = sample mean; μ = population mean; s = sample standard deviation; n = sample size; SE = standard error
Note	This is the same as the Z-test formula, but it is typically used when the sample size is small (≤ 30) or the population standard deviation is unknown

One-Sample T-Test: Example

1. Problem

- It is claimed that the average amount of coffee in a cup is 12 units
- We collect a random sample of 20 cups and measure the amount of coffee in each cup
- We found: sample mean $\bar{x} = 11.5$, standard deviation $s = 1.2$, sample size $n = 20$
- Null hypothesis H0: $\mu = 12$ units (population mean)

3. Find the critical value

- From the t-table (https://www.sjsu.edu/faculty/gerstman/StatPrimer/t-table.pdf): df = 19, $\alpha = 0.05$ (two-tailed) → critical t-value = ±2.093
- Why 0.05? Because the confidence level is 95%, so the significance level (α) is 0.05

2. Calculate the t-statistic

- $t = (\bar{x} - \mu) / (s / \sqrt{n})$
- $t = (11.5 - 12) / (1.2 / \sqrt{20}) = -0.5 / (1.2 / 4.472) = -0.5 / 0.268 = -1.87$
- Degrees of freedom: $df = n - 1 = 20 - 1 = 19$

4. Decision & conclusion

- |Calculated t-statistic| (1.87) < critical t-value (2.093)
- Decision: fail to reject the null hypothesis
- Conclusion: we do not have enough evidence to conclude that the average amount of coffee in a medium-sized cup served by the company is different from 12 units

Independent Sample T-Test

- Compare the means of two independent groups

Purpose	Compare the mean of two independent groups (i.e., samples from two different populations)
Null Hypothesis (H0)	There is no significant difference between the means of the two groups
Formula	$t = (\bar{X}1 - \bar{X}2) / \sqrt{(s1^2/n1 + s2^2/n2)}$
Degrees of freedom	$df = (s1^2/n1 + s2^2/n2)^2 / [(s1^2/n1)^2 / (n1 - 1) + (s2^2/n2)^2 / (n2 - 1)]$
Symbols	$\bar{X}1$, $\bar{X}2$: means of group 1 and 2; s1, s2: standard deviations of group 1 and 2; n1, n2: sample sizes of group 1 and 2

Two-Sample t-Test (Independent Samples) – Example

1) Data: marks obtained by two groups of students

- Group A (n1 = 25): 85, 88, 90, 92, 95, 87, 89, 91, 94, 86, 88, 92, 90, 93, 85, 89, 91, 87, 92, 90, 88, 93, 86, 89, 91
- Group B (n2 = 30): 78, 82, 80, 85, 88, 81, 83, 79, 84, 87, 80, 82, 85, 78, 81, 83, 79, 82, 80, 85, 78, 81, 83, 79, 84, 87, 80, 82, 85, 78

2) Summary statistics (rounded to two decimals)

Statistic	Group A	Group B
Sample size (n)	25	30
Sample mean ($\bar{x}$)	89.64	81.97
Sample std. deviation (s)	2.78	2.89

3) Hypotheses

- H0: $\mu A = \mu B$ (no difference in mean marks)
- H1: $\mu A \neq \mu B$ (there is a difference in mean marks)

4) Test statistic (Welch's t-test)

- $t = (\bar{x}1 - \bar{x}2) / \sqrt{(s1^2/n1 + s2^2/n2)} = (89.64 - 81.97) / \sqrt{(2.78^2/25 + 2.89^2/30)} = 10.00$
- Degrees of freedom (Welch–Satterthwaite) ≈ 51.89

5) Decision

- $\alpha = 0.05$ (two-tailed); critical t-value ≈ ±2.007 (df ≈ 52)
- Since $|t| = 10.00 > 2.007$, we reject H0
- Conclusion: there is a statistically significant difference in the mean marks of Group A and Group B at the 5% significance level
- Note: Welch's t-test is used because the standard deviations are not assumed to be equal

Paired t-Test

- Used when the same subjects are measured twice (e.g., before & after)

Data Source	Same subjects measured twice (related/dependent)
Goal	Check if there is a significant difference in the paired measurements (before vs. after)
Example	Blood sugar levels of the same patients before and after a drug
Null Hypothesis (H0)	No significant difference in the paired measurements (mean difference = 0)
Formula	$t = \bar{D} / (S_D / \sqrt{n})$ $\bar{D}$ = mean of the differences; S_D = standard deviation of the differences; n = number of pairs

Paired t-Test Example

- Same subjects measured at two different times (t1 and t2)

Subject	Time t1	Time t2	Difference (t1 − t2)
1	13	9	4
2	14	11	3
3	14	12	2
4	15	12	3
5	16	14	2
6	17	16	1
7	17	18	−1
8	18	18	0
9	19	18	1
10	20	19	1
11	22	20	2
12	23	20	3

- Mean of differences $\bar{D} = 1.75$
- Standard deviation of differences $S_D = 1.422$
- Number of pairs $n = 12$

t-statistic

- $t = \bar{D} / (S_D / \sqrt{n}) = 1.75 / (1.422 / \sqrt{12}) = 4.26$

Critical value

- For $\alpha = 0.05$ and $df = n - 1 = 11$, critical value = 2.201

Decision

- Test statistic (4.26) > critical value (2.201) → we reject H0

Conclusion

- There is a significant difference between the measurements at time t1 and time t2

ANOVA (Analysis of Variance)

Primary Question	Are means of three or more groups different?
“One-Way” ANOVA	Because we are comparing just one variable across categories (one factor with three or more levels)
Mechanism	Compare variation within each sample, relative to the variation between samples
Null Hypothesis (H0)	There is no statistically significant difference between the means of the various groups
Formula	ANOVA uses the F-ratio = Variance Between Groups ÷ Variance Within Groups

A large F-ratio suggests the group means are truly different

One-Way ANOVA – Complete Calculations

1. Data & hypotheses: test scores of students

- H0: mean test scores are the same for all three teaching methods ($\mu A = \mu B = \mu C$)
- H1: at least one teaching method has a different mean test score

Method	Scores	n	Sum	Mean
A	85, 88, 91, 78, 82	5	424	84.8
B	75, 79, 80, 82, 78	5	394	78.8
C	90, 85, 88, 92, 87	5	442	88.4

- Grand mean $\bar{X} = (84.8 + 78.8 + 88.4) / 3 = 84.0$; $N = 15$

2. Sum of squares treatment (between groups)

- $SST = \Sigma\, n_i (\bar{X}_i - \bar{X})^2 = 5(0.8)^2 + 5(-5.2)^2 + 5(4.4)^2 = 3.20 + 135.20 + 96.80 = 235.20$

3. Sum of squares error (within groups)

- $SSE = \Sigma\, \Sigma\, (X_{ij} - \bar{X}_i)^2 = 102.80\ (A) + 26.80\ (B) + 29.20\ (C) = 158.80$

4. ANOVA F-test

- $F = [SST / (k - 1)] / [SSE / (N - k)]$, $k = 3$ groups, $N = 15$
- $df1 = k - 1 = 2$; $df2 = N - k = 12$
- $F = (235.20 / 2) / (158.80 / 12) = 117.60 / 13.2333 = 8.88$
- Critical value ($\alpha = 0.05$, $df1 = 2$, $df2 = 12$): $F_{crit} = 3.89$
- Since F (8.88) > F_{crit} (3.89), reject H0

5. ANOVA summary table

Source	SS	df	MS	F
Between groups	235.20	2	117.60	8.88
Within groups	158.80	12	13.2333	
Total	394.00	14		

- Conclusion: there is sufficient evidence at the 5% level to conclude that at least one teaching method has a different mean test score

U-Test (Mann–Whitney U Test)

What is it?	A non-parametric test (no assumption of normal distribution)
Compares	Two independent groups
Alternative to	The independent t-test
Used when	Data is not normally distributed
Data type	Ordinal or continuous data, e.g. marks of class A and class B; recovery time with medicine A and medicine B
Hypotheses	H0: both groups come from the same population (no difference) H1: the two groups are different
Steps	1. Combine both groups → 2. Rank all observations → 3. Calculate U statistic → 4. Find p-value → 5. Compare p-value with α (0.05)

If p-value > α (0.05) → do not reject H0

U-Test Example

- Is there a difference in exam marks between Online and Offline students?
  - Group 1: Online students
  - Group 2: Offline students
- Sample data
  - group1 = [65, 70, 68, 72, 75]
  - group2 = [80, 82, 85, 79, 84]

U-Test Example

- Sample data
  - group1 = [65, 70, 68, 72, 75]
  - group2 = [80, 82, 85, 79, 84]
- Combine and rank

Value	Group	Rank
65	G1	1
68	G1	2
70	G1	3
72	G1	4
75	G1	5
79	G2	6
80	G2	7
82	G2	8
84	G2	9
85	G2	10

- Find sum of Ranks
  - Group 1: 1 + 2 + 3 + 4 + 5 = 15 = R1
  - Group 2: 6 + 7 + 8 + 9 +10 = 40 = R2
- Calculate U statistics
  - U1 = n1.n2 + $\frac{n1(n1+1)}{2}$ - R1 = (5)(5) + $\frac{5(6)}{2}$ - 15 = 25
  - U2 = n1.n2 + $\frac{n2(n2+1)}{2}$ - R2 = (5)(5) + $\frac{5(6)}{2}$ - 40 = 0
- Choose the smaller U, i.e. U1 = 0 = Test statistic
- U-table for n1=5, n2=5, α=0.05, critical value = 2
- If U < U Critical value -> Reject H0

Chi-Square Test: Test for Categorical Data

What is it?	A statistical test used to compare observed results with expected results of categorical variables
Data arrangement	Data is arranged in a contingency table (a cross-tabulation)
Purpose	To determine whether there is a significant difference between observed and expected frequencies
Key idea	The chi-square test helps us evaluate categorical data by comparing counts (frequencies) to see if any differences are likely due to chance or are statistically significant

Types of Chi-Square Test

- Three different chi-square tests are used to analyze categorical data

Type	Main Question	Example Scenario	Null Hypothesis (H0)	Alternate Hypothesis (H1)
1. Chi-Square Test of Independence	Are two categorical variables related or independent?	Are gender and pet preference (cat/dog) related?	Gender and pet preference are independent (no relationship)	Gender and pet preference are associated (there is a relationship)
2. Chi-Square Test of Homogeneity	Do different groups have the same distribution of a categorical variable?	Do different company servers fail at the same rate?	All groups have similar proportions of failures	At least one group has a different failure rate
3. Chi-Square Goodness of Fit Test	Does the observed data match an expected theoretical distribution?	Is a coin fair (equal heads/tails)?	Data follows the expected distribution (e.g., fair coin → 50/50)	Data does not follow the expected distribution

Creation of Contingency Table (Example)

1. Raw categorical data: survey of 20 students (Student ID, Gender, Study Preference)

- e.g. 1: Male, Online; 2: Female, Offline; 3: Male, Online; …; 20: Female, Online

2. Identify categories

- Gender: Male, Female
- Study Preference: Online, Offline

3. Contingency table (frequency counts)

Gender	Online	Offline	Total
Male	6	4	10
Female	7	3	10
Total	13	7	20

Python (pandas) – using crosstab

```
import pandas as pd
gender = ['Male', 'Female'] * 10
pref = ['Online','Offline','Online','Online',
        'Online','Offline','Offline','Online',
        'Online','Online','Offline','Offline',
        'Online','Online','Offline','Online',
        'Online','Online','Offline','Online']
df = pd.DataFrame({'Gender': gender,
                   'Study Preference': pref})
ct = pd.crosstab(df['Gender'],
       df['Study Preference'],
       margins=True, margins_name='Total')
print(ct)
```

Chi-Square Test – Formula

$\chi^2 c = \Sigma (O_i - E_i)^2 / E_i$ (sum over i = 1 … k)

- Where:
  - c = degrees of freedom: the number of values that can vary in a calculation
  - O = observed value: the value observed from the data
  - E = expected value: the value expected under the null hypothesis

Degrees of freedom for a chi-square grid

- Equal to the number of rows minus one, times the number of columns minus one:

df = (R − 1) × (C − 1)

- In a 2 × 2 grid: (2 − 1) × (2 − 1) = 1
- In general: df = (number of rows − 1) × (number of columns − 1)

Chi-square Test of Independence

- Does gender affect which pet is liked more, for the following data?

	Cat	Dog
Men	207	282
Women	231	242

Solution: first define hypotheses

- H0: gender and preference for cats or dogs are independent
- H1: gender and preference for cats or dogs are not independent

Chi-square Test of Independence: Expected Values

1. Add up all the row and column values

	Cat	Dog	Row Total
Men	207	282	489
Women	231	242	473
Column Total	438	524	962

2. Calculate the expected value for each cell: (row total × column total) / overall total

	Cat	Dog	Row Total
Men	489 × 438 / 962 = 222.64	489 × 524 / 962 = 266.36	489
Women	473 × 438 / 962 = 215.36	473 × 524 / 962 = 257.64	473
Column Total	438	524	962

Chi-square Test of Independence: $(O - E)^2 / E$

- Subtract expected from observed, square it, and divide by expected: $(O - E)^2 / E$, where O = observed (actual) value and E = expected value

	Cat	Dog
Men	$(207 - 222.64)^2 / 222.64 = 1.099$	$(282 - 266.36)^2 / 266.36 = 0.919$
Women	$(231 - 215.36)^2 / 215.36 = 1.136$	$(242 - 257.64)^2 / 257.64 = 0.950$

Chi-square Test of Independence: Decision

1. Add up the calculated values	1.099 + 0.919 + 1.136 + 0.950 = 4.10 Chi-square test statistic ≈ 4.10
2. Degrees of freedom	(row count − 1) × (column count − 1) = (2 − 1) × (2 − 1) = 1
3. Look up in the chi-square table	(https://people.richland.edu/james/lecture/m170/tbl-chi.html) for df = 1 and α = 0.05: critical value = 3.841, which is less than our chi-square value
4. Conclusion	Observed value (our test statistic) > critical value (chi-square table value), so we reject the null hypothesis

Chi-square Test of Goodness of Fit

1. What is it?

- Determine the probability of an observed frequency of events, given its expected frequency

2. Example

- We toss a coin 18 times and get 12 heads
- Is it due to chance or is the coin biased?

3. Chi-square formula

- Considers the sum of squared distances between observed values O and expected values E, divided by each expected value E
- Heads: observed 12, expected 9; Tails: observed 6, expected 9
- $\chi^2 = \Sigma\ (O - E)^2 / E = (12 - 9)^2 / 9 + (6 - 9)^2 / 9 = 2.0$

4. How to interpret the test statistic ($\chi^2 = 2.0$)

- Low chi-square value = observed and expected values are close to each other
- Degrees of freedom = number of possible outcomes − 1 = 2 − 1 = 1 (heads or tails)
- Critical value with df = 1 and $\alpha = 0.05$ is 3.841 (https://www.scribbr.com/statistics/chi-square-distribution-table/)
- H0: 12 heads in 18 tosses is reasonable for a fair coin
- Since test statistic (2.0) < critical value (3.841), we fail to reject H0

Conclusion: the difference between observed (12 heads) and expected (9 heads) is not large enough. The coin appears to be fair.

Exercise

- What if we toss a coin 18 times and get 16 heads?
- Will we reject or not reject H_0?

Chi-square Test

- What if we get 16 heads in 18 coin tosses?
- $\chi^2 = \sum \frac{(O-E)^2}{E} = \frac{(16-9)^2}{9} + \frac{(2-9)^2}{9} = 10.88$

  Heads **Tails**
- High chi-square value = Low correlation between observed and expected values (Because we are subtracting E from O and squaring it, and if it is high, it means E and O are not close to each other)
- Degrees of Freedom = Number of possible outcomes - 1 = 2 - 1 = 1
  - Because we have either Heads or Tails
- Critical value with DF = 1 and α = 0.05 is 3.841 (https://www.scribbr.com/statistics/chi-square-distribution-table/)
- H_0 = 16 heads in 18 tosses is reasonable for a fair coin
- Since Test statistic (10.88) > Critical value (3.841), we reject H0

Chi-Square Test of Homogeneity

- A company runs six identical servers for its IT infrastructure
- It wants to find out if server failure is occurring at the same rate for all the servers
- Assumptions
  - When a server fails, it does not necessarily mean that it will fail again (When we get *Heads* in a toss of a coin, it does not say anything about the next toss)
  - One server's failure is not impacting any other server in any way
  - A server is either healthy or has failed – No *in-between* states
- H_0: There is no statistically significant difference between server failure rates

Chi-Square Test of Homogeneity

1. Determine expected counts

- Server failure actual values are given; we need to calculate expected values
- We expect all servers to have an equal failure rate
- Total observed count = 240, divided by 6 servers → expected failure count for each server = 240 / 6 = 40

Result

- Chi-square test statistic = 10
- df = 6 − 1 = 5; α = 0.05; critical value = 11.07
- Since test statistic < critical value, we fail to reject H0

2. Chi-square calculations: $\chi^2 = \Sigma (O - E)^2 / E$

Server	O	E	O – E	$(O - E)^2$	$(O - E)^2 / E$
A	46	40	6	36	0.9
B	36	40	−4	16	0.4
C	52	40	12	144	3.6
D	26	40	−14	196	4.9
E	42	40	2	4	0.1
F	38	40	−2	4	0.1
Total	240	240			10

Predictive Analytics

Uses historical data and statistical models to predict future outcomes

- Example 1: retail demand forecasting
  - Predicts product demand to optimize inventory and reduce stockouts
- Example 2: customer churn prediction
  - Identifies customers at risk of leaving to enable targeted retention actions

Supervised vs Unsupervised Learning

Supervised learning (learning with labeled data)

- Dataset example: house prices

Size (sq ft)	Bedrooms	Location	Price (USD) (Label)
1200	2	Downtown	250,000
1500	3	Suburb	320,000
1800	3	Downtown	440,000
2000	4	Suburb	500,000
…	…	…	…

- → Model → predict price (for new houses)

Unsupervised learning (learning without labeled data)

- Dataset example: customer data

Age	Annual Income (USD)	Spending Score (1-100)
25	28,000	35
34	52,000	70
45	71,000	20
52	33,000	40
…	…	…

- → Model → discover groups (customer segments)

Machine Learning Algorithms

Supervised learning (learning with labeled data)

- Labeled data → model → prediction

Algorithms

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- Support Vector Machines (SVM)
- K-Nearest Neighbors (KNN)
- Artificial Neural Networks (ANN)
- Gradient Boosting Machines (GBM)

Unsupervised learning (learning without labeled data)

- Unlabeled data → model → patterns / groups

Algorithms

- K-Means Clustering
- Hierarchical Clustering
- DBSCAN
- Principal Component Analysis (PCA)
- Independent Component Analysis (ICA)
- Apriori Algorithm (Association Rule Learning)
- Autoencoders
- t-SNE (t-Distributed Stochastic Neighbor Embedding)

Steps in Machine Learning

Step	What happens
1. Load dataset	Read CSV file (salary.csv); define X and y (X = YearsExperience, y = Salary)
2. Train-test split	Create training data (80%) and testing data (20%)
3. Train model	Linear regression: fit model using training data
4. Make predictions	Predict test data (y_pred); predict new experience values (1, 3, 5, 8, 12, 15, 20, 25)
5. Visualize results	Training set graph and test set graph: regression line, salary vs experience
6. Evaluate model	MAE (Mean Absolute Error), MSE (Mean Squared Error), R^2 score (R-squared)

Types of Predictive Analytics

Type	What it does
Regression	Predicts a continuous value
Classification	Predicts a category or class label
Time Series Forecasting	Predicts future values over time
Clustering	Finds similar groups

LINEAR REGRESSION

Linear Regression: Find the best straight line that predicts Y using another variable X.

EQUATION

$$Y = a + bX$$

- a = Intercept → value of Y when X = 0
- b = Slope → how much Y changes for 1 unit change in X

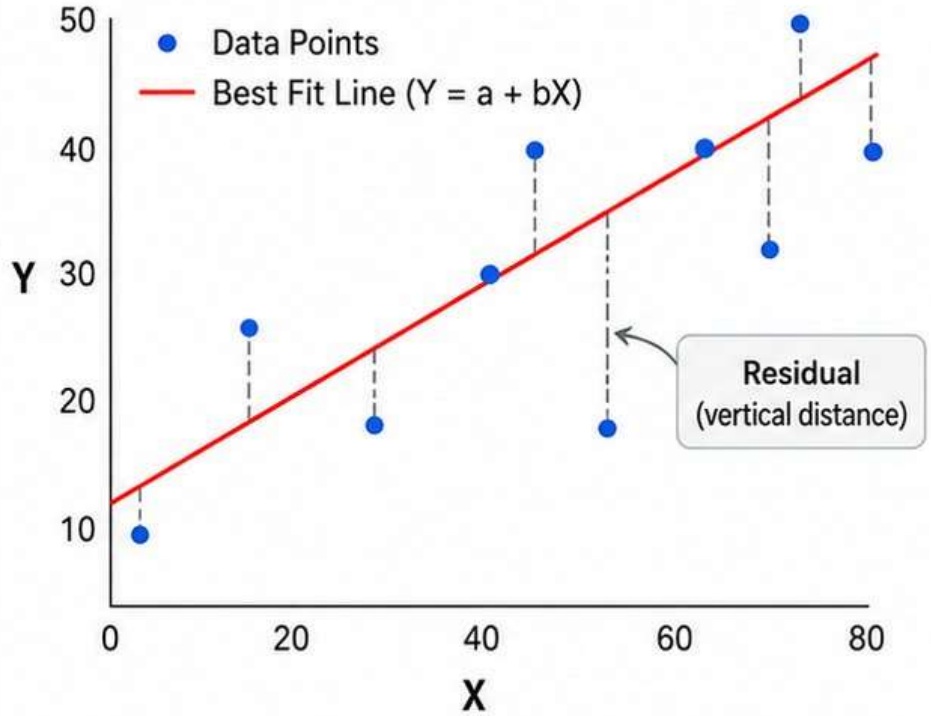
UNDERSTANDING

- Plot the data points (X, Y) on a graph.
- Draw a line that fits the points as closely as possible.

ERROR AND BEST FIT

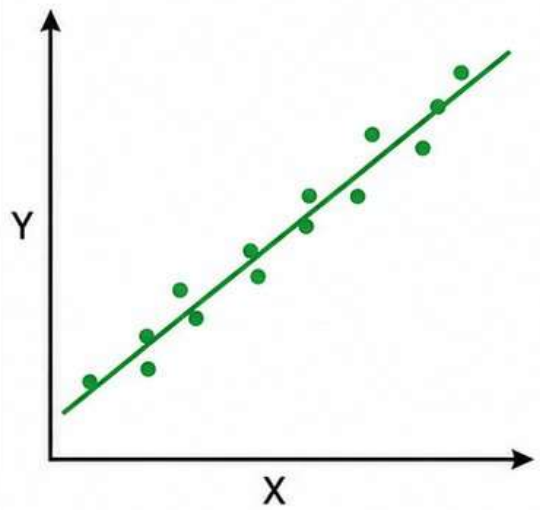
- **Error (Residual):** Vertical distance between a point and the line.
- **Least Squared Error:** Choose the line that minimizes the total squared error.

BEST FIT LINE



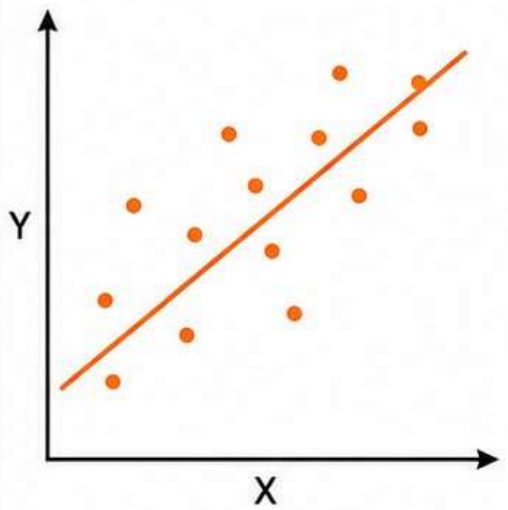
R^2 (R-Squared) — Explained Intuitively

$R^2 = 0.99$



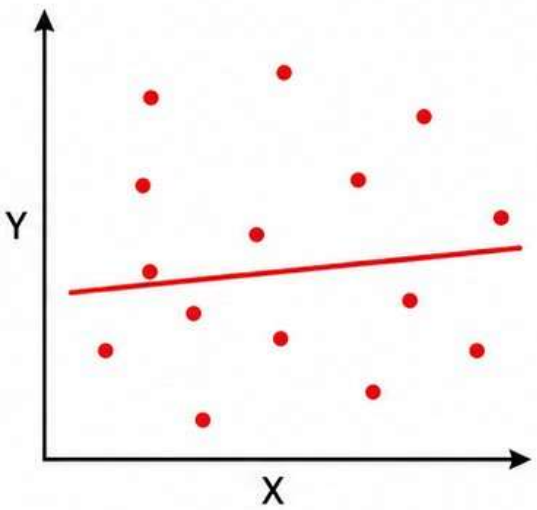
99%

$R^2 = 0.60$



60%

$R^2 = 0.02$



2%



$R^2 = 1$
Perfect prediction

Higher R^2
better fit

Least Squared Error (LSE)

- Choose the line that minimizes the sum of squared vertical errors (residuals) between actual values and predicted values

Suppose our line: y = 10 + 0.5x

X	Y (Actual)	Y (Line) = 10 + 0.5x	Error $(y - \hat{y})$	Squared Error $(y - \hat{y})^2$
10	10	15.0	−5.0	25.00
20	25	20.0	5.0	25.00
30	20	25.0	−5.0	25.00
40	40	30.0	10.0	100.00
50	15	35.0	−20.0	400.00
			Sum of Squared Error (SSE)	575.00

- We square each error to make it positive and penalize larger errors more

The best fit line is the one that minimizes the Sum of Squared Error (SSE)

Regression Metrics Evaluation

Metric	Full Form	Formula	Range / Ideal Value	Intuition (Meaning)
R^2	Coefficient of Determination	$1 - SSE / TSS$	0 to 1 (higher = better)	How well the model explains variation in Y; 1 = perfect prediction
MAE	Mean Absolute Error	$\Sigma \lvert y_i - \hat{y}_i \rvert / n$	≥ 0 (lower = better)	Average error in predictions
MSE	Mean Squared Error	SSE / n	≥ 0 (lower = better)	Penalizes large errors more (since errors are squared)
RMSE	Root Mean Squared Error	$\sqrt{MSE}$	≥ 0 (lower = better)	Same as MSE but in original units of Y – easier to interpret

Regression Metrics: Example

x	y (Actual)	$\hat{y}$ (Predicted)	$\lvert y - \hat{y} \rvert$	$(y - \hat{y})^2$
10	10	16	6	36
20	25	20	5	25
30	20	24	4	16
35	30	26	4	16
40	40	28	12	144
50	15	32	17	289
60	40	36	4	16
65	30	38	8	64
70	50	40	10	100
80	40	44	4	16
Total			74	722 (SSE)

- Mean of y: $\bar{y} = \Sigma y_i / n = 300 / 10 = 30.0$
- $TSS = \Sigma (y_i - \bar{y})^2 = (-20)^2 + (-5)^2 + (-10)^2 + 0^2 + 10^2 + (-15)^2 + 10^2 + 0^2 + 20^2 + 10^2$
- $= 400 + 25 + 100 + 0 + 100 + 225 + 100 + 0 + 400 + 100 = 1450$
- $R^2 = 1 - SSE / TSS = 1 - 722 / 1450 = 0.502$
- $MSE = SSE / n = 722 / 10 = 72.2$
- $RMSE = \sqrt{MSE} = \sqrt{72.2} = 8.49$
- $MAE = \Sigma \lvert y_i - \hat{y}_i \rvert / n = 74 / 10 = 7.4$

TSS	R^2	MSE	RMSE	MAE
1450	0.502	72.2	8.49	7.4

Scaling and Encoding

	Scaling	Encoding
What	Converting numeric features to a common scale	Converting categorical variables to numeric form
Why?	Numeric features may have very different ranges. Scaling puts them on similar scales so machine learning works better.	Machine learning models understand numbers, not text or categories.
Example	Age (0–100) and Income ($0–$1,000,000) should be scaled to similar ranges	Gender: Male = 0, Female = 1

Scaling: Bring Numeric Data on a Common Scale

Method	Formula	Output Range	When to Use
Standardization (Z-score scaling)	$z = (x - \mu) / \sigma$ x = value, μ = mean, σ = standard deviation	Typically −3 to +3, but no fixed range	When features have different means and variances; suitable for algorithms assuming normal distribution (e.g., Linear Regression, Logistic Regression, PCA)
Min–Max scaling (Normalization)	$x' = (x - x_min) / (x_max - x_min)$ x = value, x_min = minimum, x_max = maximum	[0, 1] or a chosen range	When we need features within a specific range (e.g., Neural Networks, KNN, SVM)

Standardized Scaling (Normalized Scaling or Calculating Z-Score)

Age	Income	Age Scaled = $\frac{(Age - Age\ Mean)}{Age\ SD}$	Income Scaled = $\frac{(Income - Income\ Mean)}{Income\ SD}$
25	35,000	-1.22	-1.08
40	50,000	-0.26	-0.43
55	70,000	0.70	0.43
68	1,00,000	1.54	1.73
32	45,000	-0.77	-0.65
Mean: 44	**60,000**		
SD: 15.61	**23,128.91**		

Min-Max Scaling

Age	Income	Age – Minimum Age	Income – Minimum Income	Age Normalized = $\frac{(\text{Age} - \text{Minimum Age})}{\textit{Age Range}}$	Income Normalized = $\frac{(\text{Income} - \text{Minimum Income})}{\text{Income } \textit{Range}}$
25	35,000	0	0	0	0
40	50,000	15	15,000	0.34	0.23
55	70,000	30	35,000	0.69	0.53
68	1,00,000	43	65,000	1	1
32	45,000	7	10,000	0.16	0.15
Minimum: 25	**35,000**				
Range: 68 - 25 = 43	**1,00,000 – 35,000 = 65,000**				

Working with Non-Numeric Features – Encoding

Encoding Method	Example Mapping	Type of Output	When to Use
One-Hot Encoding	Business → 100 Economy Plus → 010 Economy → 001	Binary columns (0/1)	When categories are nominal (no order). Useful for models like Linear Regression or Neural Networks
Label Encoding	Business → 0 Economy Plus → 1 Economy → 2	Single integer column	When categories are ordinal (have order). Suitable for tree-based models (e.g., Decision Trees, Random Forests)
Frequency Encoding	Business → 0.20 Economy Plus → 0.30 Economy → 0.50	Single numeric column (frequency)	When the dataset is large or has many categories. Helps models capture category frequency without high dimensionality

Key takeaway: choose the encoding method based on the nature of the categorical feature (nominal vs ordinal), the model type, and the dataset size

fit_transform() on Train and transform() on Test

Train data (before scaling)

ID	Height (cm)	Weight (kg)
1	150	50
2	160	55
3	170	65
4	180	75

fit_transform() on TRAIN → learns mean and std from TRAIN

- Mean: Height = 165.0, Weight = 61.25
- Std: Height = 11.18, Weight = 9.60

ID	Height (z)	Weight (z)
1	−1.34	−1.17
2	−0.45	−0.65
3	0.45	0.39
4	1.34	1.43

Test data (before scaling)

ID	Height (cm)	Weight (kg)
5	190	85
6	200	95

transform() on TEST → reuses the mean and std learned from TRAIN

- Test data must never be used to compute the scaling parameters (avoids data leakage)

ID	Height (z)	Weight (z)
5	2.24	2.47
6	3.13	3.51

Logistic Regression

Concept	Explanation
Technique Name	Logistic Regression
Type	Classification (not regression despite the name)
Purpose	Predicts the probability of a binary outcome (e.g., Yes/No, True/False, 0/1)
Output Range	0 to 1
Curve Shape	S-shaped (sigmoid curve)
Key Formula	First find $Z = b0 + b1 \cdot x$, and then $P(Y = 1) = 1 / (1 + e^{(-Z)})$ Z = linear combination (logit); e = Euler's number ≈ 2.71828
Decision Rule	If probability ≥ 0.5 → Class = 1; else → Class = 0
Example	Predict whether a passenger survived (1) or not (0) on the Titanic based on age, class, etc. $P(Y = 1) = 0.82$ → survived (1); $P(Y = 1) = 0.23$ → did not survive (0)

Logistic Regression: Key Points

1. Logit = z = a score	$Z = b0 + b1 \cdot x$ Logit (z) is the linear combination of input (x): b0 = intercept, b1 = coefficient of x
2. Sigmoid = converter	$P(Y = 1) = 1 / (1 + e^{(-z)})$ The sigmoid function converts z (any real number) into a probability: $0 < P(Y = 1) < 1$
3. Output = probability	Convert probability into a class label using threshold 0.5 If $P \geq 0.5$ → output class = 1; else ($P < 0.5$) → output class = 0

Logistic Regression: Predict Obesity for a Dog

Dog	Weight (kg)	Obese (y)
D1	8	0 (Not obese)
D2	10	0
D3	12	0
D4	15	1 (Obese)
D5	18	1
D6	22	1

Dog	Linear score z	p(obese)	Prediction
D1 (8 kg)	0.8 × 8 − 10 = −3.6	1 / (1 + e^3.6) ≈ 0.027	Not obese
D3 (12 kg)	0.8 × 12 − 10 = −0.4	1 / (1 + e^0.4) ≈ 0.40	Not obese
D4 (15 kg)	0.8 × 15 − 10 = 2.0	1 / (1 + e^−2) ≈ 0.88	Obese

Interpreting probability

- ≥ 0.5: more likely obese; < 0.5: more likely not obese
- Close to 0: very unlikely obese; close to 1: very likely obese

1. Linear score: z = w × weight + b (like linear regression)

- w (weight): how much the score changes for each 1 kg increase
- b (bias/intercept): the starting score when weight = 0
- After training, suppose w = 0.8, b = −10 → z = 0.8 × weight − 10

2. Convert to probability: p(obese) = 1 / (1 + e^(−z))

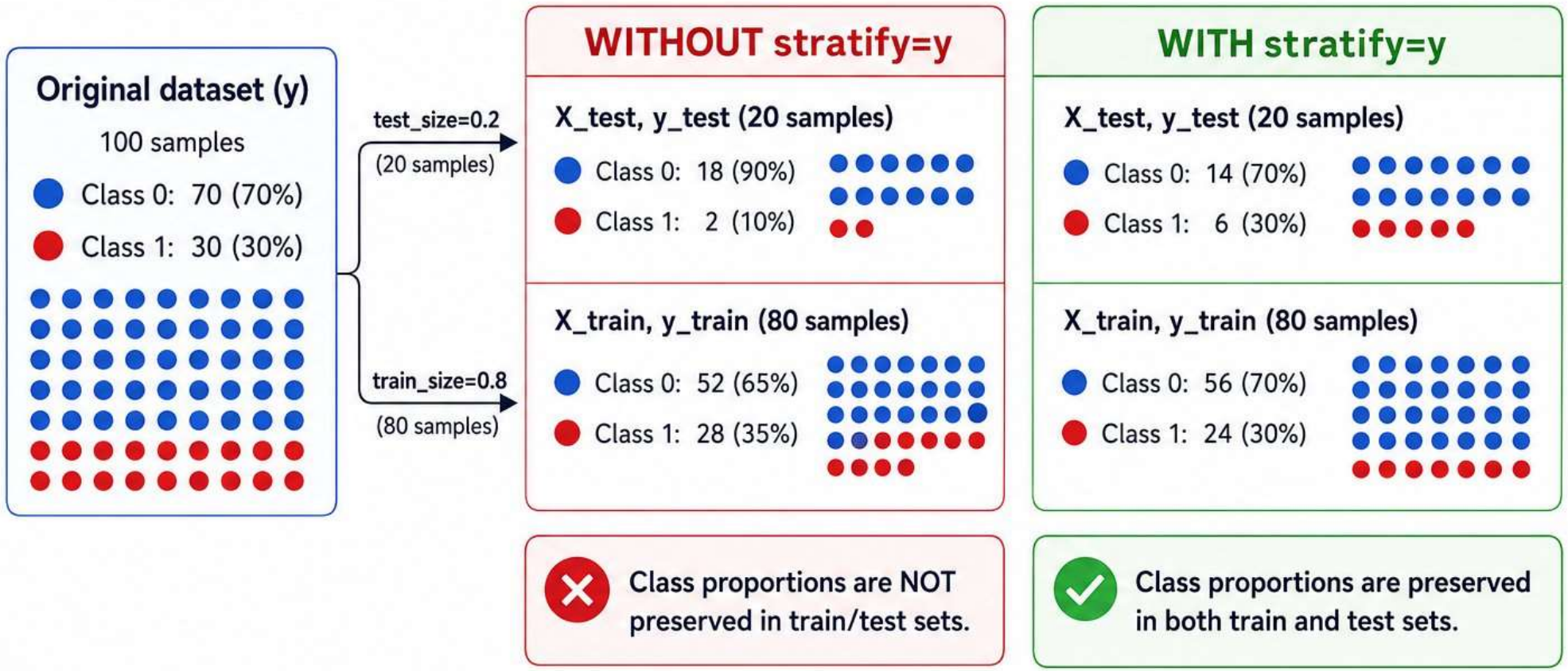
3. Predict: p ≥ 0.5 → obese (1); p < 0.5 → not obese (0)
(0.5 is the common threshold – you can change it)

What does `stratify=y` do?

It ensures that the **proportion of each class** in the target variable (y) is preserved in both the training and test sets.

```
X_train, X_test, y_train, y_test = train_test_split(
  X, y, test_size=0.2, random_state=101, stratify=y
)
```

✓ Use `stratify=y` for classification problems (not for regression).



✗ Class proportions are NOT preserved in train/test sets.

✓ Class proportions are preserved in both train and test sets.

Logistic Regression Example

Suppose in our given example

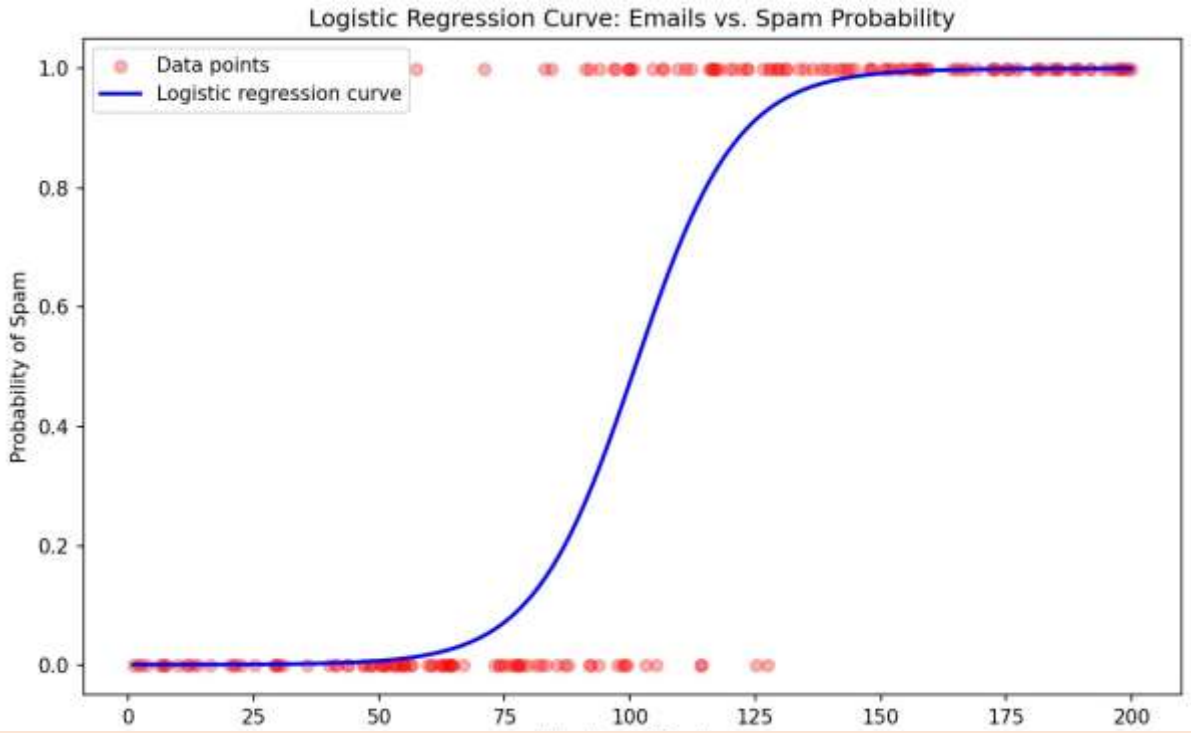
$b_0 = -10$ -> Intercept (How likely is a dog obese even before considering its weight)

$b_1 = 0.2$ -> If weight increases by 1 kg, how much does the logit change?

Dog Weight (kg)	Logit Value ($z = b_0 + b_1x$)	Probability (P(Y=1))	Result
20	-6.0	0.0025	Not Obese
30	-4.0	0.018	Not Obese
40	-2.0	0.119	Not Obese
50	0.0	0.50	Obese (boundary)
60	2.0	0.88	Obese
70	4.0	0.982	Obese

Logistic Regression Curve

- Forms an S-shaped (Sigmoid) curve between 0 and 1



Count of reasons causing email to become spam (Example: Number of suspicious words, Number of links, Percentage of CAPS, etc)

Logistic Regression Curve

- Curve explanation

Element	Explanation
X-axis	Count of reason causing email to become spam (Example: Number of suspicious words, Number of links, Percentage of CAPS, etc)
Y-axis	Represents the **predicted probability** that an email is spam (values between 0 and 1)
Red Dots (Data Points)	Actual observed data — each point shows whether a given email was spam (1) or not spam (0)
Blue Curve (Logistic Regression Line)	The fitted **logistic function** that estimates the probability of spam for each email count. It smoothly transitions from 0 → 1

Logistic Regression Curve

- Curve interpretation

Region on Curve	What Happens	Interpretation
Left side (few spam indicators, near 0–50)	The blue curve is flat near **0**	When the number of such indicators is small, the model thinks it is **very unlikely** they are spam (Maybe most legitimate users have only few such indicators)
Middle region (around 75–125)	The blue curve starts to **rise steeply**	Here, the model starts seeing a **pattern** — as the number of such indicators increases, it becomes **more likely** that the email is spam
Right side (above 125–200)	The blue curve **flattens near 1**	When the number of such indicators is very high, the model becomes **almost certain** that they are spam - It does not go above 1 because probabilities cannot exceed 100%

Example

- Logistic regression formula $P(Y=1)=\frac{1}{1+e^{-(b_0+b_1x)}}$
- Based on the training data, values of b_0 and b_1 are calculated
- Here, let us consider $b_0 = -4$ and $b_1 = 0.8$
- If x is the number of suspicious indicators, then

$$P(\text{Spam})=\frac{1}{1+e^{-(-4+0.8x)}}$$

If we have Number of Suspicious Words (x) =	Predicted Probability of Spam, P(Spam)	Decision (Spam = 1, Ham = 0)	Explanation
1	$\frac{1}{1+e^{-(-4+0.8\times1)}}=0.12$	**0 (Ham)**	Very low probability — model predicts not spam
5	$\frac{1}{1+e^{-(-4+0.8\times5)}}=0.73$	**1 (Spam)**	Probability > 0.5 → model predicts spam
9	$\frac{1}{1+e^{-(-4+0.8\times9)}}=0.97$	**1 (Spam)**	Very high probability — almost certainly spam

Logistic Regression – Python – Step 1: Predicting Probabilities

- Predict probabilities and labels: y_prob = model.**predict_proba**(X)[:, 1]
  - The *predict_proba* function gives two columns for binary classification
    - [
    - [P(class=0), P(class=1)], # First probability = Email is not spam, Second = Email is spam
    - [P(class=0), P(class=1)],
    - ...
    -]
  - Each row = One data point
- Example: model.predict_proba(X[:1])
    - [[0.95, **0.05**],
    - [0.90, **0.10**],
    - [0.40, **0.60**],
    - [0.20, **0.80**],
    - [0.05, **0.95**]]
  - This selects the second column, i.e. probability that each email is spam (1)
- So, y_prob = List of numbers between 0 and 1, such as [0.05, 0.10, 0.60, 0.80, 0.95]

Logistic Regression – Python – Step 2: Converting Probabilities into Final Decisions

- Convert probabilities into final decisions: y_pred = (y_prob >= 0.5).astype(int)
  - Logistic regression does not directly say *spam* or *no spam*
    - It gives probabilities of *how likely it is to be spam*
  - We then set a decision boundary (usually 0.5)
    - If probability >= 0.5, predict spam (1)
    - If probability < 0.5, predict ham (0)
  - The expression **(y_prob >= 0.5)** will give the following result:
    - [False, False, True, True, True]
  - Converting them into integers will give:
    - [0, 0, 1, 1, 1]

Confusion Matrix

- Confusion matrix: a table that summarizes how well a classification model performs – by comparing actual labels with predicted labels
- Tells us where our model is getting confused – i.e. which classes it is mixing up

Symbol	Meaning
H0	Null hypothesis – the “default” or “no effect” assumption
H1	Alternative hypothesis – the competing claim, suggesting a difference or effect

Hypothesis	Statement
H0: Email is NOT spam	Default assumption: the email is legitimate (ham)
H1: Email IS spam	Alternative assumption: the email is spam

Confusion Matrix Example

Case	When It Occurs	What It's Called	Short Meaning	Email Example
1. True Negative (TN)	Not spam predicted as not spam	Correct not-spam	Correctly negative	Genuine email → not spam
2. False Positive (FP)	Not spam predicted as spam	Type I error (false alarm)	False positive	Good email → spam
3. False Negative (FN)	Spam predicted as not spam	Type II error (missed spam)	False negative	Spam email → not spam
4. True Positive (TP)	Spam predicted as spam	Correct spam	Correctly positive	Spam email → spam

Confusion Matrix

Structure

	Predicted: 0 (Ham / Not Spam)	Predicted: 1 (Spam)
Actual: 0 (Ham)	TN – True Negative (correctly predicted not spam)	FP – False Positive (predicted spam but actually not spam)
Actual: 1 (Spam)	FN – False Negative (predicted not spam but actually spam)	TP – True Positive (correctly predicted spam)

Example

	Predicted: 0 (Ham)	Predicted: 1 (Spam)
Actual: 0 (Ham)	90 (TN)	10 (FP)
Actual: 1 (Spam)	20 (FN)	80 (TP)

Confusion Matrix - Exercise

- Out of 1000 emails, 800 non-spams were classified correctly, 20 were incorrectly classified as spam, 40 were incorrectly classified as non-spam, and the remaining spams were identified correctly
- Write the null hypothesis and create a confusion matrix

Confusion Matrix - Exercise

- Out of 1000 emails, 800 non-spams were classified correctly, 20 were incorrectly classified as spam, 40 were incorrectly classified as non-spam, and the remaining spams were identified correctly
- Write the null hypothesis and create a confusion matrix
- H_0: Email is not spam

n = 1000	Predicted: NO	Predicted: YES
Actual: NO	**TN = 800**	**FP = 20 (Type I Error)**
Actual: YES	**FN = 40 (Type II Error)**	**TP = 140**

Hospital Problem

- A hospital tested 200 patients for a disease.
- 60 patients actually had the disease.
- The model correctly identified 50 of them as having the disease.
- Out of the 140 healthy patients, the model incorrectly classified 20 as having the disease.

Hospital Problem

- A hospital tested 200 patients for a disease.
- 60 patients actually had the disease.
- The model correctly identified 50 of them as having the disease.
- Out of the 140 healthy patients, the model incorrectly classified 20 as having the disease.

n = 200	Predicted: NO	Predicted: YES
Actual: NO	**TN = 120**	**FP = 20 (Type I Error)**
Actual: YES	**FN = 10 (Type II Error)**	**TP = 50**

Metrics Derived From Confusion Matrix

Metric	Formula	Meaning	Example (TN = 90, FP = 10, FN = 20, TP = 80)	Value
Accuracy	(TP + TN) / (TP + TN + FP + FN)	Overall correctness	(80 + 90) / (80 + 90 + 10 + 20)	0.85
Precision	TP / (TP + FP)	Of predicted positives, how many were correct	80 / (80 + 10)	0.89
Recall (Sensitivity)	TP / (TP + FN)	Of actual positives, how many were caught	80 / (80 + 20)	0.80
Specificity	TN / (TN + FP)	Of actual negatives, how many were correct	90 / (90 + 10)	0.90
F1 Score	2 × (Precision × Recall) / (Precision + Recall)	Balance between precision and recall	2 × (0.89 × 0.80) / (0.89 + 0.80)	0.84

Python Classification Report – Key Takeaways

Metric	Definition	Class 0	Class 1	Interpretation
Support	Actual number of instances in the test set	150	81	150 instances of Class 0 and 81 of Class 1: the dataset is imbalanced (Class 0 has more samples)
Precision	Of the predicted positives, how many were correct	0.80	0.73	When predicting Class 0 the model is correct 80% of the time; for Class 1, 73%. Better precision for Class 0.
Recall	Of the actual positives, how many were correctly identified	0.88	0.60	The model finds 88% of actual Class 0 instances but only 60% of actual Class 1 instances. Significantly weaker recall for Class 1.
F1-Score	Harmonic mean of precision and recall	0.84	0.66	Overall performance is better for Class 0 (0.84) than Class 1 (0.66), reflecting higher recall for Class 0

Bottom line: the model performs well on Class 0 but struggles to identify Class 1 instances

Metrics Derived from Confusion Matrix

- **Accuracy**: Overall correctness of the model's predictions
- **Precision (Positive Predictive Value)**: Accuracy of positive predictions
- **Recall (Sensitivity or True Positive Rate)**: Ability of the model to identify all positive instances
- **Specificity (True Negative Rate)**: Ability of the model to identify all negative instances
- **F1 Score**: Harmonic mean of precision and recall and provides a balance between the two metrics

Metrics for Our Example

- $Accuracy = \frac{TP+TN}{Total} = \frac{140+800}{1000} = \frac{940}{1000} = 0.94 = 94\%$
- $Precision = \frac{TP}{TP+FP} = \frac{140}{140+20} = \frac{140}{160} = 0.875 = 87.5\%$
- $Recall = \frac{TP}{TP+FN} = \frac{140}{140+40} = \frac{140}{180} = 0.77 = 77\%$
- $Specificity = \frac{TN}{TN+FP} = \frac{800}{800+20} = \frac{800}{820} = 0.97 = 97\%$
- $F1\ Score = \frac{2\ x\ (Precision\ x\ Recall)}{Precision+Recall} = \frac{2\ x\ (0.875\ x\ 0.77)}{0.875+0.77} = 0.81 = 81\%$

Resampling

Problem	One single train-test split is often not trustworthy enough
Example dataset	100 customer records
One split	80 → training, 20 → testing
Issue	Model accuracy depends heavily on which 20 records go into the test set
Case 1	Easy test set → 95% accuracy
Case 2	Difficult test set → 78% accuracy
Observation	Same model, same data, different split → different result
Solution	Create multiple training/testing combinations
Why	Observe performance across several splits
Outcome	Get a more reliable estimate of model performance
Techniques	Bootstrapping and K-Fold Cross Validation

Bootstrapping vs K-Fold Cross Validation

Dataset: [A, B, C, D, E, F]

Feature	Bootstrapping Example	K-Fold Cross Validation Example (3 folds)
Sampling method	Random sampling with replacement	Split dataset into 3 equal folds
Sample / fold	Sample 1: [A, B, A, D, F, C] Sample 2: [B, C, E, E, A, F] … (many such samples can be drawn)	Fold 1: Test [A, B], Train [C, D, E, F] Fold 2: Test [C, D], Train [A, B, E, F] Fold 3: Test [E, F], Train [A, B, C, D]
Data point usage	Some points may appear multiple times	Each point used exactly once as test set
Repetition	Samples are repeated multiple times	Each fold is used once as test; total 3 iterations

Oversampling

Oversampling is a technique used to increase the number of instances in the minority class to balance the dataset

Why needed?

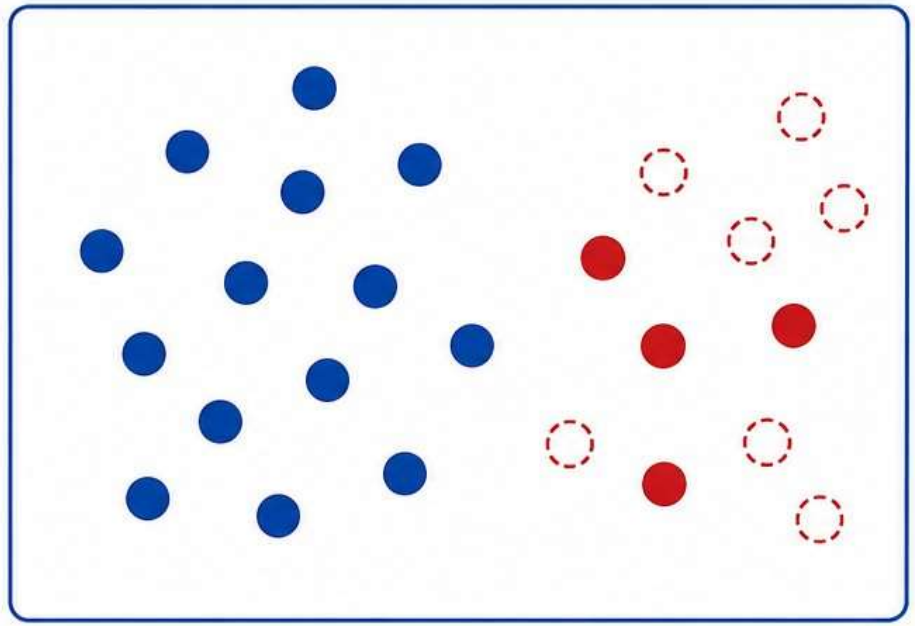
- In highly imbalanced datasets, models tend to ignore the minority class
- Oversampling gives the minority class more importance, helping the model learn its patterns better

Example: credit card fraud detection

- Dataset: 99,998 non-frauds and 2 frauds
- The data is extremely imbalanced. Oversampling is needed to increase the fraud cases so the model can learn and detect frauds effectively.

OVERSAMPLING vs SMOTE

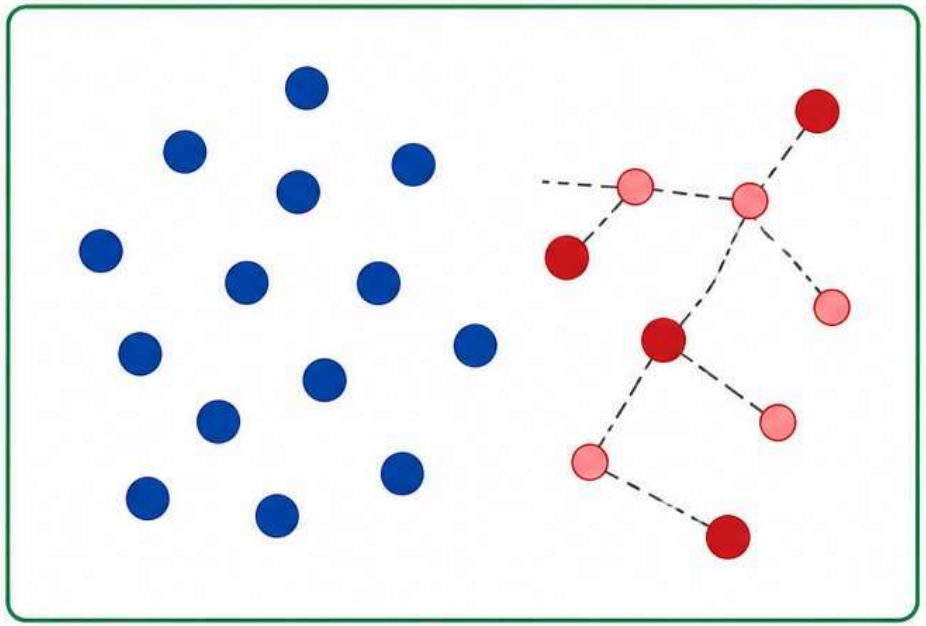
OVERSAMPLING



- Majority Class
- Minority Class (Duplicated)
- Same points duplicated

Same points duplicated

SMOTE



- Majority Class
- Minority Class (Original)
- Synthetic Minority Class (New)
- Points modified somewhat

Points modified somewhat

Oversampling and SMOTE

Original data

ID	Feature	Class
1	2	0
2	3	0
3	4	0
4	5	1
5	6	1

Oversampling

ID	Feature	Class
1	2	0
2	3	0
3	4	0
4	5	1
5	6	1
6	5	1
7	6	1

- Minority class records are duplicated

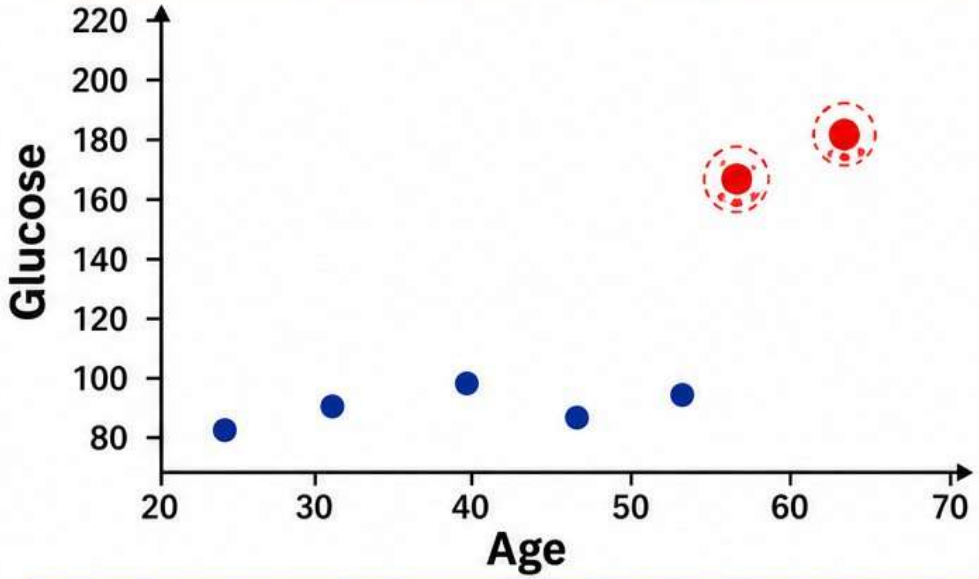
SMOTE

ID	Feature	Class
1	2	0
2	3	0
3	4	0
4	5	1
5	6	1
6	5.5	1
7	5.2	1

- New synthetic records are generated

OVERSAMPLING vs SMOTE

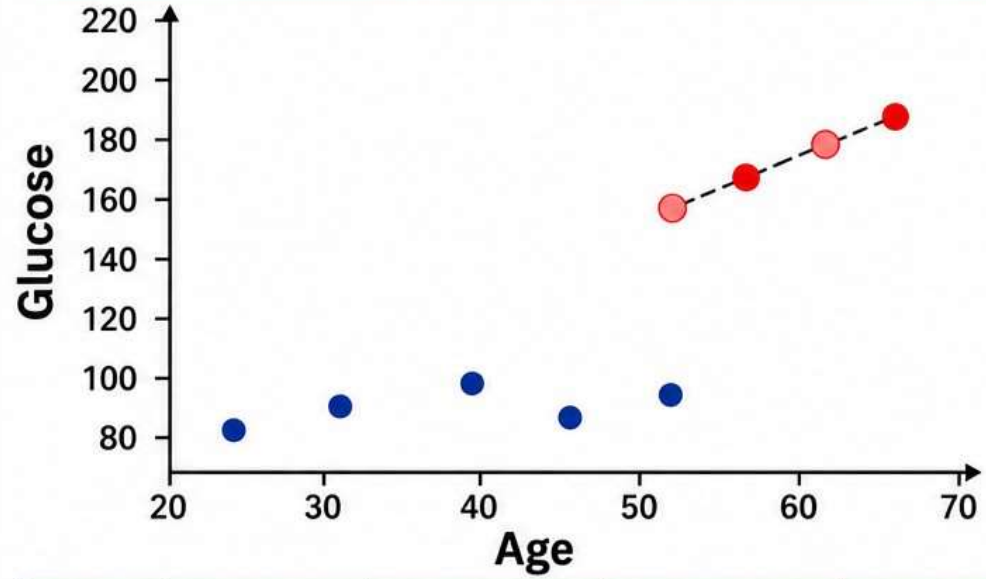
OVERSAMPLING



ID	Age	Glucose	Diabetes (0/1)	Type
1	25	90	0	Original
2	32	95	0	Original
3	40	100	0	Original
4	45	92	0	Original
5	50	98	0	Original
6	55	170	1	Original
7	60	185	1	Original
8	55	170	1	Duplicate
9	60	185	1	Duplicate

Same diabetes records copied

SMOTE



ID	Age	Glucose	Diabetes (0/1)	Type
1	25	90	0	Original
2	32	95	0	Original
3	40	100	0	Original
4	45	92	0	Original
5	50	98	0	Original
6	55	170	1	Original
7	60	185	1	Original
8	52.5	162.5	1	Synthetic
9	57.5	177.5	1	Synthetic

New diabetes records generated

Oversampling and SMOTE

- Original data

ID	Feature	Class
1	2	0
2	3	0
3	4	0
4	5	1
5	6	1

Oversampling

ID	Feature	Class
1	2	0
2	3	0
3	4	0
4	5	1
5	6	1
6	5	1
7	6	1

SMOTE

ID	Feature	Class
1	2	0
2	3	0
3	4	0
4	5	1
5	6	1
6	5.5	1
7	5.2	1

Naive Bayes Classifier

- Compares the probabilities of two outcomes and predicts the class with the higher probability

Sample of original dataset

Customer Id	Subscribed to Newsletter?	Follows Social Media?	Visited Website?	Purchase? (Outcome)
1	0	1	0	1
2	1	0	1	0
3	1	1	0	1
…	…	…	…	…
10	1	1	0	1

What Naive Bayes does

- Calculates the probability of each outcome (will buy = 1 or will not buy = 0)
- Compares the two probabilities
- Predicts the class with the higher probability

New customer (ID 11)

- Subscribed to newsletter = 1; follows social media = 0; has visited website = 1

Compare probabilities

- Class = 1 (will buy): 0.066
- Class = 0 (will not buy): 0.075

Since 0.075 > 0.066 → prediction: will NOT buy (class = 0)

Logistic Regression vs Naive Bayes

- Core idea (email spam example)

	Logistic Regression	Naive Bayes
What it does	Predicts the probability that an email is spam (class 1)	Computes probabilities for both classes (spam and not spam) and compares them
Computes	P(class 1 | email) = probability email is SPAM	P(class 1 | email) and P(class 0 | email)
Decision	If P(class 1 | email) ≥ 0.50 → predict SPAM Else → predict NOT SPAM	If P(class 1 | email) > P(class 0 | email) → predict SPAM Else → predict NOT SPAM

In simple terms

- Logistic Regression: predict SPAM if its probability is at least 50%
- Naive Bayes: predict the class (SPAM or NOT SPAM) with the higher probability

Naïve Bayes Example

Customer Id	Subscribed to the Newsletter?	Follows our Social Media Handle?	Has visited our Website?	Purchase? (Outcome)
1	0	1	0	1
2	1	0	1	0
3	1	1	0	1
4	0	0	0	0
5	1	0	0	0
6	1	1	1	1
7	0	1	1	1
8	1	0	1	0
9	0	0	1	0
10	1	1	0	1
11	1	0	1	?
12	0	1	1	?

Bayes Theorem Reminder

- $P(A|B) = \frac{P(B|A)\ x\ P(A)}{P(B)}$

- Let x = Our case (i.e. N=1, S=0, W=1) … We want to calculate

- $P(Purchase = 1|x) = \frac{P(X|Purchase = 1)\ x\ P(Purchase=1)}{P(x)}$ and
- $P(Purchase = 0|x) = \frac{P(X|Purchase = 0)\ x\ P(Purchase=0)}{P(x)}$
- And see which one is higher

Step 1: Problem Setup (Customer 11)

- We want to predict the outcome (Purchase = 1 or 0) for Customer 11, whose features are
  - Subscribed = 1
  - Follows = 0
  - Visited = 1

  } **Features**

- We want to find

  P(Purchase = 1 | features) vs P(Purchase = 0 | features)

  That is, will a customer buy or not buy, given the above features
- Then select whichever is higher

Reminder of Bayes Theorem …

$$P(A \mid B) = \frac{P(B|A)\ x\ P(A)}{P(B)}$$

Step 2: Calculate Base Probabilities

- Total customers = 10
- Count (Purchased = 1) = 5
- Count (Purchased = 0) = 5

- So:
  - P (Purchased = 1) = 5/10 = 0.5
  - P (Purchased = 0) = 5/10 = 0.5

Step 3: Compute Conditional Probabilities For Each Class

If the numerator is 0, we add 1 to it so that the value does not become 0 and we add *number of categories* to the denominator, here 2 (purchase = 0 and purchase = 1). This is called **Laplace Smoothing**.

- For Purchase = 1

Feature	**Count of feature = 1 among Purchase = 1**	**Total Purchase = 1**	**Probability**	**Meaning**
Subscribed = 1	3 (customers 3, 6, 10)	5	3/5 = 0.6	P (Subscribed = 1 | Purchase = 1)
Follows = 0	0 (No customer, so considering as 1)	5	0+1/5+2 = 0.1	P (Follows = 0 |Purchase = 1)
Visited = 1	2 (customers 6, 7)	5	2/5 = 0.4	P (Visited = 1 | Purchase = 1)

- For Purchase = 0

Feature	**Count of feature = 1 among Purchase = 0**	**Total Purchase = 0**	**Probability**	**Meaning**
Subscribed = 1	3 (customers 2, 5, 8)	5	3/5 = 0.6	P (Subscribed = 1 | Purchase = 0)
Follows = 0	5 (customers 2, 4, 5, 8, 9)	5	5/5 = 1	P (Follows = 0 | Purchase = 0)
Visited = 1	3 (customers 2, 8, 9)	5	3/5 = 0.6	P (Visited = 1 | Purchase = 0)

Step 4: Apply Naïve Bayes Formula

- We want to test for a new Customer 11, whose features are:
  - Subscribed = 1, Follows = 0, Visited = 1
- Probability that this customer will purchase …
  - P (Purchase = 1 | features) = P(1) × P(Sub = 1 | 1) × P(Follows = 0 | 1) × P(Visit = 1 | 1)
  - P (Purchase = 1 | features) = $\frac{5}{10} \times \frac{3}{5} \times \frac{1}{7} \times \frac{2}{5} = 0.017$
- Probability that this customer will NOT purchase …
  - P (Purchase = 0 | features) = P(0) × P(Sub = 1 | 0) × P(Follow = 0 | 0) × P(Visit = 1 | 0)
  - P (Purchase = 0 | features) = $\frac{5}{10} \times \frac{3}{5} \times \frac{5}{5} \times \frac{3}{5} = 0.18$
- Normalizing (So that the probabilities add to 1)
  - P(X) = 0.017 + 0.18 = 0.197
  - P(Y = 1 | X) = $\frac{0.017}{0.197} = 0.09$
  - P(Y = 0 | X) = $\frac{0.18}{0.197} = 0.91$

Since 0.91 > 0.09,
Prediction = Customer will NOT purchase

For Customer 12 (Considering Customer 11)

- Total customers = 11
- Count (Purchased = 1) = 5
- Count (Purchased = 0) = 6

- So:
  - P (Purchased = 1) = 5/11 = 0.4545
  - P (Purchased = 0) = 6/11 = 0.5455

For Customer 12 (Considering 11)

* **Laplace Soothing**: When any such case happens, add 1 to numerator and *number of categories* to denominator, so 0/5 will become 1/7

- For Purchase = 1

Feature	**Count of feature = 1 among Purchase = 1**	**Total Purchase = 1**	**Probability**	**Meaning**
Subscribed = 0	2 (customers 1, 7)	5	2/5 = 0.4	P (Subscribed = 0 | Purchase = 1)
Follows = 1	5 (customer 1, 3, 6, 7, 10)	5	5/5 = 1	P (Follows = 1 |Purchase = 1)
Visited = 1	2 (customers 6, 7)	5	2/5 = 0.4	P (Visited = 1 | Purchase = 1)

- For Purchase = 0

Feature	**Count of feature = 1 among Purchase = 0**	**Total Purchase = 0**	**Probability**	**Meaning**
Subscribed = 0	2 (customers 4, 9)	6	2/6 = 0.33	P (Subscribed = 0 | Purchase = 0)
Follows = 1	0 (customers: None)*	6	1/8 = 0.125	P (Follows = 1 | Purchase = 0)
Visited = 1	4 (customers 2, 8, 9, 11)	6	4/6 = 0.667	P (Visited = 1 | Purchase = 0)

For Customer 12 (Considering 11)

- We want to test for a new Customer 12, whose features are:
  - Subscribed = 0
  - Follows = 1
  - Visited = 1

- Probability that this customer will purchase …
- P (Purchase = 1 | features) = P(1) × P(Sub = 0 | 1) × P(Follows = 1 | 1) × P(Visit = 1 | 1)
- P (Purchase = 1 | features) = 0.4545 × 0.4 × 1 × 0.4 = **0.0727**

- Probability that this customer will NOT purchase …
- P (Purchase = 0 | features) = P(0) × P(Sub = 0 | 0) × P(Follow = 1 | 0) × P(Visit = 1 | 0)
- P (Purchase = 0 | features) = 0.5455 × 0.333 × 0.125 × 0.667 = **0.0151**

- Since 0.0727 > 0.0151, Naïve Bayes predicts that the customer WILL purchase

Naïve Bayes Exercise

Record	High_Income	Good_Credit_Rating	Own_Residence	Loan_Sanctioned
1	1	1	1	1
2	0	1	1	1
3	1	0	1	1
4	1	1	0	1
5	0	0	1	0
6	0	1	0	0
7	1	0	0	0
8	0	0	0	0
9	1	1	1	1
10	1	0	1	1
11	1	1	0	?

Step 1: Problem Setup (Customer 11)

- We want to predict the outcome (Loan = 1 or 0) for Customer 11, whose features are
  - Income = 1
  - Rating = 1
  - Residence = 0

  } **Features**

- We want to find

  P(Loan = 1 | features) vs P(Loan = 0 | features)

- Then select whichever is higher

Reminder of Bayes Theorem …

$$P(A \mid B) = \frac{P(B|A)\ x\ P(A)}{P(B)}$$

Step 2: Calculate Base Probabilities

- Total customers = 10
- Count (Loan = 1) = 6
- Count (Loan = 0) = 4

- So:
  - P (Loan = 1) = 6/10 = 0.6
  - P (Loan= 0) = 4/10 = 0.4

For Customer 11

- For Loan = 1

Feature	**Count of feature among Loan = 1**	**Total Loan = 1**	**Probability**	**Meaning**
Income = 1	5	6	5/6 = 0.86	P (Income = 1 | Loan = 1)
Rating = 1	4	6	4/6 = 0.66	P (Rating = 1 | Loan = 1)
Residence = 0	1	6	1/6 = 0.16	P (Residence = 0 | Loan = 1)

- For Loan = 0

Feature	**Count of feature among Loan = 0**	**Total Loan = 0**	**Probability**	**Meaning**
Income = 1	1	4	1/4 = 0.25	P (Income = 1 | Loan = 0)
Rating = 1	1	4	1/4 = 0.25	P (Rating = 1 | Loan = 0)
Residence = 0	3	4	3/4 = 0.75	P (Residence = 0 | Loan = 0)

For Customer 11

- We want to test for a new Customer 11, whose features are:
  - Income = 1
  - Rating = 1
  - Residence = 0

- Probability that this customer will be given a loan …
- P (Loan = 1 | features) = P(1) × P(Income = 1 | 1) × P(Rating = 1 | 1) × P(Residence = 0 | 1)
- P (Loan = 1 | features) = 0.6 × 0.86 × 0.66 × 0.16 = **0.05**

- Probability that this customer will NOT be given a loan…
- P (Loan = 0 | features) = P(0) × P(Income = 1 | 0) × P(Rating = 1 | 0) × P(Residence = 0 | 0)
- P (Loan = 0 | features) = 0.4 × 0.25 × 0.25 × 0.75 = **0.018**

- Since 0.05 > 0.01, Naïve Bayes predicts that the customer WILL be given a loan

Naïve Bayes Types

Type	Recommended Usage
Gaussian Naïve Bayes	Normally distributed numerical data
Multinomial Naïve Bayes	Discrete data, Text classification, Word frequency calculation
Bernoulli Naïve Bayes	Binary/Boolean classification
Complement Naïve Bayes	Imbalanced datasets
Text Classification	Spam detection, Sentiment analysis

Decision Tree

Model Type	Supervised machine learning
Function	Flowchart for Classification and Regression (CART) Partitions data based on feature values to make predictions at leaf nodes
Versatility	Used for both classification and regression tasks Classification: predicting if a customer will churn (Yes/No) Regression: predicting the value of a house
Interpretability	High – easy for humans to understand and explain (“white-box” model)

Entropy in Decision Trees

- Entropy measures the impurity (or randomness) in a set of training examples. It quantifies how mixed the classes are in a node.

Definition

$H(S) = -\Sigma\, p_i \log_2(p_i)$ (sum over classes i = 1 … C)

- where p_i = (number of examples in class i) / (total number of examples in S)
- $0 \le H(S) \le \log_2(C)$

Intuition

- Entropy is 0 when all examples in S belong to the same class (perfect purity)
- Entropy is maximum when classes are equally mixed
- Higher entropy means more disorder (impurity) in the node

Initial Entropy Calculation

Dataset

Temperature (°C)	Weekend	Play Tennis?
35	Yes	Yes
22	No	No
38	Yes	Yes
30	No	No
33	Yes	Yes
40	No	No
25	Yes	No
28	No	Yes
24	Yes	No
36	No	Yes

Step 1: count the classes

- Yes = 5, No = 5; N = 5 + 5 = 10

Step 2: calculate probabilities

- p(Yes) = 5/10 = 0.5; p(No) = 5/10 = 0.5

Step 3: apply the entropy formula

- $\text{Entropy}(S) = -\Sigma\, p_i \log_2(p_i)$
- $= -[0.5 \log_2(0.5) + 0.5 \log_2(0.5)]$
- $= -[0.5(-1) + 0.5(-1)] = 1$

Initial entropy = 1 (maximum impurity – classes are perfectly balanced)

Split on Temperature

- Split the dataset on “Temperature” using the threshold 30°C: left = Temperature ≤ 30°C, right = Temperature > 30°C

Temperature ≤ 30°C (5 records)

Temp (°C)	Weekend	Play Tennis?
22	No	No
30	No	No
25	Yes	No
28	No	Yes
24	Yes	No

Temperature > 30°C (5 records)

Temp (°C)	Weekend	Play Tennis?
35	Yes	Yes
38	Yes	Yes
33	Yes	Yes
40	No	No
36	No	Yes

- Yes = 1, No = 4
- Entropy = $-[(1/5)\log_2(1/5) + (4/5)\log_2(4/5)] = 0.722$

- Yes = 4, No = 1
- Entropy = $-[(4/5)\log_2(4/5) + (1/5)\log_2(1/5)] = 0.722$

- Weighted average entropy after split = $(5/10) \times 0.722 + (5/10) \times 0.722 = 0.722$

Information gain (Temperature) = 1.000 − 0.722 = 0.278

Split on Weekend

- Split the dataset on “Weekend” and calculate the entropy for each subset and the weighted average entropy

Weekend = Yes (5 records)

Temp (°C)	Play Tennis?
35	Yes
38	Yes
33	Yes
25	No
24	No

- Yes = 3, No = 2; Entropy (Weekend = Yes) = 0.971

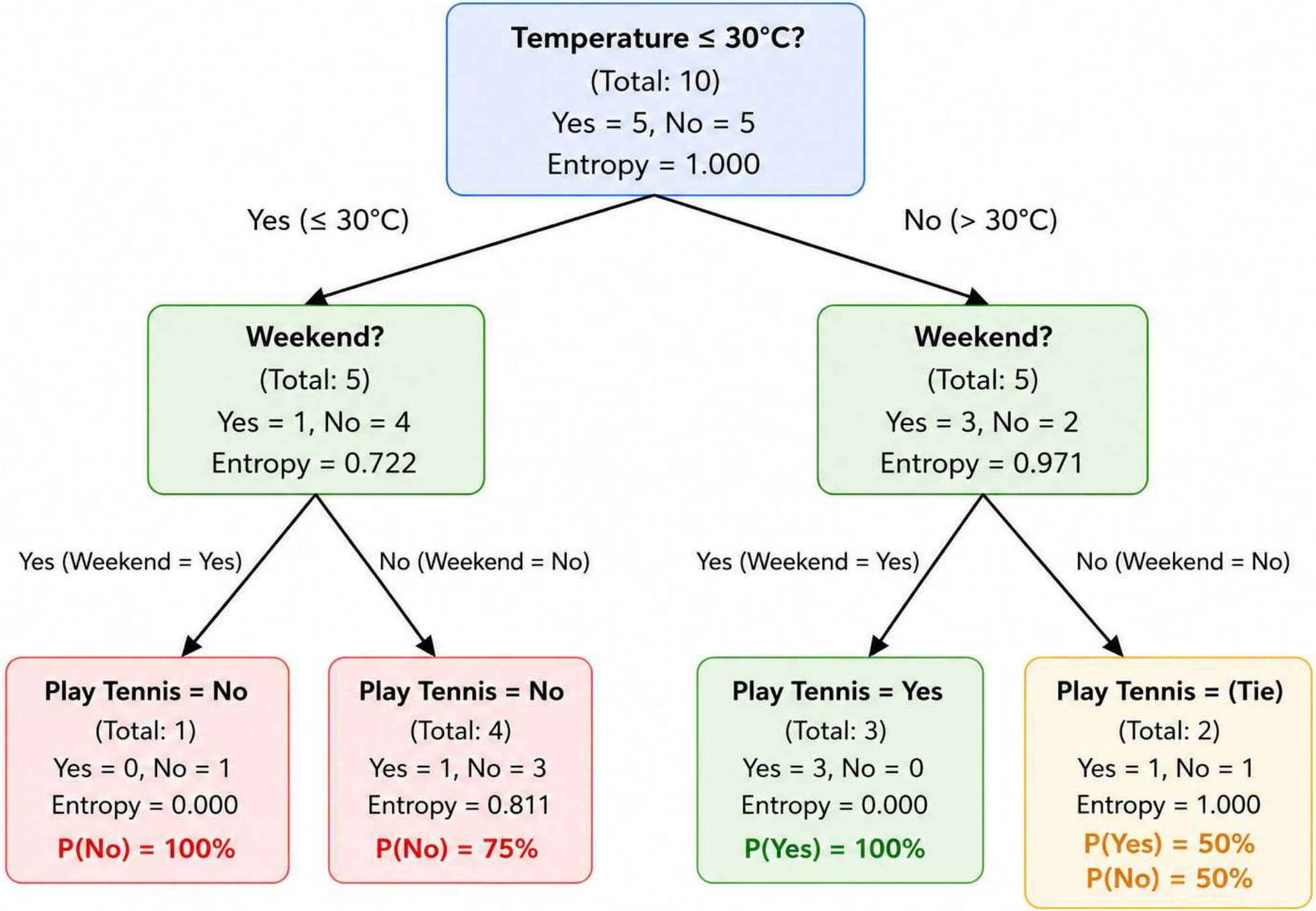
Weekend = No (5 records)

Temp (°C)	Play Tennis?
22	No
30	No
40	No
28	Yes
36	Yes

- Yes = 2, No = 3; Entropy (Weekend = No) = 0.971

- Weighted average entropy after split = (5/10) × 0.971 + (5/10) × 0.971 = 0.971

Information gain (Weekend) = 1.000 – 0.971 = 0.029 – Weekend only partially classifies the data


Temperature ≤ 30°C?
(Total: 10)
Yes = 5, No = 5
Entropy = 1.000
Yes (≤ 30°C)
No (> 30°C)
Weekend?
(Total: 5)
Yes = 1, No = 4
Entropy = 0.722
Weekend?
(Total: 5)
Yes = 3, No = 2
Entropy = 0.971
Yes (Weekend = Yes)
No (Weekend = No)
Yes (Weekend = Yes)
No (Weekend = No)
Play Tennis = No
(Total: 1)
Yes = 0, No = 1
Entropy = 0.000
P(No) = 100%
Play Tennis = No
(Total: 4)
Yes = 1, No = 3
Entropy = 0.811
P(No) = 75%
Play Tennis = Yes
(Total: 3)
Yes = 3, No = 0
Entropy = 0.000
P(Yes) = 100%
Play Tennis = (Tie)
(Total: 2)
Yes = 1, No = 1
Entropy = 1.000
P(Yes) = 50%
P(No) = 50%

Entropy

- Calculate entropy

Income	Age	Loan Granted?
100000	25	1
50000	40	0
200000	30	1
150000	35	0
300000	28	1
110000	38	0

Income	Age	Loan Granted?
200000	25	1
375000	40	1
200000	30	1
150000	35	1
300000	28	1
110000	38	0

Entropy

Income	Age	Loan Granted?
100000	25	1
50000	40	0
200000	30	1
150000	35	0
300000	28	1
110000	38	0

Income	Age	Loan Granted?
200000	25	1
375000	40	1
200000	30	1
150000	35	1
300000	28	1
110000	38	0

$$\boldsymbol{Entropy = -\Sigma pi.\log_2(p_i) \text{ for i} = 1 \text{ to n}}$$

$\boldsymbol{Entropy}$ = -[($\frac{3}{6}$ x $\log_2$ ($\frac{3}{6}$) + ($\frac{3}{6}$ x $\log_2$ ($\frac{3}{6}$)]

= - [(0.5 x -1) + (0.5 x -1)]

= - [(-0.5) + (-0.5)] = **1**

$\boldsymbol{Entropy}$ = -[($\frac{1}{6}$ x $\log_2$ ($\frac{1}{6}$) + ($\frac{5}{6}$ x $\log_2$ ($\frac{5}{6}$)]

= - [(0.1666 x -2.5849) + (0.8333 x -0.2630)]

= - [(-0.4308) + (-0.2191)] = **0.6500**

Entropy and Decision Tree

Income	Age	Loan Granted?
100000	25	1
50000	40	0
200000	30	1
150000	35	0
300000	28	1
110000	38	0

- Consider the first dataset only
- **Choice 1: Use Age as the criteria**

Criteria	Total cases	Loan Granted
Age <= 30	3	Yes = 3
	3	No = 0

$$\boldsymbol{Entropy} = -[(\frac{3}{3} \text{ x } \log_2(\frac{3}{3}) + (\frac{0}{3} \text{ x } \log_2(\frac{0}{3})]$$
$$= 0$$

Criteria	Total cases	Loan Granted
Age > 30	3	Yes = 0
	3	No = 3

$$\boldsymbol{Entropy} = -[(\frac{0}{3} \text{ x } \log_2(\frac{0}{3}) + (\frac{3}{3} \text{ x } \log_2(\frac{3}{3})]$$
$$= 0$$

Weighted average = 0

Entropy and Decision Tree

- Consider the first dataset only
- **Choice 2: Use Income as the criteria**

Income	Age	Loan Granted?
100000	25	1
50000	40	0
200000	30	1
150000	35	0
300000	28	1
110000	38	0

Criteria	Total cases	Loan Granted
Income > 1.5L	2	Yes = 2
	2	No = 0

$$\boldsymbol{Entropy} = -[(\frac{2}{2} \text{ x } \log_2(\frac{2}{2}) + (\frac{0}{2} \text{ x } \log_2(\frac{0}{2})]$$

= 0

Criteria	Total cases	Loan Granted cases
Income <= 1.5L	4	Yes = 1
	4	No = 3

$$\boldsymbol{Entropy} = -[(\frac{1}{4} \text{ x } \log_2(\frac{1}{4}) + (\frac{3}{4} \text{ x } \log_2(\frac{3}{4})]$$

= - [(0.25 x -2) + (0.75 x -0.415)]

= - [(-0.5) + (-0.311)] = 0.811

Weighted average = $(0 \text{ x } \frac{2}{6}) + (0.811 \text{ x } \frac{4}{6}) = 0 + 0.5407 = 0.54$

Information Gain

- **Information Gain**: Change in entropy

Data	Entropy	Change in entropy (Information Gain)
Original	1	NA
Split on Age	0	1 – 0 = 1
Split on Income	0.54	1 – 0.54 = 0.46

- Conclusion: Age is a better indicator of prediction than original data or Income

Entropy

- $Entropy = -\Sigma \mathrm{pi}.\log_2(\mathrm{pi})\ \mathrm{for}\ \mathrm{i} = 1\ \mathrm{to}\ \mathrm{n}$
  - n=number of classes in data, p = proportion of data points belonging to that class
- D1: [A, B, C, D, E, F, G, H] -> Not at all homogeneous -> 8 classes -> Entropy = 3
- Entropy(D1) = -[(1/8 x $\log_2$ (1/8) + (1/8 x $\log_2$ (1/8) …] 8 times
  = -[(1/8 x -3) + (1/8 x -3) …] 8 times
  = -[-3/8 + -3/8 …] 8 times = -[-24/8] = 3

- D2: [A, A, A, A, B, B, B, B] -> Partially homogeneous -> 2 classes -> Entropy = 1
- Entropy(D2) = -[(4/8 x $\log_2$ (4/8) + (4/8 x $\log_2$ (4/8)]
  = -[(1/2 x -1) + (1/2 x -1)] = -[-1/2 – 1/2] = -[-2/2] = 1

- D3: [A, A, A, A, A, A, A , A] -> Highly homogeneous -> Only 1 class -> Entropy = 0
- Entropy(D2) = -[(8/8 x $\log_2$ (8/8)]
  = -[(8/8 x 0)] = 0

Entropy

$$\boldsymbol{Entropy} = -\Sigma \text{pi}.\log_2(\text{pi}) \text{ for i} = 1 \text{ to n}$$

Here, n=number of classes in data, p = proportion of data points belonging to that class

$$\boldsymbol{Entropy} = -\Sigma \text{pi}.\log_2(\text{pi}) \text{ for i} = 1 \text{ to n}$$

Example: Case 2 below …

$$[(\tfrac{4}{8} \text{ x } \log_2(\tfrac{4}{8}) + (\tfrac{4}{8} \text{ x } \log_2(\tfrac{4}{8})]$$

$$= -[(\tfrac{4}{8} \text{ x } -1) \quad + (\tfrac{4}{8} \text{ x } -1)]$$

$$= -([-\tfrac{1}{2}) \quad + (-\tfrac{1}{2})] = -(-1) = 1$$

Data	Observation	Number of classes	Entropy calculation	Entropy value
A, B, C, D, E, F, G, H	Not homogeneous at all	8	-[(1/8 x $\log_2$ (1/8) + (1/8 x $\log_2$ (1/8) …] 8 times	3
A, A, A, A, B, B, B, B	Partially homogeneous	2	[(4/8 x $\log_2$ (4/8) + (4/8 x $\log_2$ (4/8)]	1
A, A, A, A, A, A, A , A	Fully homogeneous	1	-[(8/8 x $\log_2$ (8/8)]	0

Exercise: Split at 4.5 hours of study and 77.5% attendance

Hours Studied	Attendance (%)	Pass Exam?
8	95	Yes
7	90	Yes
6	85	Yes
5	75	Yes
4	70	No
3	65	No
2	60	No
1	55	No
6	80	Yes
3	68	No

DECISION TREE – SOLUTION

We compute entropy and Information Gain for both attributes and choose the best split.

1. ORIGINAL DATASET

Hours Studied	Attendance (%)	Pass Exam?
8	95	Yes
7	90	Yes
6	85	Yes
5	75	Yes
4	70	No
3	65	No
2	60	No
1	55	No
6	80	Yes
3	68	No

Overall (S)

Yes = 5, No = 5, N = 10

$\text{Entropy}(S) = -[0.5 \log_2(0.5) + 0.5 \log_2(0.5)]$

$= -[0.5(-1) + 0.5(-1)]$

$= 1.000$

2A. SPLIT ON HOURS STUDIED

Split at 4.5 hours (≤ 4.5 vs > 4.5)

≤ 4.5 hours (5 records)

Hours Studied	Attendance (%)	Pass Exam?
4	70	No
3	65	No
2	60	No
1	55	No
3	68	No

Yes = 0, No = 5, N = 5

$\text{Entropy} = -[0 \log_2(0) + 1 \log_2(1)]$

$= 0$

> 4.5 hours (5 records)

Hours Studied	Attendance (%)	Pass Exam?
8	95	Yes
7	90	Yes
6	85	Yes
5	75	Yes
6	80	Yes

Yes = 5, No = 0, N = 5

$\text{Entropy} = -[1 \log_2(1) + 0 \log_2(0)]$

$= 0$

Weighted Average Entropy (Hours Studied)

$= \left(\frac{5}{10} \times 0\right) + \left(\frac{5}{10} \times 0\right) = 0$

Information Gain (Hours Studied)

$= 1.000 - 0 = 1.000$

2B. SPLIT ON ATTENDANCE (%)

Split at 77.5% (≤ 77.5% vs > 77.5%)

≤ 77.5% (6 records)

Hours Studied	Attendance (%)	Pass Exam?
5	75	Yes
4	70	No
3	65	No
2	60	No
1	55	No
3	68	No

Yes = 1, No = 5, N = 6

$\text{Entropy} = -[(1/6) \log_2(1/6) + (5/6) \log_2(5/6)]$

$= 0.650$

> 77.5% (4 records)

Hours Studied	Attendance (%)	Pass Exam?
8	95	Yes
7	90	Yes
6	85	Yes
6	80	Yes

Yes = 4, No = 0, N = 4

$\text{Entropy} = -[1 \log_2(1) + 0 \log_2(0)]$

$= 0$

Weighted Average Entropy (Attendance)

$= \left(\frac{6}{10} \times 0.650\right) + \left(\frac{4}{10} \times 0\right) = 0.390$

Information Gain (Attendance)

$= 1.000 - 0.390 = 0.610$

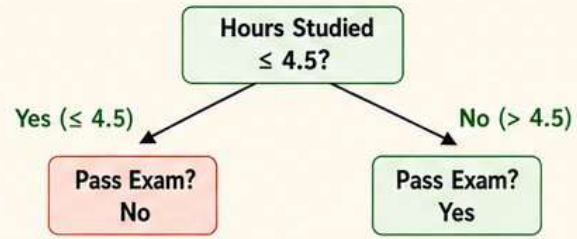
SUMMARY

Attribute	Weighted Avg. Entropy	Information Gain
Hours Studied (split at 4.5)	0.000	1.000
Attendance (split at 77.5%)	0.390	0.610

BEST SPLIT:

Hours Studied (split at 4.5 hours)

It gives the higher Information Gain (1.000) and perfectly separates the data.



Entropy and Information Gain

- We have data about some students. Decide the branching/split by calculating **entropy** and **information gain**

Theory Study	Lab Study	Pass Exam
High	High	Yes
High	Medium	Yes
High	Medium	Yes
High	Low	Yes
Medium	High	Yes
Medium	Medium	No
Medium	Medium	Yes
Medium	Low	No
Low	Low	No

Solution

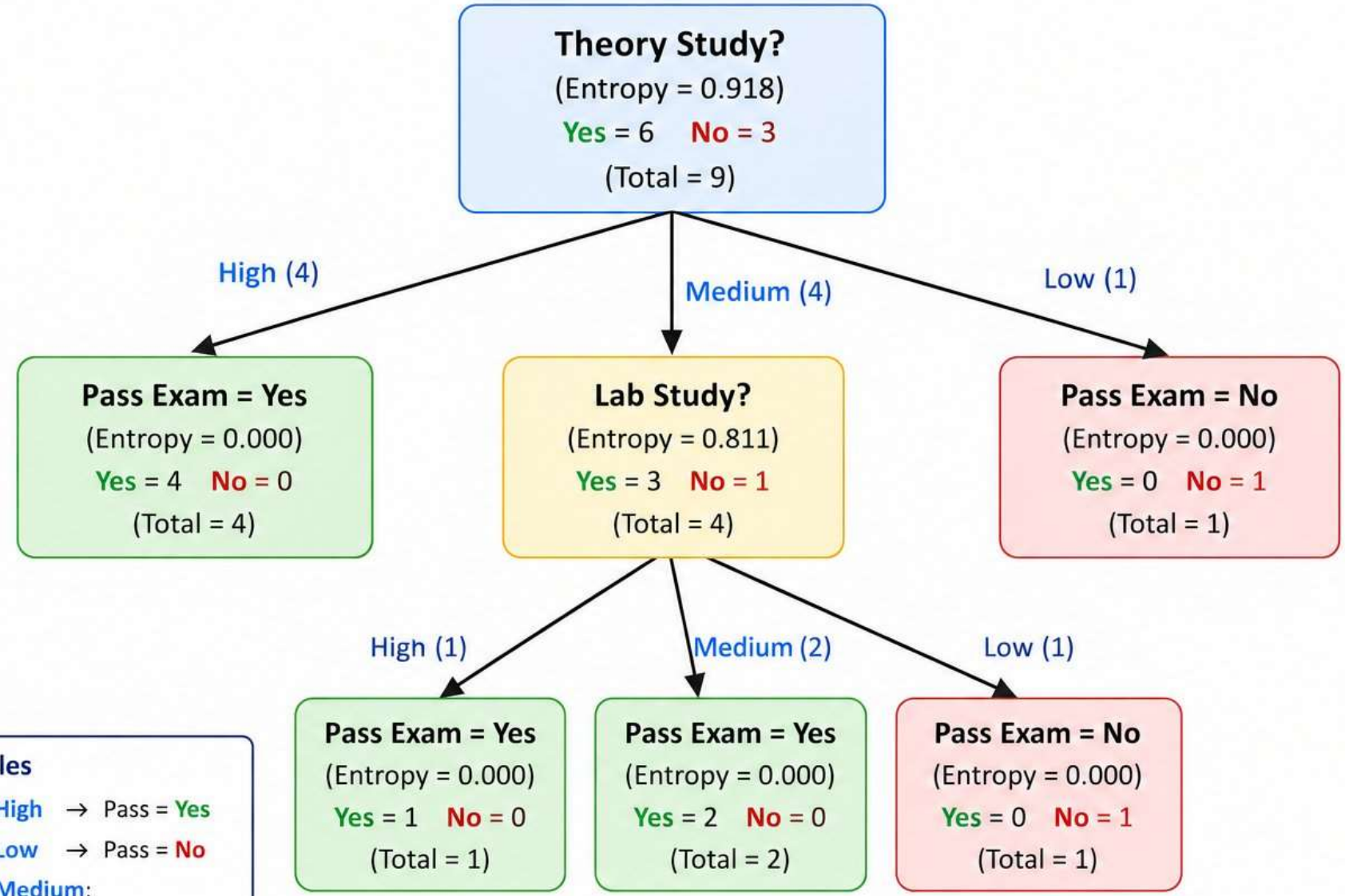
- Overall entropy = $-(\frac{6}{9}\log_2\frac{6}{9}+\frac{3}{9}\log_2\frac{3}{9}) = 0.91$
- Splitting on Theory study
  - E(High) = $-(\frac{4}{4}\log_2\frac{4}{4}+\frac{0}{4}\log_2\frac{0}{4}) = 0$
  - E(Medium) = $-(\frac{2}{4}\log_2\frac{2}{4}+\frac{2}{4}\log_2\frac{2}{4}) = 1$
  - E(Low) = $-(\frac{0}{1}\log_2\frac{0}{1}+\frac{1}{1}\log_2\frac{1}{1}) = 0$
- Splitting on Lab study
  - E(High) = $-(\frac{2}{2}\log_2\frac{2}{2}+\frac{0}{2}\log_2\frac{0}{2}) = 0$
  - E(Medium) = $-(\frac{3}{4}\log_2\frac{3}{4}+\frac{1}{4}\log_2\frac{1}{4}) = 0.81$
  - E(Low) = $-(\frac{1}{3}\log_2\frac{1}{3}+\frac{2}{3}\log_2\frac{2}{3}) = 0.91$
- Conclusion: Theory Study is a better predictor

Theory Study	Lab Study	Pass Exam
High	High	Yes
High	Medium	Yes
High	Medium	Yes
High	Low	Yes
Medium	High	Yes
Medium	Medium	No
Medium	Medium	Yes
Medium	Low	No
Low	Low	No

Weighted average of E(Theory Study) =
$\frac{4}{9}$ x 0 + $\frac{4}{9}$ x 1 + $\frac{1}{9}$ x 0 = 0.444
Information gain = 0.91 – 0.444 = 0.47

Weighted average of E(Lab Study) =
$\frac{2}{9}$ x 0 + $\frac{4}{9}$ x 0.81 + $\frac{3}{9}$ x 0.91 = 0.66
Information gain = 0.91 – 0.66 = 0.24

Decision Tree (using Entropy & Information Gain)



Decision Rules

1. If Theory = High → Pass = Yes
2. If Theory = Low → Pass = No
3. If Theory = Medium:
   - If Lab = High → Pass = Yes
   - If Lab = Medium → Pass = Yes
   - If Lab = Low → Pass = No

Exercise

Customer Id	Subscribed to the Newsletter?	Follows our Social Media Handle?	Has visited our Website?	Purchase? (Outcome)
1	0	1	0	1
2	1	0	1	0
3	1	1	0	1
4	0	0	0	0
5	1	0	0	0
6	1	1	1	1
7	0	1	1	1
8	1	0	1	0
9	0	0	1	0
10	1	1	0	1

Solution

- Overall entropy = $-\left(\frac{5}{10}\log_2\frac{5}{10}+\frac{5}{10}\log_2\frac{5}{10}\right)$ = - (-0.5 – 0.5) = +1
- Split on Newsletter
  - E(Not Subscribed) = $-\left(\frac{2}{4}\log_2\frac{2}{4}+\frac{2}{4}\log_2\frac{2}{4}\right)$ = -[(0.5 x -1) + (0.5 x -1)] = 1
  - E(Subscribed) = $-\left(\frac{3}{6}\log_2\frac{3}{6}+\frac{3}{6}\log_2\frac{3}{6}\right)$ = -[(0.5 x -1) + (0.5 x -1)] = 1
  - Weighted average = $\left(\frac{4}{10}\text{ x }1\right)+\left(\frac{6}{10}\text{ x }1\right)=1$ … Information Gain = 1 – 1 = 0
- Split on Social Media
  - E(Not Following) = $-\left(\frac{5}{5}\log_2\frac{5}{5}+\frac{0}{5}\log_2\frac{0}{5}\right)=0$
  - E(Following) = $-\left(\frac{0}{5}\log_2\frac{0}{5}+\frac{5}{5}\log_2\frac{5}{5}\right)=0$
  - Weighted average = 0 … Information Gain = 1 – 0 = 1
- Split on Visited Web Site
  - E(Not Visited) = $-\left(\frac{3}{5}\log_2\frac{3}{5}+\frac{2}{5}\log_2\frac{2}{5}\right)$ = -[(0.6 x -0.73) + (0.4 x -1.32)] = 0.97
  - E(Visited) = $-\left(\frac{2}{5}\log_2\frac{2}{5}+\frac{3}{5}\log_2\frac{3}{5}\right)$ = -[(0.4 x -1.21) + (0.6 x -0.73)] = 0.97
  - Weighted average = 0.97 … Information Gain = 1 – 0.97 = 0.03
- Conclusion: Split on Social Media

Gini Index (Impurity)

Gini impurity: measures how mixed (impure) a node is. Lower is better: 0 = pure (all one class); maximum impurity for 2 classes = 0.5

Example: predict fruit A or B

#	Color	Class
1	Red	A
2	Yellow	B
…	…	…
20	Yellow	B

- 10 of A, 10 of B; total = 20
- P(A) = 10/20 = 0.5; P(B) = 10/20 = 0.5

1. Entropy (for reference only)

- Measures randomness (higher = more uncertain)
- $H = -\Sigma\, p_i \log_2(p_i)$; for 2 classes: $H = -[p(A) \log_2 p(A) + p(B) \log_2 p(B)]$
- If p(A) = p(B) = 0.5 → H = 1 (maximum)

2. Gini impurity

- Measures impurity (lower = better)
- $G = 1 - \Sigma\, p_i^2$; for 2 classes: $G = 1 - [p(A)^2 + p(B)^2]$
- If p(A) = p(B) = 0.5 → $G = 1 - (0.5^2 + 0.5^2) = 0.5$

Key takeaway: lower Gini = better split | 0 = pure node | 0.5 = max impurity (for 2 classes)

HOW RANDOM FOREST WORKS (DIABETES DATASET)

1 DATASET

Feature 1	...	Feature 8	Outcome
...	...	...	1
...	...	...	0
...	...	...	1
. . .		. . .	. . .

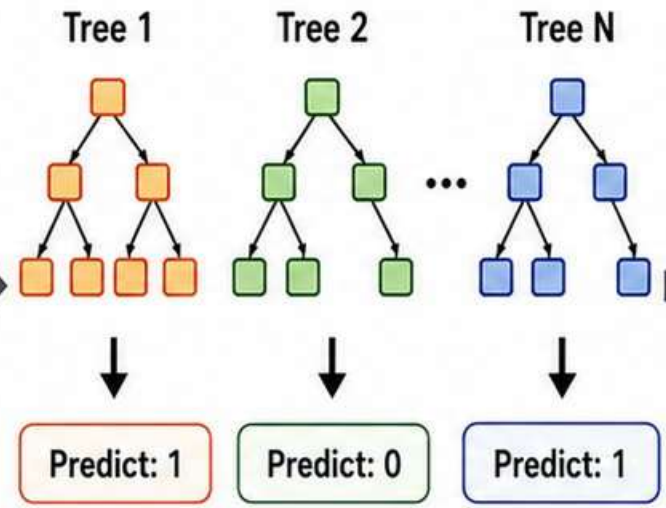
Outcome:
0 = No Diabetes
1 = Diabetes

2 BOOTSTRAP SAMPLING

Sample 1
Sample 2
Sample 3
⋮
Sample N

Random samples (with replacement)

3 TRAIN MULTIPLE DECISION TREES



Each tree is trained on a bootstrap sample and random features.

4 PREDICTIONS FROM EACH TREE

Tree 1 → 1
Tree 2 → 0
⋮
Tree N → 1

(1 = Diabetes, 0 = No Diabetes)

5 MAJORITY VOTING (FINAL PREDICTION)

Votes
1, 0, 1, 1, 0, 1, ...
↓
Majority = 1

FINAL PREDICTION
1 = DIABETES

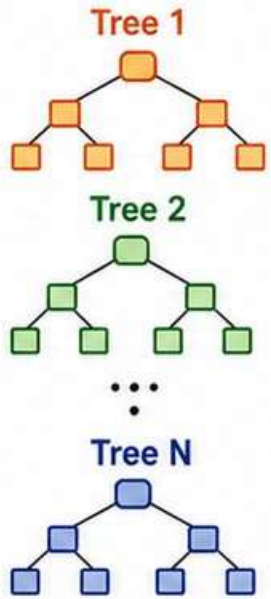
HOW RANDOM FOREST WORKS (DIABETES)

1 NEW PATIENT



Glucose = 142
BMI = 28.7
Age = 33
...

2 GOES TO ALL TREES



3 EACH TREE PREDICTS

Tree 1 → 1
Tree 2 → 0
⋮
Tree N → 1

(1 = Diabetes, 0 = No Diabetes)

4 MAJORITY VOTING

VOTES
1, 0, 1, 1, 0, 1, ...
↓
MAJORITY = 1
(Diabetes)

5 FINAL PREDICTION



PREDICTION = 1
(DIABETES)



Random Forest combines many decision trees trained on different samples and features.
The final prediction is the majority vote of all trees.

When We Have More Classes

- Consider we have three classes with probabilities of 0.4, 0.4, and 0.2 respectively
- Gini impurity = 1 - (p1^2 + p2^2 + p3^2)
- Gini impurity = 1 - (0.4^2 + 0.4^2 + 0.2^2)
- Gini impurity = 1 - (0.16 + 0.16 + 0.04)
- Gini impurity = 1 - 0.36
- Gini impurity = 0.64

- Conclusion: More classes, higher Gini impurity

Classification Techniques: Comparison

Aspect	Logistic Regression	Naive Bayes	Decision Tree
Type	Linear model	Probabilistic model	Rule-based model
Speed	Fast	Very fast	Medium
Assumptions	Linear boundary	Features independent	No assumptions
Overfitting	Low	Low	High (needs pruning)
Interpretability	High	Medium	Very high
Scaling Needed	Yes	No	No
Works Well For	Simple classification	Text/NLP	Complex decision rules

Time Series Forecasting

Time Series

Term	Meaning
Time Series	A sequence of data points collected over time (e.g., daily sales, monthly temperature, yearly GDP)
Purpose	To study how data changes over time and make predictions about the future
Examples	Stock prices, weather records, website traffic, electricity demand
Key Features	Trend, seasonality, cycles, and random noise
Goal	Understand past patterns and forecast future values

Time Series Components

Component	Meaning	Example
Trend	Long-term movement or direction in data	Sales increasing steadily over years
Seasonality	Regular repeating pattern within a fixed period	Ice cream sales rise every summer
Cyclic Variation	Irregular, long-term fluctuations (not fixed like seasons)	Business cycles or economic booms and recessions
Random / Irregular Component	Unpredictable, short-term variations	Sudden drop due to a power outage or strike

Time Series Steps

Step	What We Do	Purpose
1. Collect Data	Gather data recorded over regular time intervals	Build the foundation for analysis
2. Visualize	Plot the data to see patterns, trends, and outliers	Understand overall behavior
3. Check Stationarity	Test if mean and variance remain constant over time	Required for many models (e.g., ARIMA)
4. Transform Data	Apply differencing, log, or scaling to stabilize variance	Make the series stationary
5. Identify Model	Use ACF/PACF plots to select AR, MA, or seasonal terms	Choose the right model form
6. Fit Model	Train ARIMA/SARIMAX or other models	Capture patterns mathematically
7. Validate Model	Compare predictions vs. actual data	Ensure accuracy and reliability
8. Forecast	Predict future time points	Final goal of analysis

What We See In the Output Plot

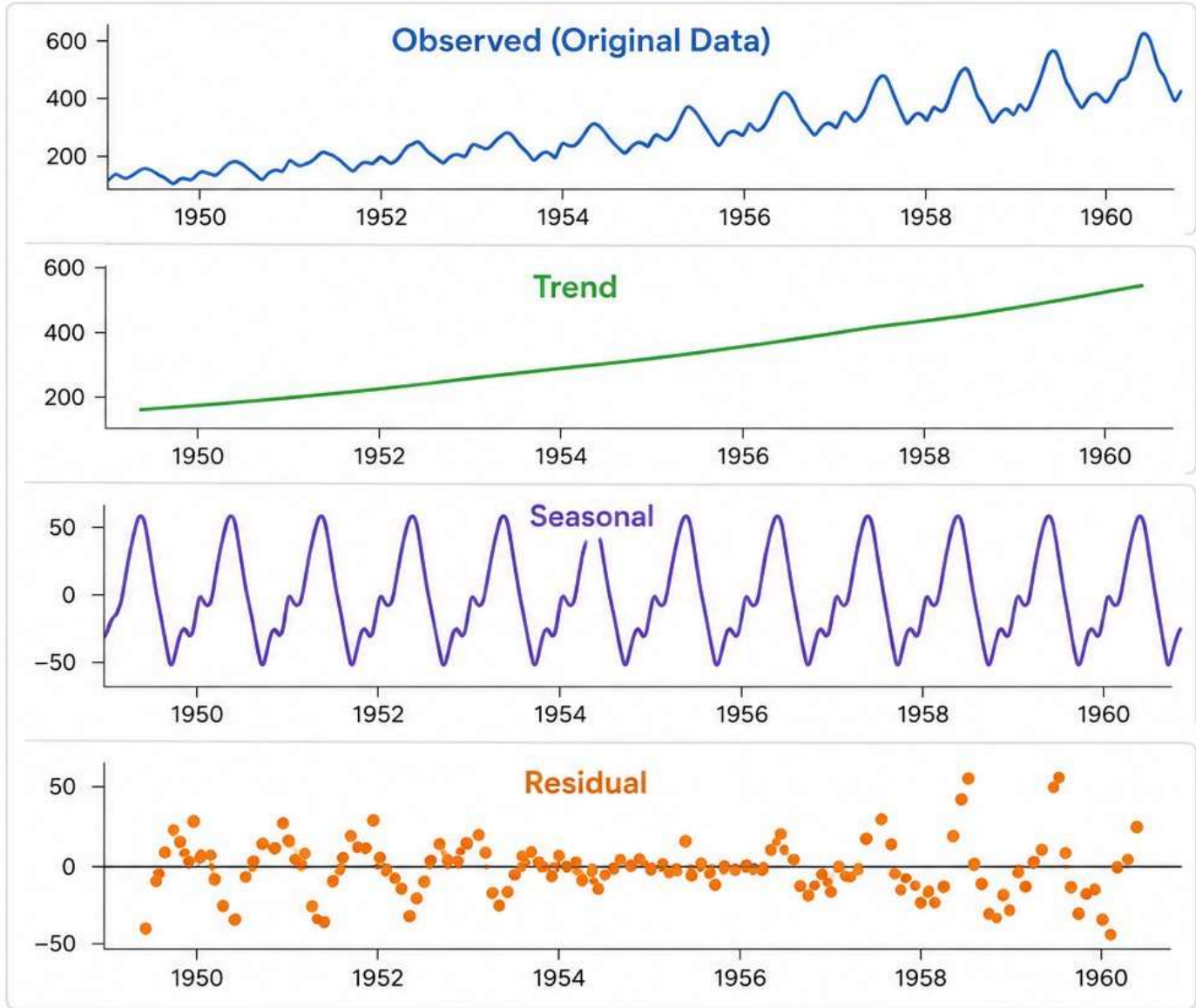
Observed (Original Data)
The actual data we started with.

Trend
Long-term movement or direction in the data.

Seasonal
Repeating pattern that occurs at regular intervals.

Residual
Noise or irregularities left after removing trend and seasonality.

Note
To see cyclicality, we need ADF (TBD).



WHAT IS STATIONARY DATA?

Stationary data looks **stable** over time.

NON-STATIONARY DATA

Changing Mean



Changing Variance



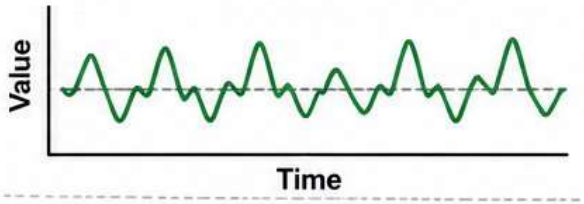
Seasonal Pattern



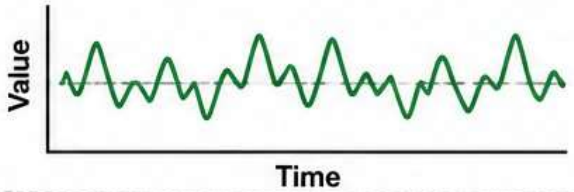
Not stable over time

STATIONARY DATA

Constant Mean



Constant Variance



No Clear Seasonal Pattern



Stable over time

Stationary data = No trend, constant mean and variance, and no repeating seasonal pattern.

Stationary versus Non-Stationary Data

Month	Original Series	Mean up to that Point	First Difference	Diff Mean up to that Point
1	100	100.00	—	—
2	105	102.50	5	5.00
3	111	105.33	6	5.50
4	116	108.00	5	5.33
5	122	110.80	6	5.50
6	127	113.50	5	5.40
7	133	116.29	6	5.50
8	138	119.00	5	5.43
9	144	121.78	6	5.50
10	149	124.50	5	5.44
11	155	127.27	6	5.50
12	160	130.00	5	5.45

Month	Original Series	Mean up to that Point	First Difference	Diff Mean up to that Point
13	166	132.77	6	5.50
14	171	135.50	5	5.46
15	177	138.27	6	5.50
16	182	141.00	5	5.47
17	188	143.76	6	5.50
18	193	146.50	5	5.47
19	199	149.26	6	5.50
20	204	152.00	5	5.47
21	210	154.76	6	5.50
22	215	157.50	5	5.48
23	221	160.26	6	5.50
24	226	163.00	5	5.48

- The running mean of the original series keeps rising (non-stationary); the running mean of the first difference settles around 5.5 (stationary)

Stationary Time Series

- Code: C:\code\Data Analytics\implementation\time_series\3_Check_stationary.py

Concept	Description
Why check stationarity?	Before performing time series analysis (like ARIMA or SARIMAX), we must ensure that the data is stationary
What is stationarity?	A stationary time series is one whose statistical properties – such as mean, variance, and autocorrelation – do not change over time
Non-stationary series	Shows trends, seasonality, or changing variance over time
Goal of preprocessing	Remove trend and seasonality components to make the series stationary
Stationary series example	The data fluctuates around a constant mean and has a constant standard deviation over time

How to Check Stationarity?

Method	Description	Interpretation
1. Visual Inspection	Plot the time series and look for trend or seasonality	If the series shows upward/downward movement or repeating patterns, it is likely non-stationary
2. Rolling Statistics	Calculate and plot rolling mean and rolling standard deviation over time	If they remain constant, the series is stationary
3. Augmented Dickey-Fuller (ADF) Test	A statistical test that checks for a unit root in the series (H0: time series is NOT stationary)	p-value < 0.05 → stationary p-value ≥ 0.05 → non-stationary

1. Shift (Lag) – What & How

What it does: moves the entire series forward or backward in time. It does NOT change the values – only their positions in time.

- Example: shift by 1 (lag = 1)

Time (t)	t1	t2	t3	t4	t5
Original series	100	105	111	116	122
Shifted series (lag = 1)	–	100	105	111	116

- How: for lag = 1, each value at time t moves to time t + 1. The first position has no previous value, so it becomes blank (or NA).

Key idea: shift aligns the series with its past (or future) values so we can compare values at different times. Use shift to align data.

2. Diff (Differencing) – What & How

What it does: calculates the change between consecutive observations. It removes trend (long-term movement) and helps make the series stationary.

- Example: diff with lag = 1

Time (t)	t1	t2	t3	t4	t5
Original series (Xt)	100	105	111	116	122
		105 – 100	111 – 105	116 – 111	122 – 116
Differenced series (Diff Xt)	–	5	6	5	6

- How: for lag = 1, subtract each value from its previous value: Diff Xt = Xt − Xt−1. The first position has no previous value, so it becomes blank (or NA).
- Higher differencing: diff with lag = k compares values k periods apart: Diffk Xt = Xt − Xt−k

Key idea: diff focuses on how much the series changes from one time point to the next. Use diff to remove trend and make the series stationary.

Differencing (Diff) – Removing Trend to Expose the Core Pattern

Trend: Sales increasing every month, Temperatures rising yearly

Differencing: How much did it change from the last point?

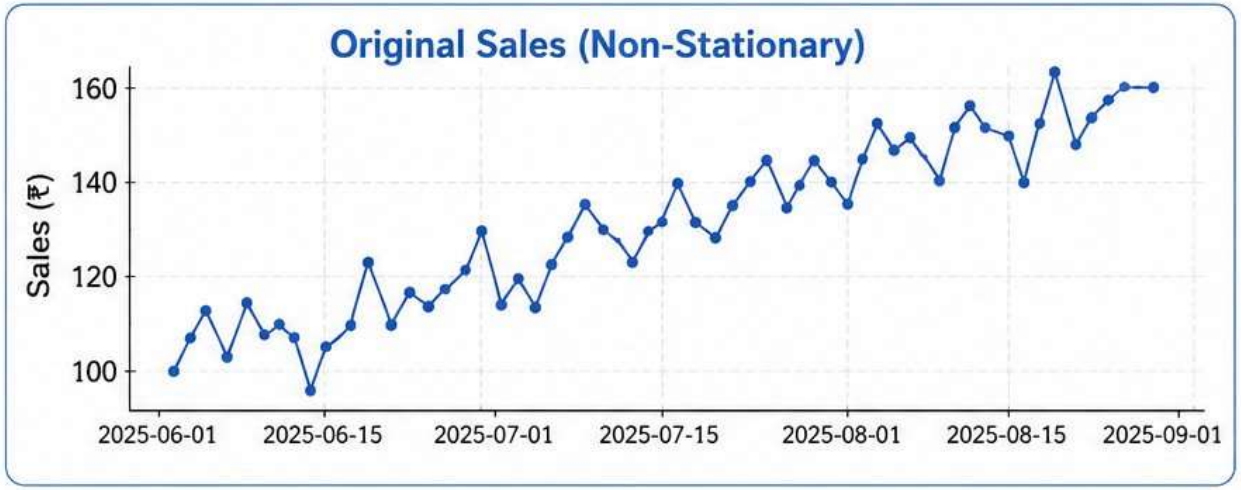
That is, instead of looking at the time *t*, we look at the change ***t* — *t*–1**

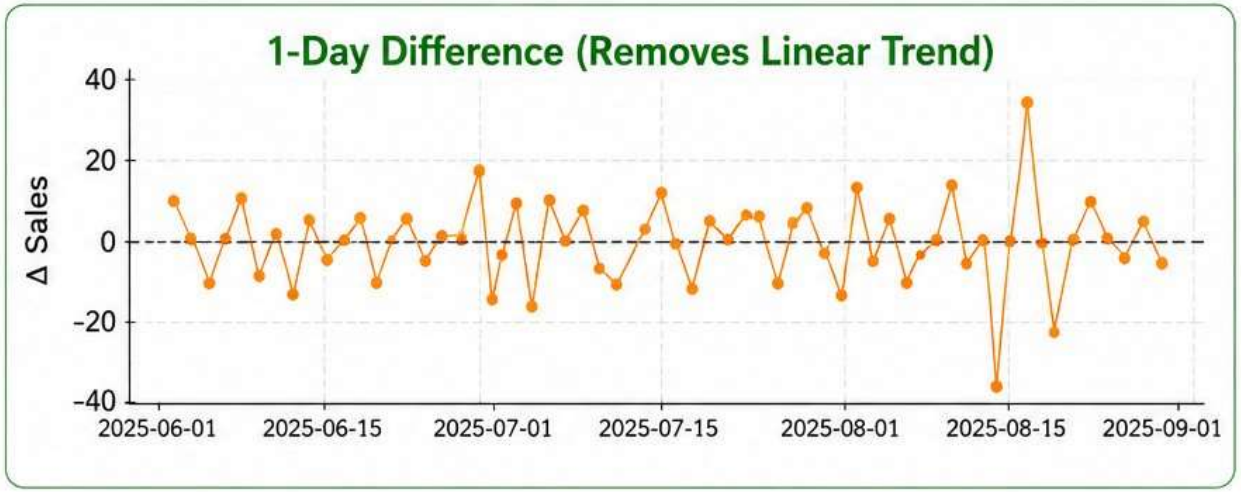
Removes the trend – Making the data stationary

Example: Monthly revenue -> 100, 120, 140, 160, ... -> **Rising trend**

After differencing -> 20, 20, 20, ...

Now it is flat, the ***trend*** is gone -> **Stable pattern** of data



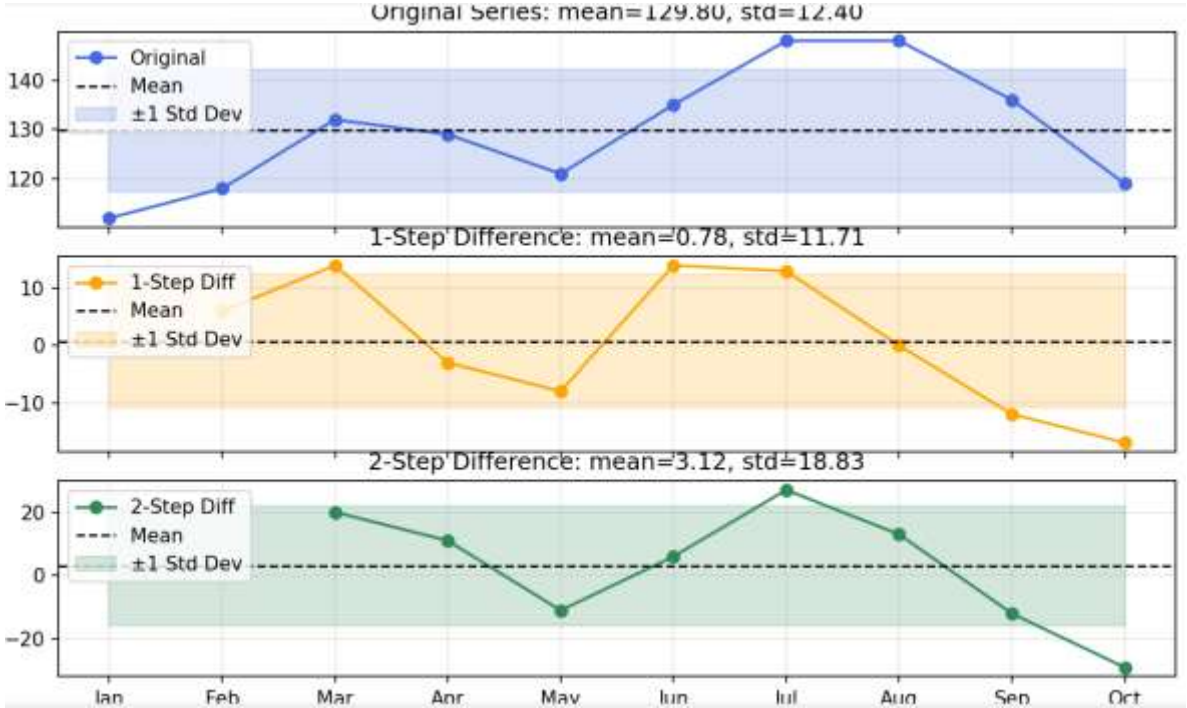


Differencing **removes the trend** and exposes the true underlying pattern, making the series **stationary** and easier to model.

Stationary Time Series

- **Stationary Time Series**: No trend, No changing variance, No seasonality (Ideally)
- Why? ARIMA assumes stationarity
- How?
- Try shift + diff(1), shift + diff(2)

Month	passengers	shift(1)	shiftDiff	shift(2)	shiftDiff2
Jan	112	NaN	NaN	NaN	NaN
Feb	118	112	6	NaN	NaN
Mar	132	118	14	112	20
Apr	129	132	-3	118	11
May	121	129	-8	132	-11
Jun	135	121	14	129	6
Jul	148	135	13	121	27
Aug	148	148	0	135	13
Sep	136	148	-12	148	-12
Oct	119	136	-17	148	-29



ARIMA

Aspect	Description
Definition	A statistical time series model that uses past data to understand patterns and predict future values
Full Form	ARIMA – Auto Regressive Integrated Moving Average
Core Idea	Combines three components – AR (Auto Regression), I (Integration / Differencing), MA (Moving Average) – to capture trend and autocorrelation in time series data
Autoregressive Nature	The model predicts future values based on past observations and past forecast errors
Example	Predicting airline passenger counts or stock prices using only their past values
Best Suited For	Short-term forecasting of non-seasonal time series data after making it stationary

What is ARIMA?

ARIMA is a method to predict future values in a time series using only its past values

- ARIMA stands for AutoRegressive Integrated Moving Average. It combines 3 ideas:
  - AR (AutoRegressive): use past values to predict the next value
  - I (Integrated): make the data stable (remove trend/changes) if needed
  - MA (Moving Average): use past prediction errors to improve accuracy

Example: suppose we have daily sales (in units)

Day	1	2	3	4	5	6	7	8	9	10
Sales	120	132	118	139	150	148	160	158	170	165

- ARIMA looks at this past data, learns the patterns (like increase, decrease, ups and downs), and then predicts the next value

In simple words: ARIMA = learn from the past to predict the future

ARIMA Components: AR, I and MA

Component	Idea	Example	How it works
AR (AutoRegressive)	Uses past values to predict the next value. “The next value depends on its own past values.”	Time 1–5: values 10, 12, 13, 15, 14. To predict Y6, AR looks at the previous values and finds a pattern.	$Y6 = a1 \cdot Y5 + a2 \cdot Y4 + \ldots + ap \cdot Y6-p + c$ (uses past values – lags)
I (Integrated)	Makes the data stable (stationary) by removing trend or changes. “Transform the data to make it stable.”	Original (not stable): 10, 12, 14, 16, 18 After differencing once: 2, 2, 2, 2 – the trend is removed	$Y't = Yt - Yt-1$ If needed, difference again until the series becomes stable
MA (Moving Average)	Uses past prediction errors (mistakes) to improve accuracy. “Learn from past mistakes.”	Actual: 10, 12, 13, 15, 14 Predicted: –, 11, 12.5, 14, 14.5 Error ($Y - \hat{Y}$): –, 1, 0.5, 1, −0.5	$e6 = b1 \cdot e5 + b2 \cdot e4 + \ldots + bq \cdot e6-q + c$ (uses past prediction errors)

AR looks at past values + I makes data stable + MA learns from past errors. Together, ARIMA gives a better understanding of the data and more accurate predictions.

PACF PLOT AND WHY IT GIVES US VALUE OF *p*

PACF = Partial AutoCorrelation Function

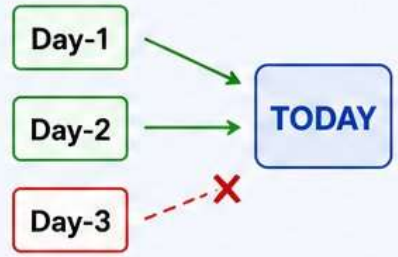
1. WHAT DOES PACF MEAN?

PACF tells us how many previous values **directly** affect the current value.

EXAMPLE

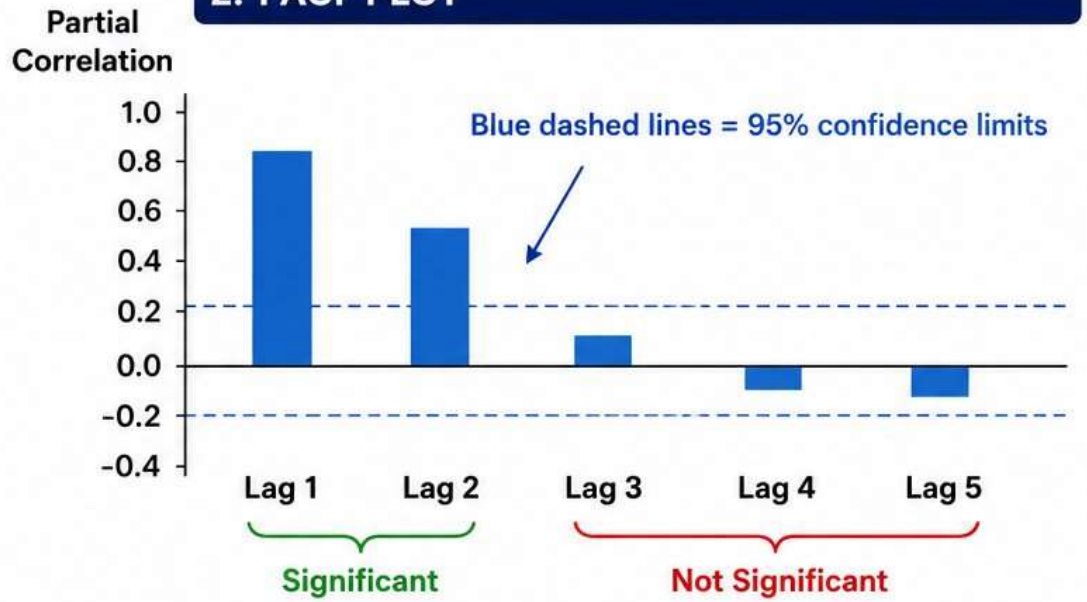
Suppose today's sales are affected by:

- ✓ Yesterday's sales (1 day ago)
- ✓ Sales 2 days ago
- ✗ Sales 3 days ago and earlier



➡ Only 2 direct lags (Day-1 and Day-2) matter.

2. PACF PLOT



3. NUMBER OF SIGNIFICANT BARS = *p*

- ✓ Lag 1 → significant
- ✓ Lag 2 → significant
- ✗ Lag 3 and onward → not significant

➡ **p = 2**

➡ So, the value of **p** (AR order) is **2**.

4. SIMPLE RULE

PACF Pattern	p Value
Only first 1 bar is significant	p = 1
First 2 bars are significant	p = 2
First 3 bars are significant	p = 3

IN SIMPLE WORDS:

PACF tells ARIMA how many **past observations (AR terms)** to use.

ACF vs. PACF – THE SIMPLE TRUTH

Example: Yesterday affects today, and the day before yesterday affects today only through yesterday.

ACF = TOTAL CORRELATION (DIRECT + INDIRECT)

ACF looks at EVERY path from the past to today.



Indirect path: Day -2 → Day -1 → Day 0

Example (Temperature °C)

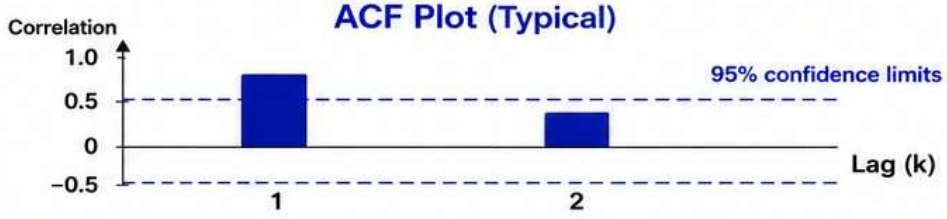
Day -2	Day -1	Day 0 (Today)
28	30	32

Day -2 → Day -1: 28 → 30 (+2°C)
Day -1 → Day 0: 30 → 32 (+2°C)

Overall change Day -2 → Day 0 = +4°C

★ **ACF at lag 2 is HIGH**
because Day -2 and Day 0 move together THROUGH THE CHAIN (indirect path).

ACF Plot (Typical)



ACF counts DIRECT + INDIRECT effects.
It asks: "Are they related in any way?"

PACF = DIRECT CORRELATION ONLY

PACF removes the middle step and checks for a UNIQUE direct effect.



Example (Temperature °C)

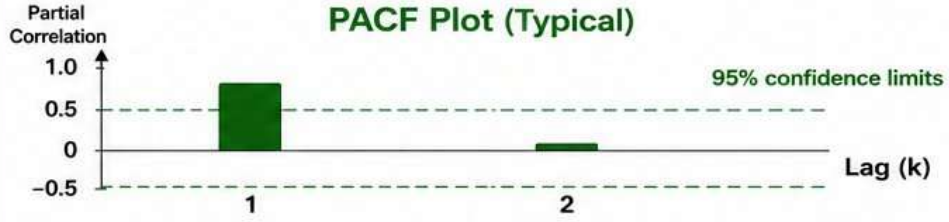
Day -2	Day -1	Day 0 (Today)
28	30	32

1. We already know Day -1 (30°C).
2. After accounting for Day -1, does Day -2 (28°C) add any new information about Day 0 (32°C)?

No. Once we know Day -1, Day -2 does not help predict Day 0.

★ **PACF at lag 2 is ~ 0 (NOT significant)**
because Day -2 has NO DIRECT effect on Day 0.
Its effect was only THROUGH Day -1.

PACF Plot (Typical)



PACF counts ONLY DIRECT effects.
It asks: "Is the relationship direct?"

In Short: ACF = Full Story (Direct + Indirect) | PACF = New, Unique Chapter (Direct Only)

SARIMAX

Full Form	SARIMAX – Seasonal Auto Regressive Integrated Moving Average with eXogenous variables
Purpose	Extends ARIMA to handle seasonal patterns that repeat at regular intervals (e.g., monthly, quarterly, yearly)
When to Use	When the time series shows both trend and seasonality (trend + seasonality + noise)
Model Notation	SARIMAX (p, d, q) × (P, D, Q, s): non-seasonal (p, d, q) × seasonal (P, D, Q, s) + exogenous variables
Seasonal Parameters	P – seasonal autoregressive order D – seasonal differencing order Q – seasonal moving average order s – seasonal period: number of time steps in one full seasonal cycle (e.g., 12 for monthly data)
Example	Monthly airline passengers, where seasonal effects repeat every 12 months
Advantage	Captures both short-term (AR & MA) and seasonal long-term (seasonal AR & MA) dependencies in one unified model

Monte Carlo Simulation

Concept	Explanation	Example / Notes
Simulation	Creating a hypothetical model of a real-world process to test outcomes under different conditions (a “what-if” analysis)	Designing a car crash test virtually instead of physically crashing cars
Why Simulation?	Real-world testing may be impossible, unsafe, or too expensive	Weather forecasting, financial risk modeling, supply chain planning
Types of Simulation	Deterministic: fixed inputs, same output every time Stochastic: includes random inputs, variable output	Deterministic → physics model Stochastic → dice rolling or demand uncertainty
Monte Carlo Simulation	A statistical technique that uses random sampling to estimate possible outcomes of uncertain events	Used to model investment returns, project risks, or queue lengths
Why “Monte Carlo”?	Named after the Monte Carlo Casino in Monaco – randomness and luck symbolize the random sampling nature	—
Key Idea	Repeat random experiments many times → observe the distribution of results to estimate probabilities	e.g., simulate 10,000 outcomes of stock returns to estimate profit chance

Linear Programming Problem (LPP)

Linear Programming Problem (LPP)

Section	Key Description	Example/Role
Overview	A **mathematical optimization technique** for finding the best result (optimal solution) under specific constraints.	Used for **resource allocation** decisions.
Goal	To achieve the absolute best value—either **Maximize** a desired outcome or **Minimize** an undesired outcome.	Maximize **Profit** (Z) or Minimize **Cost**.
Importance	Widely applied in fields like **Manufacturing, Logistics, Finance, and Agriculture** for efficiency.	Scheduling, Inventory Management, Budgeting.

LPP – Key Components

Component	Description	Example	Role in the LPP
1. Objective Function	A **linear equation** that represents the **goal** to be optimized (maximized or minimized). It defines the quantity we want to achieve the best value for.	**Maximize:** Z = 5x + 3y (Z = total profit)	**The Goal**
2. Constraints	A system of **linear inequalities or equations** that represent the **limitations** (e.g., resources, time, budget). The variables must satisfy these conditions.	x + y <= 100 (e.g., limited raw material or total production capacity)	**The Limitations**
3. Decision Variables	The **quantities** that can be controlled or changed by the decision-maker (e.g., number of products to produce, hours to allocate). These are the unknowns the LPP solves for.	**x and y** (quantities of product A and product B, respectively)	**The Unknowns**
4. Non-Negativity Restrictions	A standard requirement in most practical LPPs stating that the decision variables cannot take negative values (e.g., you can't produce a negative quantity of a product).	x >= 0 and y >= 0	**Practicality**

LPP Example

- **Problem:** We run a factory making chairs (Rs 300 profit per chair sold) and tables (Rs 1,000 profit per table sold). We have 400 units of wood: One chair needs 5 units, one table needs 10 units. We have 300 hours of worker time: One chair takes 2 hours, one table needs 6 hours. How many tables and chairs should we make to maximize profit?

Component	Description	Mathematical Expression	Notes
Decision Variables	The quantities we need to determine to maximize profit.	x = Number of Chairs to Produce	Profit: Rs 300 per chair
		y = Number of Tables to Produce	Profit: Rs 1,000 per table
Objective Function	The goal is to maximize the total profit (Z).	**Maximize** Z = 300x + 1000y	This is the equation we optimize.
Constraint 1	Limitation on **Wood** (Total available: 400 units).	**5x + 10y <= 400**	5 units of wood per chair, 10 units per table.
Constraint 2	Limitation on **Worker Time** (Total available: 300 hours).	**2x + 6y <= 300**	2 hours per chair, 6 hours per table.
Non-Negativity	Physical constraint: production quantities cannot be negative.	**x >= 0, y >= 0**	Required for all practical LPPs.

Minimization Objective

- You need a loan of ₹10,00,000 and have 3 options
  - Bank A: 9% interest, 5 years (EMI: ₹2,076 per lakh)
  - Bank B: 10% interest, 3 years (EMI: ₹3,227 per lakh)
  - Bank C: 12% interest, 2 years (EMI: ₹4,707 per lakh)

- Constraints
  - Total loan must be ₹10 lakhs
  - Minimum ₹2 lakhs from Bank A (lower risk)
  - Maximum ₹4 lakhs from Bank C (higher interest)

- Objective: Minimize total monthly EMI payment

Car Example

Overview

- 30 working days in a month
- Two cars: A and B
- Resources available: 1 robot, 2 engineers, 1 detailer
- The detailer works 21 days
- Profit per car: Car A €30,000; Car B €45,000
- Goal: maximize profit (Z)

Decision variables

- x = number of A cars to produce
- y = number of B cars to produce

Objective function

- Maximize $Z = 30000x + 45000y$

Constraints

- Robot time: $3x + 4y \leq 30$
- Engineer time: $5x + 6y \leq 30$
- Detailer time: $1.5x + 3y \leq 21$
- Non-negativity: $x \geq 0, y \geq 0$

Problem: find maximum Z

Student Course Planning Problem

- A student needs to prepare for 3 modules: Python programming (3 credits), SQL (2 credits), and advanced analytics (5 credits)
- Requirement: the student must get at least one credit in every subject
- Time available: 120 hours of study
- Hours required for one credit: Python 15 hours, SQL 18 hours, Analytics 20 hours
- Decision: how much to study in each subject (in credits)

Question: how can the student study to get maximum total credits?

Student Course Planning Problem: Formulation

Problem

- Prepare for three modules: Python Programming (3 credits), SQL (2 credits), Advanced Analytics (5 credits)

Decision variables

- x = credits earned in Python
- y = credits earned in SQL
- z = credits earned in Analytics

Hours required per credit

- Python 15 h; SQL 18 h; Analytics 20 h

Objective

- Maximize the total credits earned: maximize x + y + z

Constraint	Meaning	Formula
1. Study time limit	Only 120 hours are available	$15x + 18y + 20z \leq 120$
2. Minimum Python	At least one credit in Python	$x \geq 1$
3. Minimum SQL	At least one credit in SQL	$y \geq 1$
4. Minimum Analytics	At least one credit in Analytics	$z \geq 1$
5. Non-negativity		$x \geq 0, y \geq 0, z \geq 0$

Goal: maximize the total number of credits earned within 120 study hours

Problem: Loan Planning

You need a loan of ₹10,00,000 and have 3 options:

Bank	Interest	Tenure	EMI per lakh
Bank A	9%	5 years	₹2,076
Bank B	10%	3 years	₹3,227
Bank C	12%	2 years	₹4,707

Constraints

- Total loan must be ₹10 lakhs
- Minimum ₹2 lakhs from Bank A (lower risk)
- Maximum ₹4 lakhs from Bank C (higher interest)

Objective: minimize total monthly EMI payment

Loan Minimization Problem

Problem

- You need a loan of ₹10,00,000 and have 3 options

Decision variables

- x = loan amount from Bank A (in lakhs)
- y = loan amount from Bank B (in lakhs)
- z = loan amount from Bank C (in lakhs)

Loan options (EMI per lakh)

- Bank A: 9% interest, 5 years (₹2,076 per lakh)
- Bank B: 10% interest, 3 years (₹3,227 per lakh)
- Bank C: 12% interest, 2 years (₹4,707 per lakh)

Objective

- Minimize the total monthly EMI: minimize 2076x + 3227y + 4707z

Constraint	Meaning	Formula
1. Total loan requirement	Total loan must be ₹10 lakhs	$x + y + z = 10$
2. Minimum from Bank A	Minimum ₹2 lakhs from Bank A (lower risk)	$x \geq 2$
3. Maximum from Bank C	Maximum ₹4 lakhs from Bank C (higher interest)	$z \leq 4$
4. Non-negativity		$x \geq 0, y \geq 0, z \geq 0$

Goal: minimize the total monthly EMI for a loan of ₹10 lakhs by choosing the best mix from Bank A, Bank B, and Bank C

Dimensionality Reduction

Reduce features, retain what matters

What is it?	Reducing the number of features (variables) while retaining as much important information as possible
Why needed?	Too many features → redundant, correlated data → slower models and risk of overfitting
Popular method	Principal Component Analysis (PCA)
How PCA works	Finds relationships among features and creates new independent variables (principal components). Each component = weighted combination of original features and captures maximum variance (information).
Important note	PCA is a transformation, not feature selection – component ≠ feature
Outcome	A smaller set of uncorrelated components that summarize most of the data's variability

What Pattern Does PCA See?

The data we have (5 variables)

Person	Height (cm)	Weight (kg)	Waist (cm)	BMI	Body Fat (%)
P1	160	55	72	21.5	22
P2	165	60	76	22.0	23
P3	170	65	80	22.5	24
P4	175	70	84	22.9	25
P5	180	75	88	23.1	26
P6	185	80	92	23.4	27
P7	190	85	96	23.5	27
P8	195	90	100	23.7	28

Step 1: notice what moves together

- Look down each column
- As people go from smaller to larger, all 5 measurements increase together
- Because everything increases together, PCA creates PC1 = overall body size
- This single component captures the common “overall size” pattern across all 5 variables

PC1 (overall body size): people with higher PC1 values have higher height, weight, waist size, BMI and body fat %

How Different Loadings (Weights) Change PC1

- PC1 is a weighted combination of all variables (same data as the previous slide). Changing the weights changes what PC1 represents.

Example	Height	Weight	Waist	BMI	Body Fat %	What PC1 represents
1. Equal weights	0.20	0.20	0.20	0.20	0.20	Overall body size – all measurements contribute equally; people are ordered from smallest overall size to largest
2. More weight on weight	0.10	0.50	0.15	0.15	0.10	Heaviness – driven mainly by weight; a person can have a higher PC1 even if other measures are lower, as long as their weight is higher

Changing the weights changes what PC1 captures: equal weights → overall size; more weight on a variable → that variable drives PC1 more

PCA in 2D: Why PC1 Captures the Maximum Variance

The line that keeps the projected points the most spread out has the maximum variance

	1. Project onto PC1 (diagonal line)	2. Project onto horizontal line	3. Project onto vertical line
Variance	MAXIMUM	SMALLER	SMALLEST
What happens	Points remain far apart after projection; large spread	Only height values matter (x-coordinates); some points get closer after projection	Only weight values matter (y-coordinates); points pile up after projection
Information kept	Both height AND weight information are captured together	All weight differences are ignored	All height differences are ignored
Result	High variance	Captures ONLY height variance	Captures ONLY weight variance

PCA - Data

- Consider the following data

Student	Math	Science
A	50	52
B	60	65
C	70	68
D	80	78

- Observation: Students who are good in Maths are also good in science
- Problem: Why keep two columns when they almost convey the same thing?
- PCA: Create a new score to replace both

How PCA Works?

- Creates a new column
  - Overall Score = Combination of Math + Science
  - Something like: A -> 51, B -> 63, C -> 69, D - > 79
- Uses weights, here w1 and w2
  - w1: Weight given to Math
  - w2: Weight given to Science
- If both are equally important, say w1 = 0.7 and w2 = 0.7
- New score = 0.7 x Math + 0.7 x Science
- PCA finds w1 and w2 such that
  - Students are spread out nicely
  - Differences are clearly visible
- We want w1 and w2 showing maximum variance in the data

Finding Weights: Example

Student	Math (x_1)	Science (x_2)
A	50	52
B	60	65
C	70	68
D	80	78

	Balanced	PC1
A	0.7×50 + 0.7×52 = 71.4	71.4
B	0.7×60 + 0.7×65 = 87.5	87.5
C	0.7×70 + 0.7×68 = 96.6	96.6
D	0.7×80 + 0.7×78 = 110.6	110.6

Suppose w1 = 0.7, w2 = 0.7
Variance = 202.7

	Math-Heavy	PC1
A	0.9×50 + 0.3×52 = 60.6	60.6
B	0.9×60 + 0.3×65 = 73.5	73.5
C	0.9×70 + 0.3×68 = 83.4	83.4
D	0.9×80 + 0.3×78 = 95.4	95.4

Suppose w1 = 0.9, w2 = 0.3
Variance = 163.7

	Science-Heavy	PC1
A	0.3×50 + 0.9×52 = 61.8	61.8
B	0.3×60 + 0.9×65 = 76.5	76.5
C	0.3×70 + 0.9×68 = 82.2	82.2
D	0.3×80 + 0.9×78 = 94.2	94.2

Suppose w1 = 0.3, w2 = 0.9
Variance = 135.7

In this example, the first set of weights will be chosen, since variance is maximum -> Maximum separation between students … Of course, we have just taken three samples out of many possible

Why 0.7? Why Not 0.9 or Why Not 900?

- PCA rule: All candidate directions must have unit length
- Meaning: $w1^2 + w2^2$ must be = 1
- Here $0.7^2 + 0.7^2 = 0.98 \approx 1$
- But $0.9^2 + 0.9^2 = 1.62 \neq 1$

Variance Calculations

Suppose w1 = 0.7, w2 = 0.7

Item	Balanced	PC1
A	0.7 × 50 + 0.7 × 52	71.4
B	0.7 × 60 + 0.7 × 65	87.5
C	0.7 × 70 + 0.7 × 68	96.6
D	0.7 × 80 + 0.7 × 78	110.6

Step 1: mean

- $\bar{x}$ = (71.4 + 87.5 + 96.6 + 110.6) / 4 = 366.1 / 4 = 91.525

Step 2: compute $(x - \bar{x})^2$

Item	x	$x - \bar{x}$	$(x - \bar{x})^2$
A	71.4	−20.125	405.016
B	87.5	−4.025	16.201
C	96.6	5.075	25.756
D	110.6	19.075	363.856
Sum			810.828

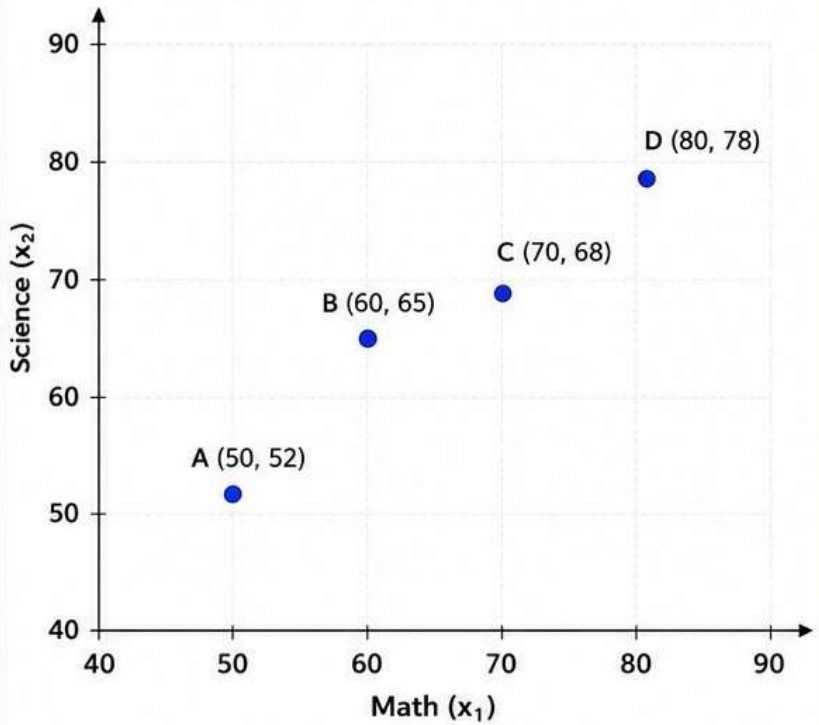
- Population variance (divide by N = 4): σ^2 = 810.828 / 4 = 202.707
- Sample variance (divide by N − 1 = 3): s^2 = 810.828 / 3 = 270.276

Variance ≈ 202.7

Step 1: Original student marks data

Student	Math (x_1)	Science (x_2)
A	50	52
B	60	65
C	70	68
D	80	78

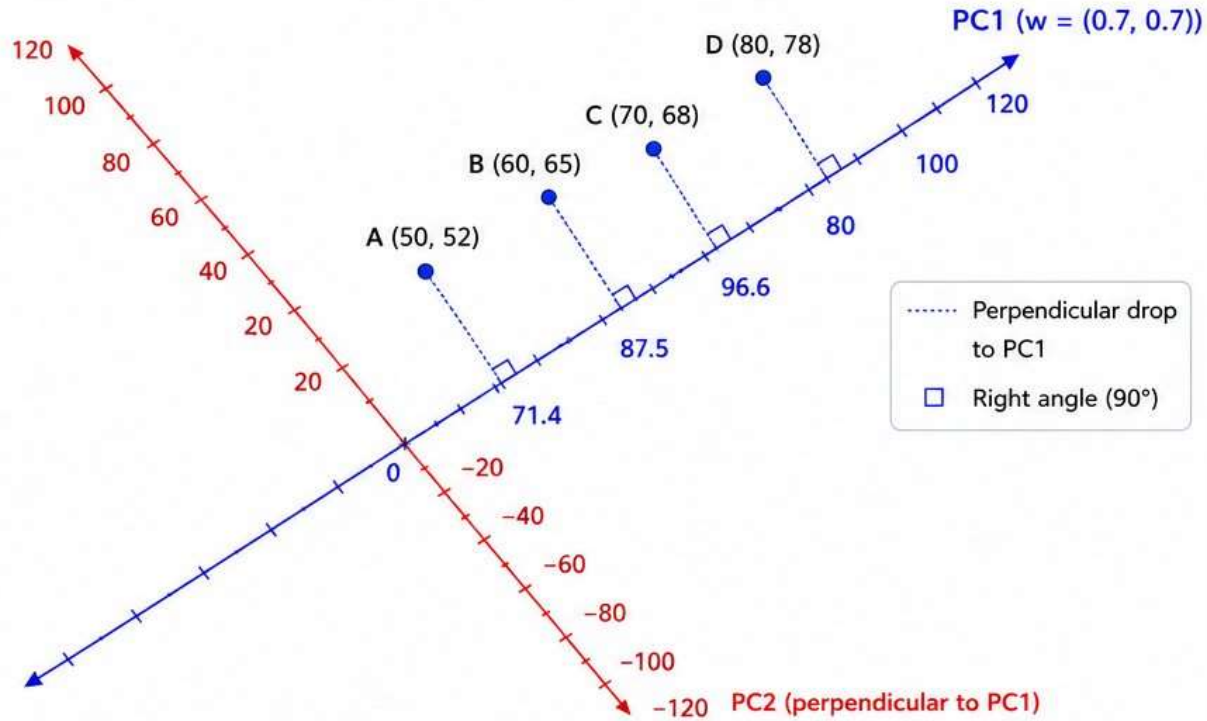
Scatter plot in original (Math, Science) axes



The points roughly lie along a line with positive slope.
PCA will find the direction of maximum spread.

Step 2: Move axes to the direction of weights w = (0.7, 0.7)

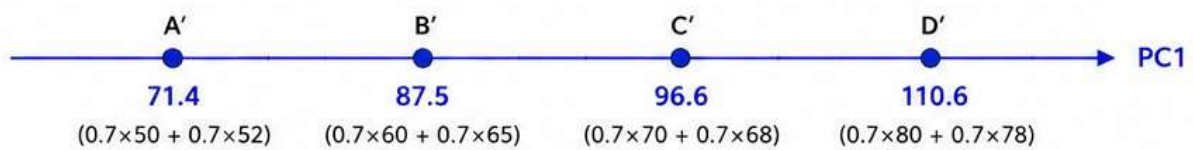
The new blue axis (PC1) has direction w = (0.7, 0.7) (45° line).
The origin (0,0) stays the same. Only the axes rotate.



Step 3: Points projected on the new blue axis (PC1)

The 1D coordinate on PC1 is computed as:

projection = 0.7 × (Math) + 0.7 × (Science) = 0.7 x_1 + 0.7 x_2

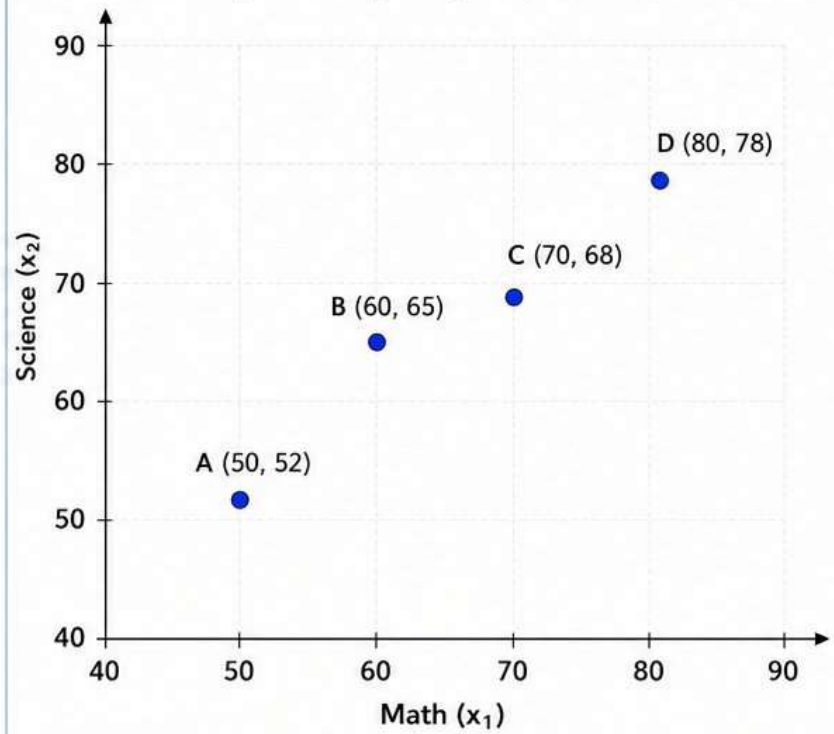


These are the 1D projected values (positions on PC1).
They match the "Balanced (w_1=0.7, w_2=0.7)" column: 71.4, 87.5, 96.6, 110.6.

Step 1: Original student marks data

Student	Math (x_1)	Science (x_2)
A	50	52
B	60	65
C	70	68
D	80	78

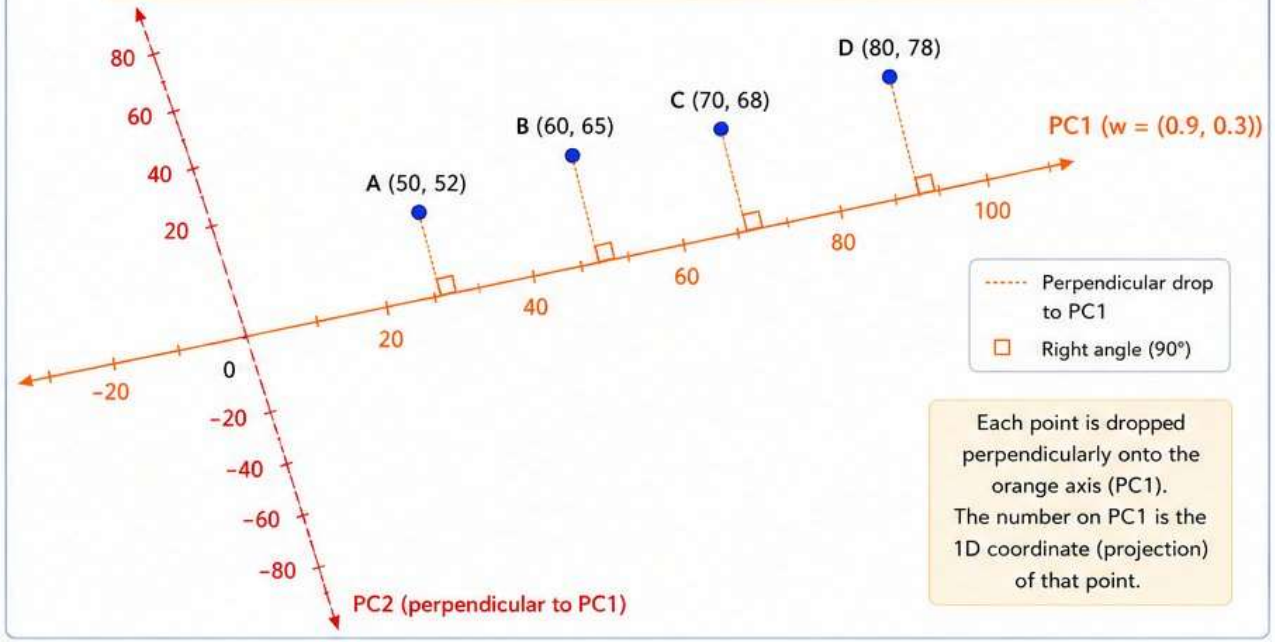
Scatter plot in original (Math, Science) axes



The points roughly lie along a line with positive slope.
PCA finds the direction of maximum spread.

Step 2: Move axes to the direction of weights w = (0.9, 0.3)

The new orange axis (PC1) has direction w = (0.9, 0.3)
(a shallow line: mostly along Math direction).
The origin (0,0) stays the same. Only the axes rotate.

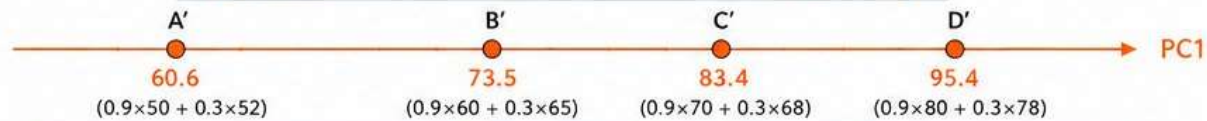


Each point is dropped perpendicularly onto the orange axis (PC1).
The number on PC1 is the 1D coordinate (projection) of that point.

Step 3: Points projected on the orange axis (PC1)

The 1D coordinate on PC1 is computed as:

projection = 0.9 × (Math) + 0.3 × (Science) = 0.9 x_1 + 0.3 x_2



These are the 1D projected values (positions on PC1).
They match the "Math-Heavy (w_1=0.9, w_2=0.3)" column: 60.6, 73.5, 83.4, 95.4.

Why this is a poorer PCA line?

- The projected values (60.6, 73.5, 83.4, 95.4) are closer together.
- The spread (range = 95.4 − 60.6 = 34.8) is smaller than with w = (0.7, 0.7) (range = 110.6 − 71.4 = 39.2).
- Therefore, this direction captures less of the data's variation.
- PCA prefers the direction with maximum spread (w = (0.7, 0.7)).

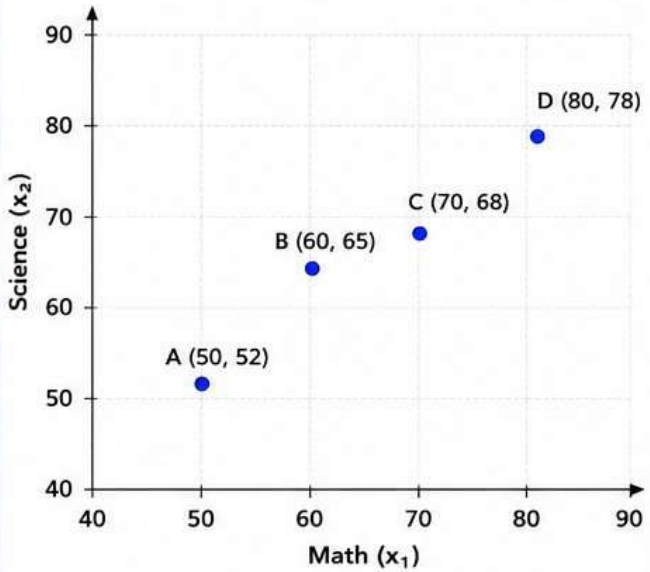
WHY THE SAME POINTS HAVE DIFFERENT PROJECTIONS

Projecting onto different lines measures "movement" in different directions.

1. Original data (same for both cases)

Points (Math, Science):

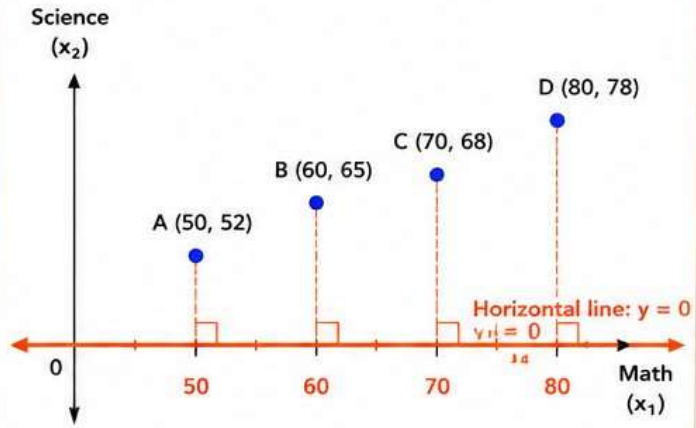
A (50, 52) B (60, 65) C (70, 68) D (80, 78)



These points are the SAME in all cases.
Only the projection line changes.

2. Project onto a HORIZONTAL line (y = 0)

This line measures ONLY horizontal movement (Math direction).



Where the points land (their 1D projections):

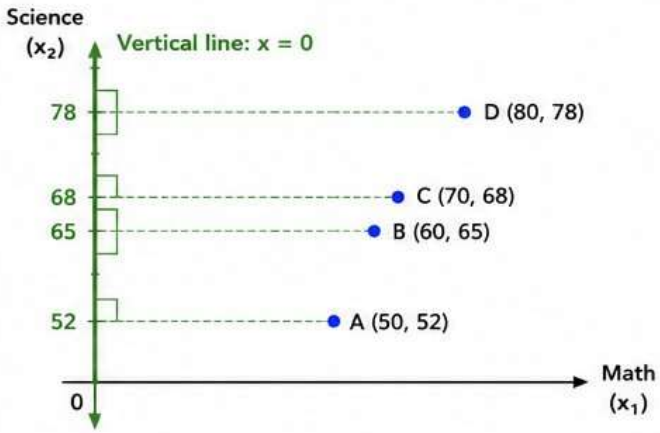


Only the x (Math) values are kept.
The vertical (Science) part is ignored.

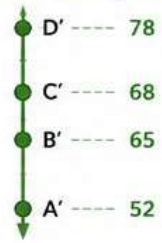
Spread on this line = 80 − 50 = 30

3. Project onto a VERTICAL line (x = 0)

This line measures ONLY vertical movement (Science direction).



Where the points land (their 1D projections):



Only the y (Science) values are kept.
The horizontal (Math) part is ignored.

Spread on this line = 78 − 52 = 26

Why do the same points have different relative places/distances?

- Each line defines its own 1D ruler.
- A point's projection is "how far you travel along that line from the origin (0,0)".
- Moving mostly horizonatly changes your position a lot on the horizontal line, but hardly changes it on the vertical line.
- Moving mostly vertically changes your position a lot on the vertical line, but hardly changes it on the horizontal line.

💡 **Key idea:** Different PCA lines look at the data from different directions.
So the same points end up at different positions and with different spreads on those lines.

PCA Visualized

- Imagine dropping all points onto each line
- Now look at how spread out they are on that line
- We choose the line, where points are most spread out
- That line captures the maximum information

